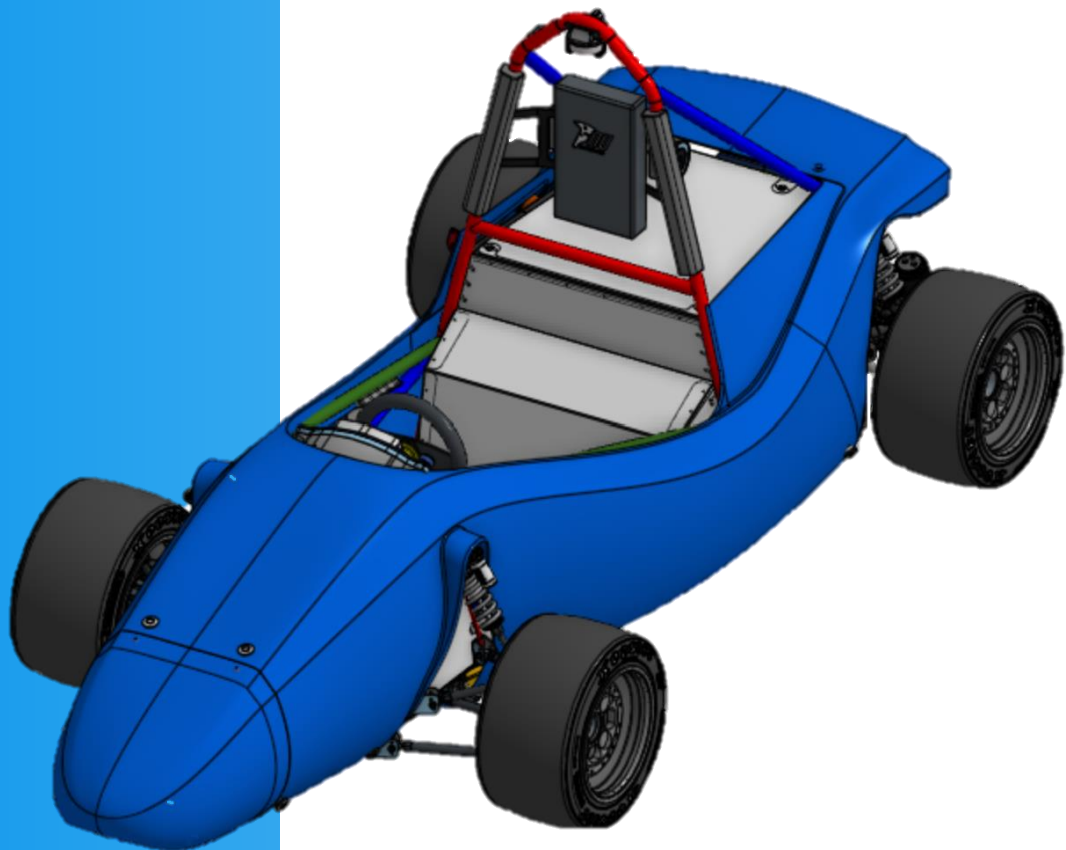




Mechatronic Department Lead Engineer

Directed Reading
NU25



Jacob Lukes
2025





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1. Abstract

NU Racing is the Formula Society of Automotive Engineers – Australasia (FSAE-A) team from the University of Newcastle. 2025 saw the design, manufacture and commission of NU25, a single seat electric vehicle, which participated in the 2025 Formula SAE-A competition in Calder Park, Melbourne.

Building upon the overall finish of 4th place in 2024, NU Racing pushed for a reliable and traction limited vehicle, while also improving data collection methods and maintaining a similar weight to NU24. The team maintained a ‘build first, design second’ philosophy, adopted from previous years due to the team’s small member count and low budget. Alongside this, NU Racing changed to a dual cycle development approach, allowing for time to be spent on projects for future NU Racing cars. Projects included designs for a Pouch Cell Accumulator, a Monocoque Chassis, Hub Motor Gearboxes, and AVONE.

NU Racing achieved 3rd place in the 2025 FSAE-A competition, the highest the team has ever achieved since it started in 2002.

The 2025 leadership team was made up of the Team Leader, Cooper Chomicki, the Chief Engineer, Alec Chapman, the Mechanical Department Lead Engineer, Cameron Higgins, and the Mechatronic Department Lead Engineer, the Author.

Technical Inspection is vital to ensure compliance with the FSAE-A ruleset and safety during the competition. Due to the extensive work from 2023 and 2024, as well as many mock inspections and preparation in 2025, NU25 passed EV Static Technical Inspection first try, Accumulator Technical Inspection second try, and EV Functional Technical Inspection first try. Both Technical Inspection and Parc fermé at the 2025 competition was extremely smooth and successful, which lead to the team receiving the Technical Inspection Specialty Award.



Figure 1. NU25, NU Racing's 2025 Competition Platform

2. Introduction

2025 was the year for validation and reliability, two areas that were identified as weak points for the team in 2024. 2025 saw a 25% budget cut from 2024, with increased Final Year Project (FYP), Directed Reading and Extra Curricular member counts. The team was asked to do more with less, which was a struggle at times, but due to the comradery of the team members, the team pulled through, resulting in a very successful year both technically and socially.

2.1. Purpose of This Report

This report details the role of the Mechatronic Department Lead Engineer, their responsibilities, goals, constraints and decisions. It includes the combined work of the Author from the 27th of January 2025 to the 14th of December 2025, amounting to 1,531 hours of time and effort. It also incorporates a technical overview of NU25 with respect to the electrical systems, recommendations and advice for future Mechatronic Department Lead Engineers. As well as overall performance of the Mechatronic team during the year and at the 2025 FSAE-A competition.

2.2. Overview of NU Racing and the FSAE-A Competition

Due to NU Racing's small team size, its success is largely reliant on the hard work, determination, and the special chemistry that the team creates, as well as the support from the University's valued staff members.

Formula Society of Automotive Engineers – Australasia (FSAE-A) is a student design competition where university teams design, build and race small formula-style open-wheel race cars. The purpose is for students to gain real-world engineering and project management experience. [1]

The competition is split into Static and Dynamic events. Each consisting of their own document submissions prior to the competition. A summary of the events and documents for the 2025 season is provided below. How NU Racing performed in each event compared to 2024 is shown in Figure 2. NU Racing Point Analysis 2025 vs 2024.

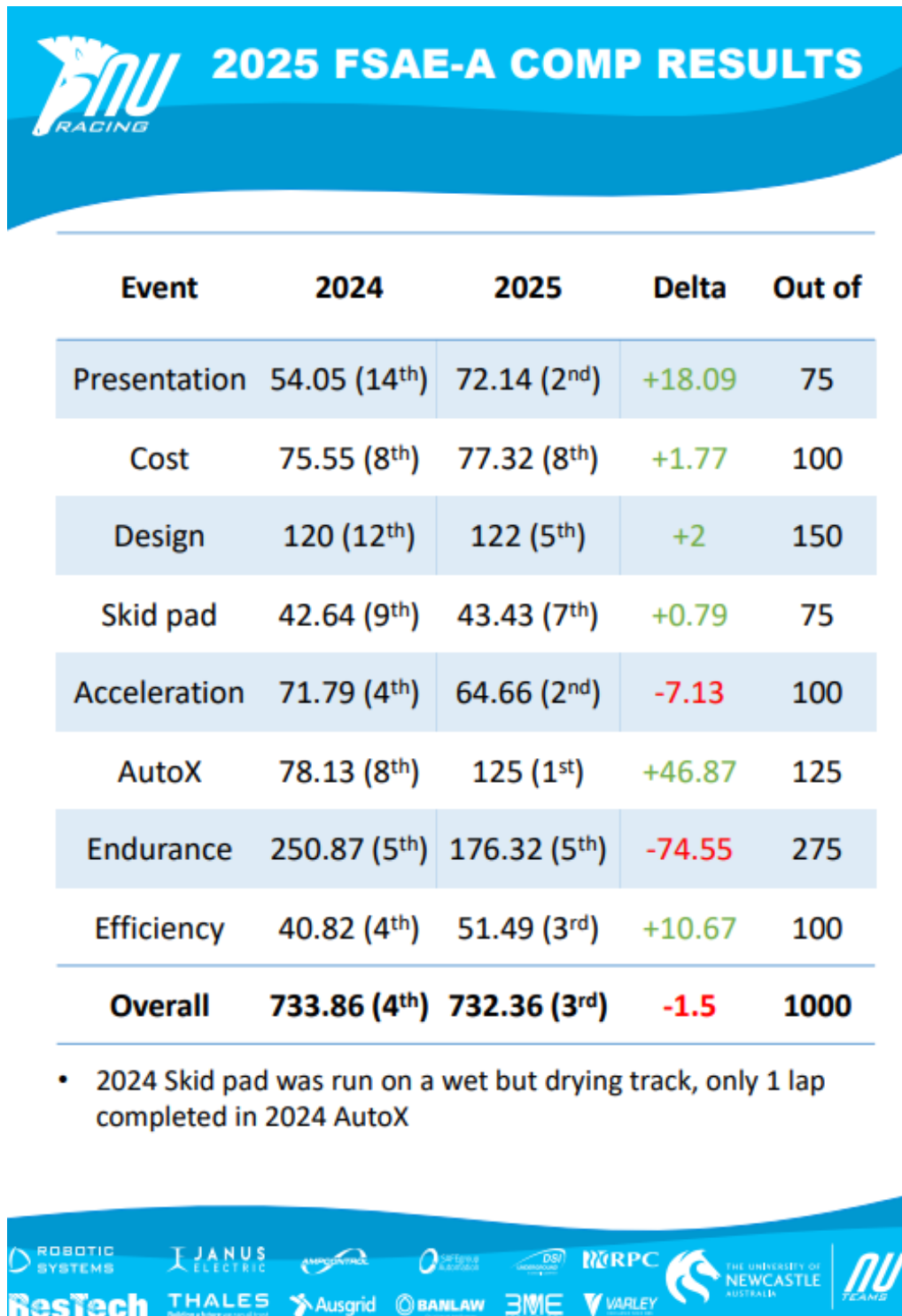


Figure 2. NU Racing Point Analysis 2025 vs 2024

2.2.1. Competition Schedule and Layout

The schedule for the competition, that spans across four days in December of the given year, is made available to teams through the Event Handbook, in 2025, the final version of this document was released on the 26th of November.

2.2.1.1. Competition Handbook

This Handbook gives critical information about competition, such as the run time for Technical Inspections, Static Events, and Dynamic Events. As well as what pits teams will share, where each Technical Inspection will take place, etc.

2.2.1.2. Accumulator Clean Rooms

One important note is that Accumulators can only be worked on in ‘Accumulator Clean rooms’, areas where the competition can easily monitor high-risk work being conducted, and to isolate any battery related accidents. Teams must gain permission from the Technical Inspection Captain before using these rooms.

2.2.1.3. Bump-in & Energy Meter

Before the first day of competition, teams are allowed to ‘bump-in’ to the event to get their pits set up early. During this time, the team is also handed the Energy Meter. A critical competition device that is mounted to the car.

It is the way the competition measures the energy used by the car during the Efficiency event, as well as to see if the car goes over the hard power limit of 80 kW for a specified amount of time during the competition. The rules surrounding the Energy Meter can be found under the Power and Voltage section of the Formula SAE Rules for the given year. For 2025, these rules are shown in Figure 3. Energy Meter and Power Limit Rules.

EV.3.3 Power and Voltage

- EV.3.3.1 The maximum power measured by the Energy Meter must not exceed **80 kW**
- EV.3.3.2 The maximum permitted voltage that may occur between any two points must not exceed **600 V DC**
- EV.3.3.3 The powertrain must not regenerate energy when vehicle speed is between 0 and 5 km/hr

EV.3.4 Violations

- EV.3.4.1 A Violation occurs when one or two of these exist:
 - a. Use of more than the specified maximum power **EV.3.3.1**
 - b. Exceed the maximum voltage **EV.3.3.2**for one or the two conditions:
 - Continuously for 100 ms or more
 - After a moving average over 500 ms is applied
- EV.3.4.2 Missing Energy Meter data may be treated as a Violation, subject to official discretion
- EV.3.4.3 Tampering, or attempting to tamper with the Energy Meter or its data may result in Disqualification (DQ)

Figure 3. Energy Meter and Power Limit Rules

2.2.1.4. Competition Packing & Pit Setup

Efficient packing for competition is critical. It can ease the stress of all individuals if done well. In 2025, the team wrote out what was packed into each Pelican case, then stuck this list on the outside and inside of each case. The Author recommends in future years to put this print out on the inside of the case, then put the contents of each case on one main piece of paper and position it somewhere in the pits.

In 2024 and 2025, NU Racing has brought industrial-style shelving to help organise the pit. This has proved to be majorly helpful and should be repeated in future years.

2.2.2. Competition Documents

2.2.2.1. *Electrical Systems Form (ESF)*

The Electrical Systems Form (ESF) is an excel spreadsheet that attempts to ensure the Mechatronic systems are compliant with the rules. An important note is that if the systems are compliant with the ESF, it does not mean they are compliant with Technical Inspection requirements. It is provided as a starting point for compliance.

2.2.2.2. *Safety Officer (ESO) and Electrical System Adviser (ESA) Form*

The Electrical Safety Officer (ESO) and Electrical System Adviser (ESA) form outlines who will be the point of contact if there are issues at competition with any of the electrical systems, namely the Accumulator.

The team requires a system adviser, one who has the relevant professional experience and/or qualifications to supervise students working on high-risk electrical systems (Anything that is high voltage (HV), Accumulator, Inverter, Motor, etc). The Electrical Safety Officer (ESO) is a list of team members who have been purposely trained to work on the high voltage systems of the car. At competition, an ESO must always be present with the Accumulator, whether it is charging, being taken to Technical Inspection, etc, one is required to be present. Historically the chosen members have received Low Voltage Rescue (LVR) training provided by the university.

2.2.2.3. Main ESO

Of the members listed as ESOs, the Mechatronic Department Lead Engineer is referred to as the main ESO, one who is responsible for the following items while at competition:

- Main point of contact for Technical Inspection related issues and questions while at competition.
- Has their name and phone number stickered to the top of the Accumulator.
- Present for all Technical Inspections.
- Presents the car to Event Coordinators before entering a Dynamic Event. In 2025, this process consisted of rolling the car up to the given Dynamic Event, measuring Tractive System voltage with the push-bar multi-meter, then once instructed, inserted the Manual Service Disconnect (MSD), turn on the car with Grounded Low Voltage Master Switch (GLVMS) and Tractive System Master Switch (TSMS).

2.2.2.4. Structural Equivalency Spreadsheet (SES)

The Structural Equivalency Spreadsheet (SES) is an excel spreadsheet that is a check for all the Mechanical systems on the car. It is the Mechanical version of the ESF. More information can be found in Grace Nimmo's FYP report [2].

2.2.2.5. Design Event Spec Sheet

The Design Event Spec Sheet is a technical summary of the car. It includes both Mechatronic and Mechanical properties. It serves as a summary of the car for the Design Event judges.

2.2.3. Static Events (325 Points)

Static Events refer to any event that contributes to the overall point rating of the competition (apart from Technical Inspection) that does not involve the car racing. They constitute 325 of the 1000 available points.

2.2.3.1. Technical Inspection

Passing Technical Inspection is a requirement for the car to compete in the Dynamic Events. There are six Technical Inspections for the Electric Vehicle (EV) class; after passing each one, the team receives a sticker to place on the front of the car to show that the team has passed an inspection. Shown in Figure 4. NU25 Technical Inspection Stickers. The Technical Inspections are listed below:

- EV Static Technical Inspection (also referred to as EV1)
- Mechanical Technical Inspection
- Accumulator Technical Inspection
- EV Functional Technical Inspection (also referred to as EV2)
- Tilt Test
- Rain Test
- Brake Test
- Parc fermé



Figure 4. NU25 Technical Inspection Stickers

2.2.3.2. *Cost Report, Scenario and Event (100 Points)*

The Cost Event challenges teams to cost the car in the most efficient way possible. It contributes 100 of the total 1000 points that can be earned at competition. The team who can most cost their car the cheapest and most accurately wins the event. The purpose is to be able to create bulk manufacturing methods to reduce costs wherever possible. More information can be found in Tom McKeown's report [3].

The event is broken up into three parts, the report, scenario and event. The report is submitted prior to the competition for the judges' review, while the scenario and event are presented to cost judges at the competition.

- The report is an excel spreadsheet that contains a sheet for every part on the car. From wire looms to uprights. Each sheet contains the materials to manufacture the part, the cost of labour for manufacturing, and the fasteners/tooling required. It is extremely in-depth and is a large time sink.
- The scenario is a theoretical situation proposed to teams. It changes year to year but often consists of a design change requirement of a system on the car, while remaining under a specified cost.
- The 'cost event' is an on-the-spot event where the judge asks the team to cost a part on the car. From material, processes, tooling and fasteners required. In 2025 they asked the team to cost an upright assembly, whereas in 2024 they asked the team to cost the steering assembly. The judge chooses the system on the spot. They usually choose a Mechanical system just because it is easy to see, but two people who are familiar with both the Mechanical and Mechatronic systems should be present in case.

2.2.3.3. Presentation (Business) (75 Points)

The Presentation (also called Business) event challenges teams to create a business proposal to investigate the potential market and develop a proposal to manufacture and sell a vehicle to compete within the market. More information can be found in Joshua Garbett's FYP report [4]. The event contributes 75 of the total 1000 points that can be earned at competition.

2.2.3.4. Design (150 Points)

The Design Event is the highest weighting of the Static Events, contributing 150 of the total 1000 points that can be earned at competition. It challenges teams to justify their design decisions during the year. It is split into six different sections, outlined below.

- Concept Integration Management
- Mechanical Structural Engineering
- Vehicle Performance
- Driver Interface
- EV Tractive System
- LV Electronics

At competition, the team presents the sections of the vehicle to judges in small groups of around 2-4 students. It is a conversational style event where students are justifying their design decisions with the aid of visual stimuli. Historically, NU Racing has visually conveyed designs with booklets, the judges suggested in 2025 to use more of a bill-board style presentation method.

The main thing that the judges are looking for is the individual's understanding of the system they are presenting. The judges aim to find the limit of the individual's knowledge, if the individual tries to lie, the judge will pick up on it and score the team badly.

In the past, NU Racing has most struggled with the validation part of this event, lacking proof of if a system change had made a positive or negative impact on the overall vehicle and why. The word used to describe this is validation, something that NU Racing strived for in 2025, and was successful.

2.2.4. Dynamic Events (675 Points)

Dynamic Events refer to any event that contributes to the overall point rating of the competition where the car races. They constitute 675 of the 1000 available points. The team can enter a maximum of eight drivers in the Dynamic Events. Each driver is allowed to be entered in two events each, with Endurance counting as two events (Endurance + Efficiency).

2.2.4.1. Acceleration (100 Points)

Acceleration is a 75-meter straight line event that shows how fast the car can go. It is ranked in order of the fastest time. It contributes 100 of the total 1000 points that can be earned at competition.

The event allows two drivers to have two attempts each at the track. The team is scored as described in the Acceleration section of the Formula SAE Rules for the given year. For 2025, it is shown in Figure 5. 2025 Acceleration Scoring.

D.9.3 Acceleration Penalties

D.9.3.1 Cones (DOO)

Two second penalty for each DOO (including entry and exit gate cones) on that run

D.9.3.2 Off Course (OC)

DNF for that run

D.9.4 Acceleration Scoring

D.9.4.1 Scoring Term Definitions:

- Corrected Time = Acceleration Run Time + (DOO * 2)
- T_{your} - the best Corrected Time for the team
- T_{min} - the lowest Corrected Time recorded for any team
- T_{max} - 150% of T_{min}

D.9.4.2 When T_{your} < T_{max}, the team score is calculated as:

$$\text{Acceleration Score} = 95.5 \times \frac{(T_{\text{max}} / T_{\text{your}}) - 1}{(T_{\text{max}} / T_{\text{min}}) - 1} + 4.5$$

D.9.4.3 When T_{your} > T_{max}, Acceleration Score = 4.5

Figure 5. 2025 Acceleration Scoring

2.2.4.2. Skid Pad (75 Points)

Skid Pad is a figure eight event, that shows how well the car handles under tight cornering. It contributes 75 of the total 1000 points that can be earned at competition. The track layout is shown in Figure 7. Skid Pad Track Layout, and can also be found in the Event Handbook for the year.

The team is scored as described in the Skidpad section of the Formula SAE Rules for the given year. For 2025, it is shown in Figure 6. 2025 Skidpad Scoring.

D.10.3 Skidpad Penalties

D.10.3.1 Cones (DOO)

A 0.125 second penalty for each DOO (including entry and exit gate cones) on that run

D.10.3.2 Off Course (OC)

DNF for that run. Vehicles that stall or spin out may continue if they have not gone Off Course.

D.10.3.3 Incorrect Laps

Vehicles that run an incorrect number of laps or run the laps in the wrong sequence will be DNF for that run.

D.10.4 Skidpad Scoring

D.10.4.1 Scoring Term Definitions

- Corrected Time = (Right Lap Time + Left Lap Time) / 2 + (DOO * 0.125)
- T_{your} - the best Corrected Time for the team
- T_{min} - is the lowest Corrected Time recorded for any team
- T_{max} - 125% of T_{min}

D.10.4.2 When T_{your} < T_{max}, the team score is calculated as:

$$\text{Skidpad Score} = 71.5 \times \frac{(T_{\text{max}} / T_{\text{your}})^2 - 1}{(T_{\text{max}} / T_{\text{min}})^2 - 1} + 3.5$$

D.10.4.3 When T_{your} > T_{max}, Skidpad Score = 3.5

Figure 6. 2025 Skidpad Scoring

A 'run' for Skid Pad is defined as the driver having two attempts at the track. An attempt consists of the driver entering through the 'IN Entry', then driving two laps of the right-hand circle, then two laps of the left-hand circle. Once complete, the driver will exit through the 'Exit OUT' gate.

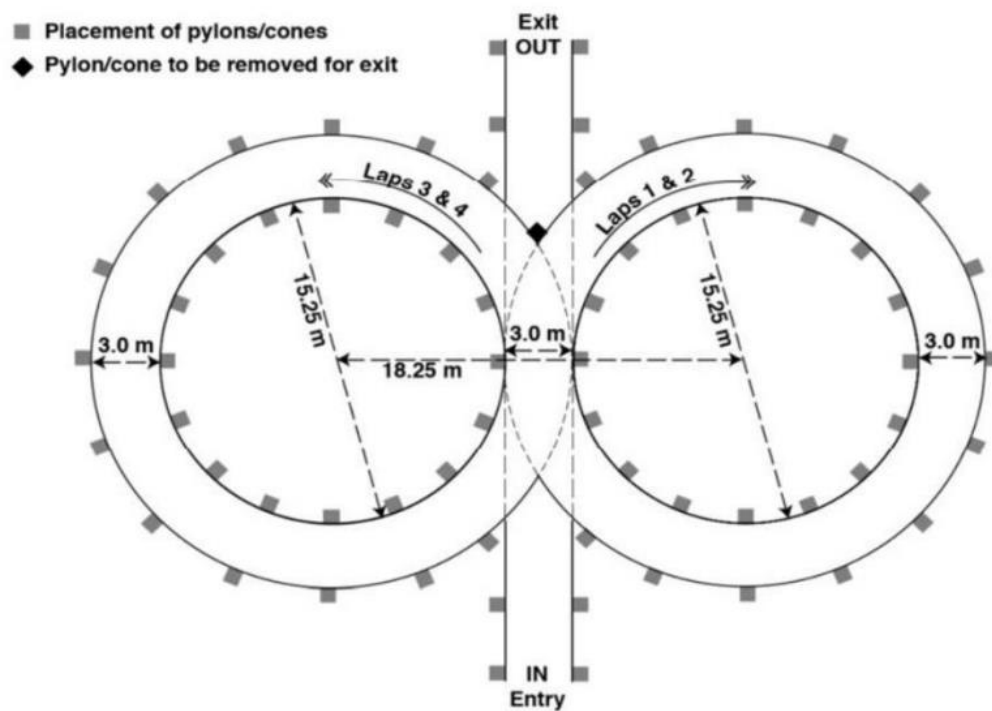


Figure 7. Skid Pad Track Layout

2.2.4.3. Autocross (AutoX) (125 Points)

Autocross (also written as AutoX) aims to see how fast a driver can take the car around the same track used for Endurance (shown in Figure 9. 2025 AutoX and Endurance Track, also available in the 2025 section of the FSAE-A website, titled 'FASE-A 2025 Track Map') in two laps. After the two laps, the car is brought back to the pits for a driver change, then an hour or so later, the second driver is allowed the same two laps.

Autocross contributes 125 of the total 1000 points that can be earned at competition.

The team is scored as described in the Auto Cross section of the Formula SAE Rules for the given year.

For 2025, it is shown in Figure 8. 2025 Autocross Scoring.

D.11.3 Autocross Penalties

D.11.3.1 Cones (DOO)

Two second penalty for each DOO (including cones after the finish line) on that run

D.11.3.2 Off Course (OC)

- When an OC occurs, the driver must reenter the track at or prior to the point of exit or receive a 20 second penalty
- Penalties will not be assessed to bypass an accident or other reasons at the discretion of track officials.

D.11.3.3 Missed Slalom

Missing one or more gates of a given slalom will be counted as one Off Course

D.11.4 Autocross Scoring

D.11.4.1 Scoring Term Definitions:

- Corrected Time = Autocross Run Time + (DOO * 2) + (OC * 20)
- T_{your} - the best Corrected Time for the team
- T_{min} - the lowest Corrected Time recorded for any team
- T_{max} - 145% of T_{min}

D.11.4.2 When T_{your} < T_{max}, the team score is calculated as:

$$\text{Autocross Score} = 118.5 \times \frac{(T_{\text{max}} / T_{\text{your}}) - 1}{(T_{\text{max}} / T_{\text{min}}) - 1} + 6.5$$

D.11.4.3 When T_{your} > T_{max}, Autocross Score = 6.5

Figure 8. 2025 Autocross Scoring

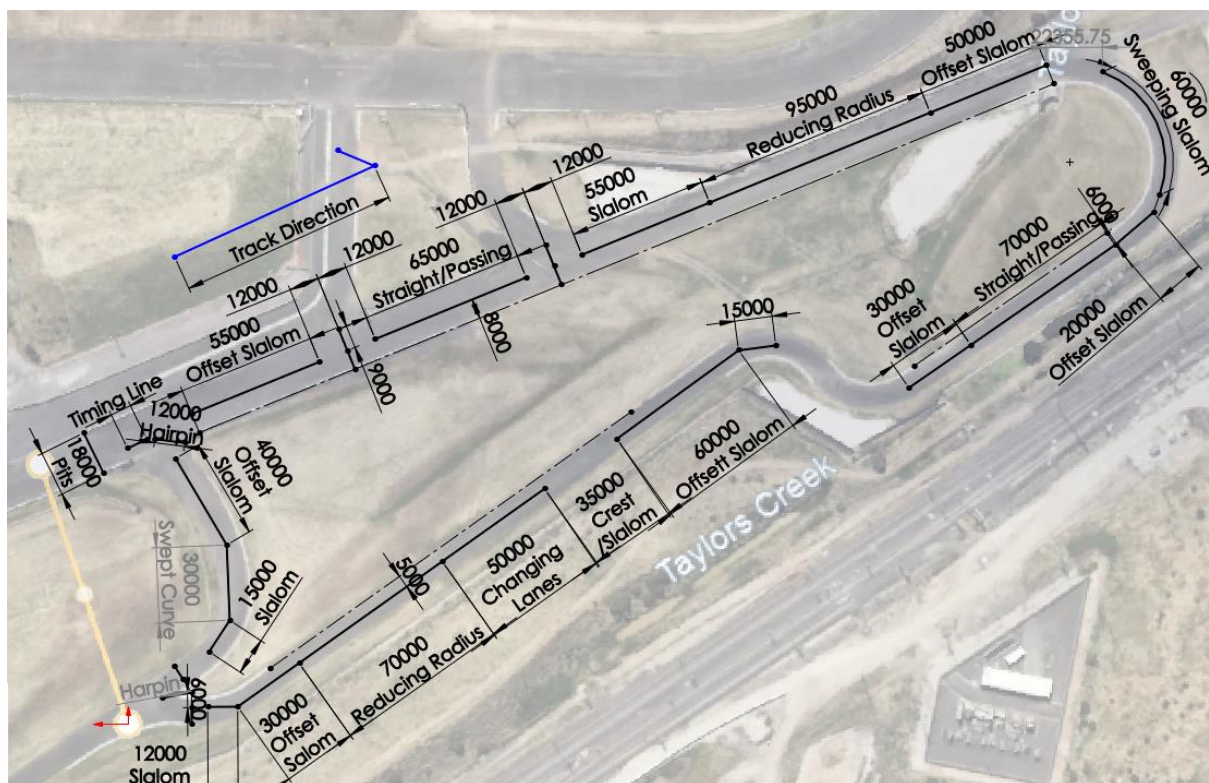


Figure 9. 2025 AutoX and Endurance Track

2.2.4.4. *Endurance (275 Points)*

Endurance is a 22-kilometre event with a timed driver change half-way through. It is the most nerve-racking event of the entire competition. It runs on the same track as AutoX, shown in Figure 9. 2025 AutoX and Endurance Track.

Endurance alone contributes 275 of the total 1000 points that can be earned at competition. To make it worse, it is also combined with the Efficiency event, meaning that a staggering 375 points can be earned from a single event, it weighs the same as all three of the Static Events combined.

The team is scored as described in the Endurance section of the Formula SAE Rules for the given year. For 2025, it is shown in Figure 10. 2025 Endurance Scoring.

D.12.13 Endurance Scoring

D.12.13.1 Scoring Term Definitions:

- Endurance Run Time - Total Time for the two Drivers, minus the Driver Change lap, minus any Mechanical Black Flag Time, plus any Penalty time **D.14.2**
- Corrected Time = Endurance Run Time + (DOO * 2) + (OC * 20)
- T_{your} - the Corrected Time for the team
- T_{min} - the lowest Corrected Time recorded for any team

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- T_{max} - 145% of T_{min}

D.12.13.2 The vehicle must complete the Endurance Event to receive a score based on their Corrected Time

- a. If T_{your} < T_{max}, the team score is calculated as:

$$\text{Endurance Time Score} = 250 \times \frac{(T_{\text{max}} / T_{\text{your}}) - 1}{(T_{\text{max}} / T_{\text{min}}) - 1}$$

- b. If T_{your} > T_{max}, Endurance Time Score = 0

D.12.13.3 The vehicle receives points based on the laps and/or parts of Endurance completed.

The Endurance Laps Score is worth up to 25 points

D.12.13.4 The Endurance Score is calculated as:

$$\text{Endurance Score} = \text{Endurance Time Score} + \text{Endurance Laps Score}$$

Figure 10. 2025 Endurance Scoring

2.2.4.5. Efficiency (100 Points)

Combined with Endurance, Efficiency aims to see which team can use the least amount energy during the endurance run. It contributes 100 of the total 1000 points that can be earned at competition.

The team is scored as described in the Efficiency section of the Formula SAE Rule for the given year.

For 2025, it is shown in Figure 11. 2025 Efficiency Scoring.

D.13.4 Efficiency Scoring

D.13.4.1 Conversion Factors

Each fuel or energy used is converted using the factors:

- a. Gasoline / Petrol 2.31 kg of CO₂ per liter
- b. E85 1.65 kg of CO₂ per liter
- c. Electric 0.65 kg of CO₂ per kWh

D.13.4.2 (EV only) Full credit is given for energy recovered through regenerative braking

D.13.4.3 Scoring Term Definitions:

- CO₂ min - the smallest mass of CO₂ used by any competitor who is eligible for Efficiency
- CO₂ your - the mass of CO₂ used by the team being scored
- Tmin - the lowest Endurance time of the fastest team which is eligible for Efficiency
- Tyour - same as Endurance (D.12.13.1)
- Lapyours - the number of laps driven by the team being scored
- Laptotal Tmin and Latptotal CO₂ min - be the number of laps completed by the teams which set Tmin and CO₂min, respectively

D.13.4.4 The Efficiency Factor is calculated by:

$$\text{Efficiency Factor} = \frac{T_{\min} / \text{LapTotal } T_{\min}}{T_{\text{your}} / \text{Lap your}} \times \frac{\text{CO}_2 \min / \text{LapTotal } \text{CO}_2 \min}{\text{CO}_2 \text{ your} / \text{Lap your}}$$

D.13.4.5 EfficiencyFactor min is calculated using the above formula with:

- CO₂ your (IC) equivalent to 60.06 kg CO₂/100km (based on gasoline 26 ltr/100km)
- CO₂ your (EV) equivalent to 20.02 kg CO₂/100km
- Tyour 1.45 times Tmin



D.13.4.6 When the team is eligible for Efficiency, the team score is calculated as:

$$\text{Efficiency Score} = 100 \times \frac{\text{Efficiency Factor your} - \text{Efficiency Factor min}}{\text{Efficiency Factor max} - \text{Efficiency Factor min}}$$

Figure 11. 2025 Efficiency Scoring

2.3. Role of the Mechatronic Department Lead Engineer

The Mechatronic Department Lead Engineer aims to bridge the gap between high level overarching team goals and the work that must be done to achieve these goals. The role operates within constraints set by the Team Leader and Chief Engineer, ensuring consistent communication with the Mechanical Department Leader and effective collaboration with members of the Mechatronic team.

The Department Lead is responsible for monitoring technical performance, member wellbeing, budget usage, and the safety and quality of the team's work. Additional key duties include ensuring compliance with competition rules, technical inspection requirements, accuracy of Mechatronic components in the cost report, overseeing correct validation of designs, and maintaining high standards in all Mechatronic systems.

Through the integration of management oversight with technical guidance, the role ensures that departmental output consistently supports the team's overall performance goals.

3. State of the Art

NU24 was built in 2024 by NU Racing. It completed 372 km of testing and was an extremely competitive vehicle to be given.

3.1. NU24 Tractive System

The car's power train featured a 7.2 kWh 453.6 V Accumulator, capable of discharging 180 A, a 124 kW EMRAX 228 MV AC Motor, a Cascadia CM200DX inverter. An overview of NU24's Tractive System is shown in Figure 12. NU24 Tractive System. The inverter allowed for regenerative braking, which Joshua Hayward implemented during 2024 and resulted in 0.92 kWh to be regenerated during the 2024 Endurance, scoring the team 4th place in the efficiency event.

To protect the cells from being over discharged or charged when at a low or high State of Charge (SOC), dynamic current limiting was implemented by Joshua Hayward in 2024, more information can be found in his FYP report [5], available on the NU Team's SharePoint.

In 2024, the CM200DZ Inverter the team was using experienced a major fault, more information can be found in Joshua Hayward's FYP report [5]. Towards the end of 2024, this inverter was sent back to the manufacturer for diagnosis. At the beginning of 2025, the team was informed that there were no faults or issues with it. The manufacturer sent it back to the team, the team did a track day with it and there were indeed no faults or issues. The issue was that an EEPROM was enabled that allowed for a variable PWM setting. This was turned off, and it worked without issue.

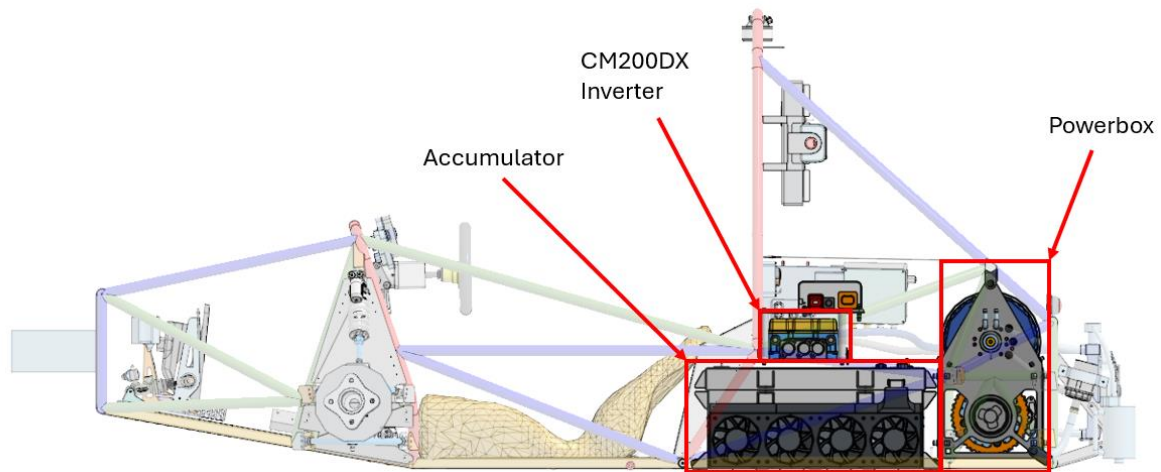


Figure 12. NU24 Tractive System

3.2. NU24 Accumulator

The Accumulator was comprised of 108 ENEPAC Li1x6pVTC6 modules (one module consists of 6 18650 cells connected in parallel) connected in series, monitored by an Orion BMS2 and the team's own temperature monitoring device, the CANaMon, a PCB which converts Analogue temperatures onto the CAN BUS. It also features the Low Voltage Distribution board (LVD), Precharge Board and Insulation Monitoring Device, the BENDER ISOMETER IR155-3203. The NU24 Accumulator is shown in Figure 13. NU24 Accumulator Assembly (taken from the 2024 Cost Report).

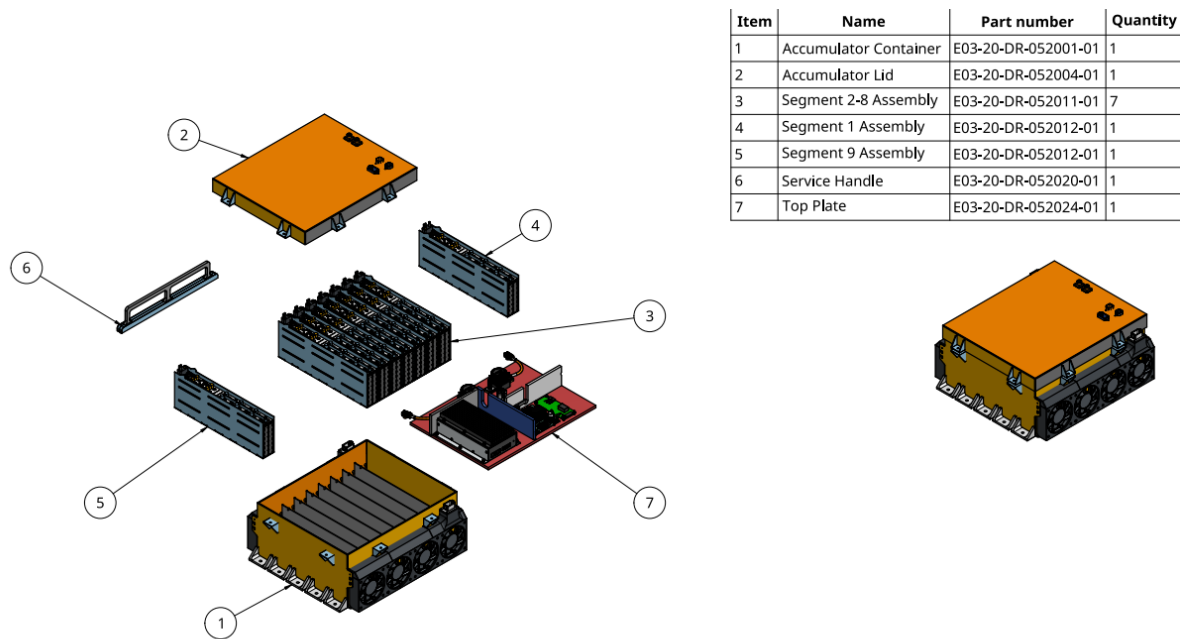


Figure 13. NU24 Accumulator Assembly

3.3. NU24 Low Voltage System

3.3.1. Low Voltage System

The low voltage (LV) system on NU24 had a nominal voltage of 12 V, supplied from a LH4LK motorcycle battery, and a TDC-IY-320-12 454 V to 12 V DCDC. A distributed topology of multiple individual nodes with Teensy 4.0 microcontrollers positioned across the car was used. Overview shown in Figure 14. NU24's LV Topology. List of nodes included:

- The Pedal Electronic Node (PEN): Monitors the Brake Pressure Sensors, Accelerator Pedal Position Sensors (APPS), and Brake Over Travel Switch (BOTS). Work conducted by Eliza O'Donnell.
- The Dash Electronic Node (DEN): Monitors the Dash E-Stop and Ready-To-Drive (RTD) button, MoTeC connection, MoTeC button and relays PEN signals. Work conducted by Jackson Boustany, more information can be found in his FYP report [6] available in the NU Team's SharePoint.

- The Central Electronic Node (CEN): Monitors the Hard Fault Latch (HFL) breakout board, Top E-Stops, Brake System Plausibility Device (BSPD) breakout board, OK High Signals (OKHS), Cooling and brake light switching, sounder and LV Power distribution. Work on the CEN Motherboard was conducted by the Author, more information can be found in the ENGG3200 report [7] also created by the Author in the NU Team's SharePoint. Information on the HFL and BSPD breakout boards can be found in Jacob Bush's ENGG4200 report [8], available in the NU Teams SharePoint.
- The Human Interface Panel (HIP): Switches on LV Power, Tractive System Master Switch (TSMS), HV Disconnect (HVD) Interlock, Test Points and Tractive System Active Light (TSAL) and Discharge Relay Breakout board. (This board also is HV). Work was conducted by Marisa Mclean.
- Low Voltage Distribution (LVD): Relays Power to the Accumulator Fans and Communicates Accumulator Temperatures from the Battery Monitor System (BMS) to the CEN as well as BMS OKHS. (This board also is HV). Work conducted by Daniel Iveson, more information can be found in his FYP report [9], available on the NU Team's SharePoint.

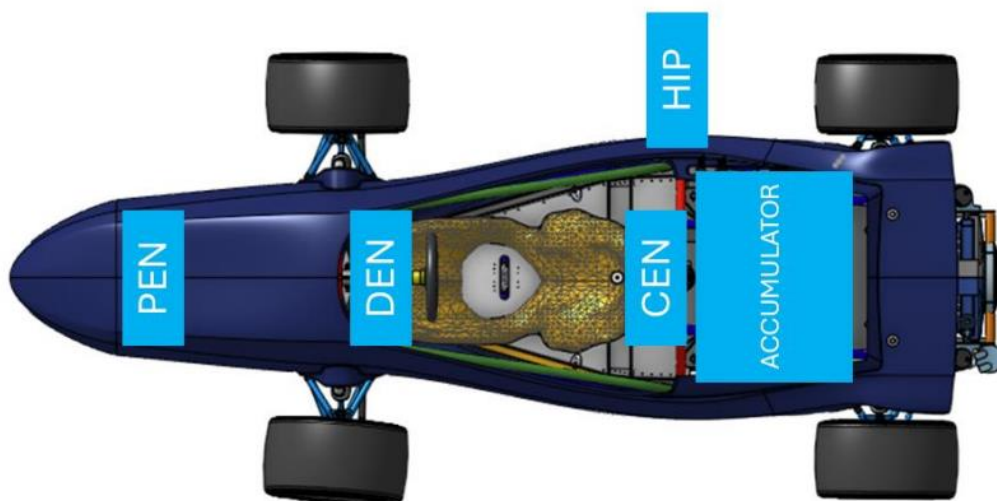


Figure 14. NU24's LV Topology

Once completely tested, this system was reliable, apart from the issue that stopped the car during the second driver stint of Auto X during the 2024 competition, detailed explanation found in the Testing, Validation and Compliance section.

3.3.2. Shutdown System

The shutdown circuit is the car's safety signal; it is a 12 V digital on-off signal that passes through many checkpoints through-out the car. It is designed to keep the driver, individuals nearby the car and the car's components safe in the case of an emergency or fault. When all checkpoints are good, the car is allowed to switch into High Voltage (HV). If something trips the circuit, the car will exit HV and depending on the fault, cannot be reactivated by the driver until the car is power cycled (Low Voltage (LV) is turned off as well). NU24's shutdown circuit is depicted in Figure 15. NU24 Shutdown Circuit. A high-resolution version can be found in the Racing-NU24 GitHub Repository.

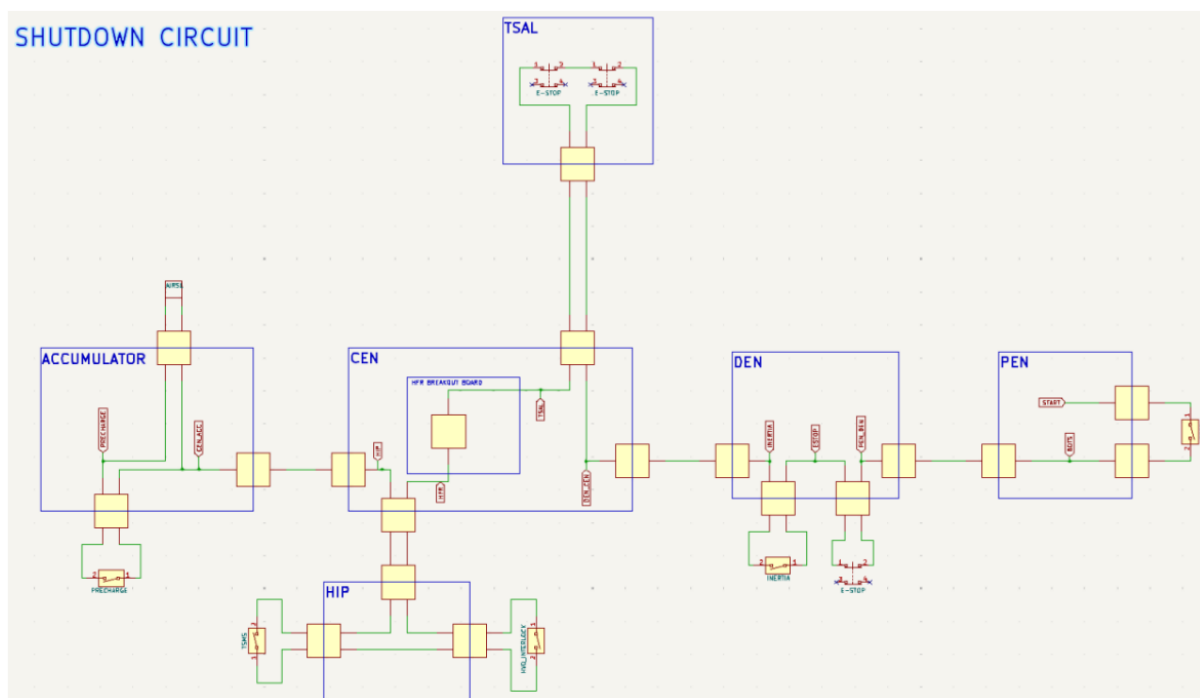


Figure 15. NU24 Shutdown Circuit

3.3.3. Live Telemetry

The Dash Electronic Node (DEN) housed the MoTeC C125 Dash, a datalogger that is used to read all desired signals on the car. It logs its data in high resolution for collection after the vehicle has run. This is done by connecting a RJ45 cable to the access point on the DEN. Paired with a Parani SD1000, the dash can transmit its data live, at a reduced resolution. The dash and data logging was implemented by Robert Stanley in 2023 and has saved many hours of fault finding, as well as effectively highlighting bottlenecks in the vehicles electrical system, allowing for informed data driven decisions to be made.

The mounting of these devices is shown in Figure 16. NU24 Live Telemetry Devices (taken from the 2025 Cost Report).

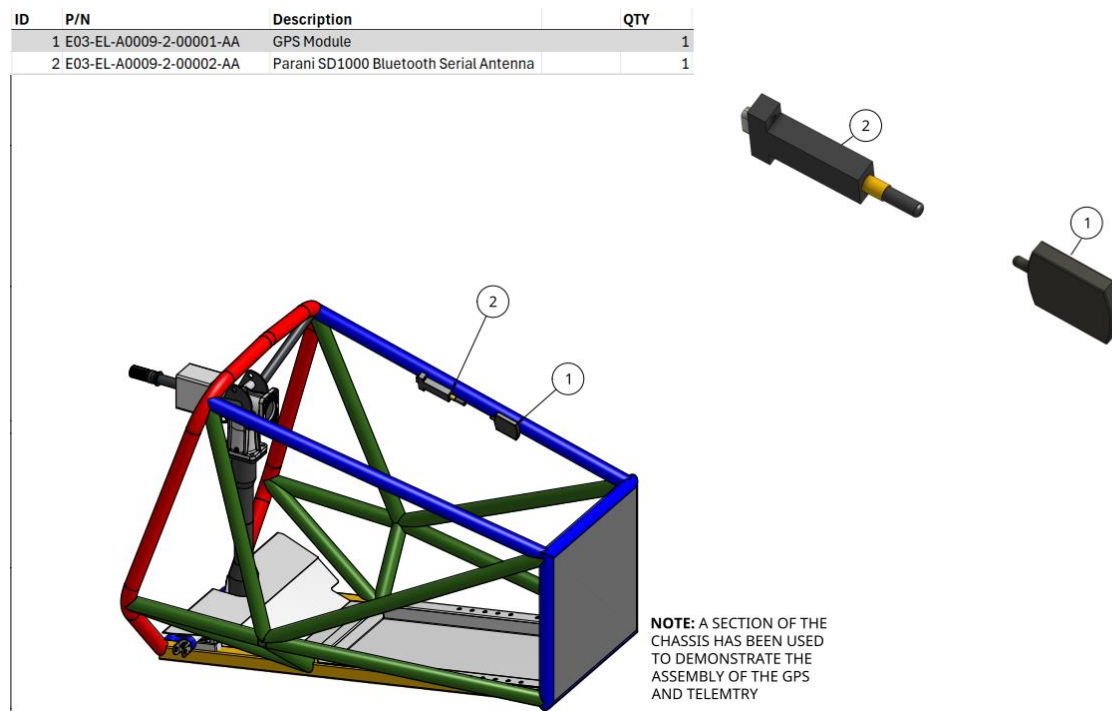


Figure 16. NU24 Live Telemetry Devices

4. NU25 Mechatronic System Architecture

NU25 was built in 2025 by NU Racing. It completed 523 km of testing and turned out to be an incredible vehicle.

The biggest change for NU25 was the implementation of a Limited Slip Differential (LSD), which would allow for efficient cornering at low to medium speeds and increased drivability when driving at the limit of the car. To maximise testing time for both the differential and other vehicle dynamic tests, it was decided to keep the Mechatronic system changes to a minimum.

4.1. NU25 Tractive System

Aligning with the minimal system changes NU25 kept the same 7.2 kWh 453.6 V Accumulator, Cascadia CM200DX inverter, and 124 kW EMRAX 228 MV AC Motor. Samuel Connett built upon Josh Hayward's multivariable current limiting in 2025; by including both cell temperature and voltage limits, the car would successfully not allow the driver to request high current during periods of high cell temperatures and cell voltages outside of their limits. More information can be found in Samuel Connett's FYP report [10], available on the NU Team's SharePoint.

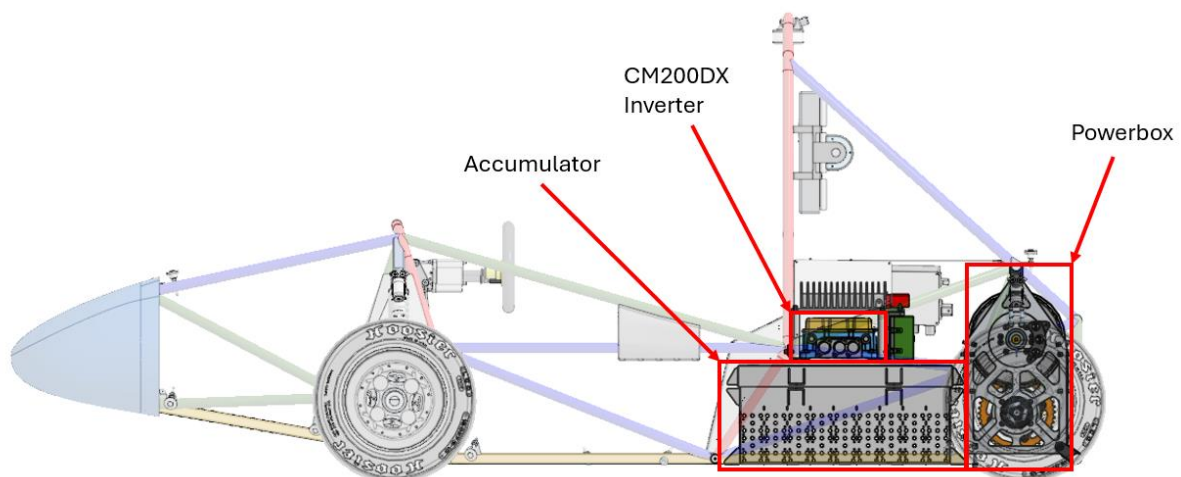


Figure 17. NU25 Tractive System

4.2. NU25 Accumulator

NU25 kept the same cell arrangement as NU24's Accumulator. Improvements in weight reduction were gained by changing:

- Accumulator insulating material from FR4 to Nomex paper
- Top plate material from GPO3 to PEEK
- Segment housing material from FR4 to a PC/ABS 3D printed material

More information can be found in Ryan Bird's FYP report [11], available on the NU Team's SharePoint. Serviceability of the Accumulator also increased with the implementation of the new PEEK top plate, due to a rearrangement of the components, allowing the Precharge Teensy 4.0 programming port and the BMS wire harnesses to be in a more accessible location.

The NU25 Accumulator is shown in Figure 18. NU25 Accumulator Assembly (taken from the 2025 Cost Report).

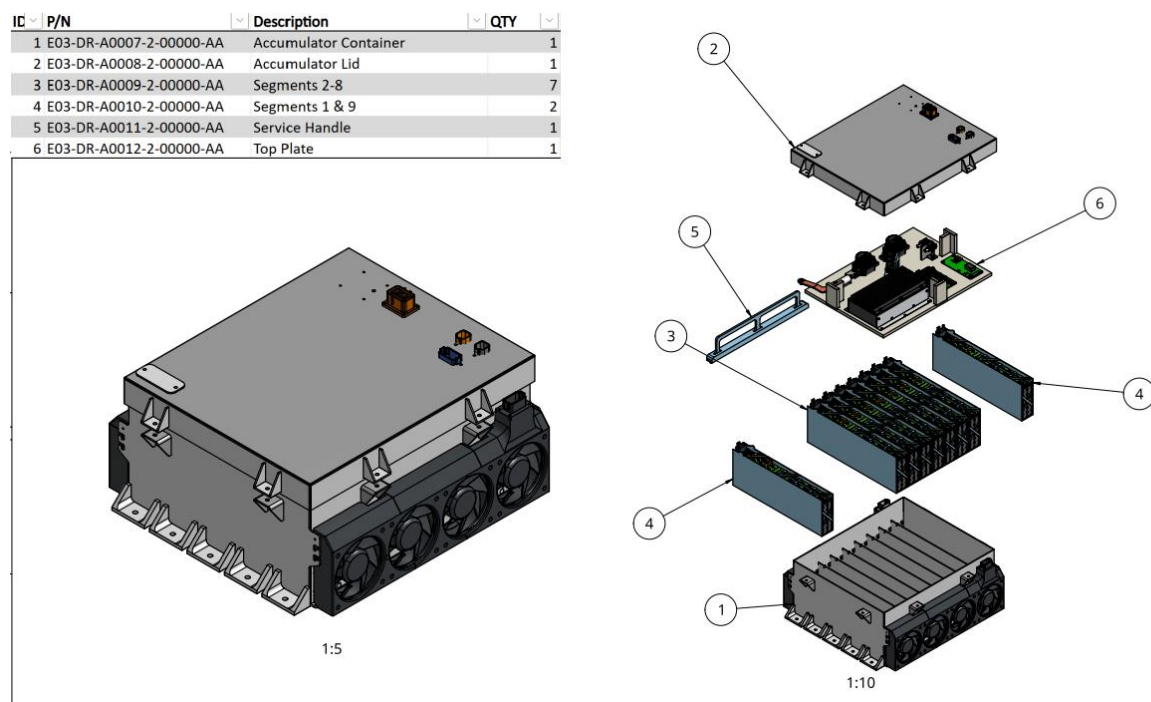


Figure 18. NU25 Accumulator Assembly

4.3. NU25 Low Voltage System

4.3.1. Low Voltage System

NU25's LV system maintained the same 12 V nominal voltage battery and DCDC, with the same distributed topology of multiple individual nodes using Teensy 4.0 microcontrollers. These nodes maintained largely the same functionality, changes include:

- The Pedal Electronic Node (PEN): Rearrangement of components, increasing fault finding ability when bench testing and suppressing potential noise on sensor inputs. Work was conducted by Jayson Mills, more information can be found in his ENGG3200 report [12], available on the NU Team's SharePoint.
- The Dash Electronic Node (DEN): No changes.
- The Central Electronic Node (CEN): Changed mounting location to be opposite the Human Interface Panel (HIP) on the chassis. This significantly increased the serviceability of the node and moved it away from the high switching noise of the inverter and DCDC. Work was conducted by Zak Lobko, more information can be found in his ENGG3200 report [13], available on the NU Team's SharePoint.
- The Human Interface Panel (HIP): Saw improvements to reliability and rules compliance. Higher quality switches were used for the Grounded Low Voltage (GLV) and Tractive System (TS) master switches. Different HV connectors with dedicated interlock contacts were used, as well as other component changes. Work was conducted by Mia Cosentino, more information can be found in her ENGG3200 report [14], available on the NU Team's SharePoint.

- Low Voltage Distribution (LVD): Saw improvements to both reliability and serviceability. By implementing a double switching MOSFET circuit from the NU Teams Standard Circuits, previous cooling concerns of the board were completely negated. NU24's LVD had a large 24 pin MOLEX connector to interface with the Precharge, Isolation Relays, BMS and IMD, it effectively reduced the part count of the board but over time the connector became harder to disconnect from the board, which significantly impacted the serviceability of the Accumulator. This was replaced with a dedicated MOLEX connector for each interface (Precharge, BMS, etc), minimising the time it took to service the Accumulator. Work was conducted by Oscar Rudman, more information can be found in his FYP report [15], available on the NU Team's SharePoint.

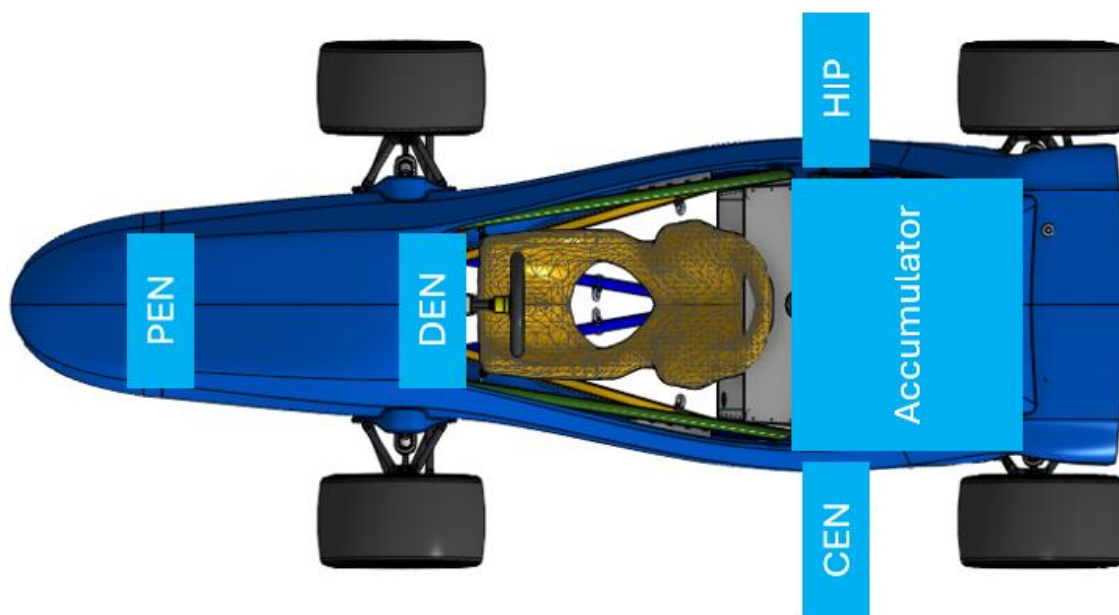


Figure 19. NU25's LV Topology

4.3.2. Shutdown System

NU25 kept a similar shutdown circuit to NU24. Notable changes being the increased number of interlocks on the HIP with the introduction of HV connectors with dedicated interlock lines, as well as moving the interpose relay from the LVD to the CEN, making it more accessible. Depicted in Figure 20. NU25 Shutdown Circuit. A high-resolution version can be found on the NU Teams SharePoint, Racing-NU25 GitHub Repository, and in Appendix C - NU25 Electrical System Overview.

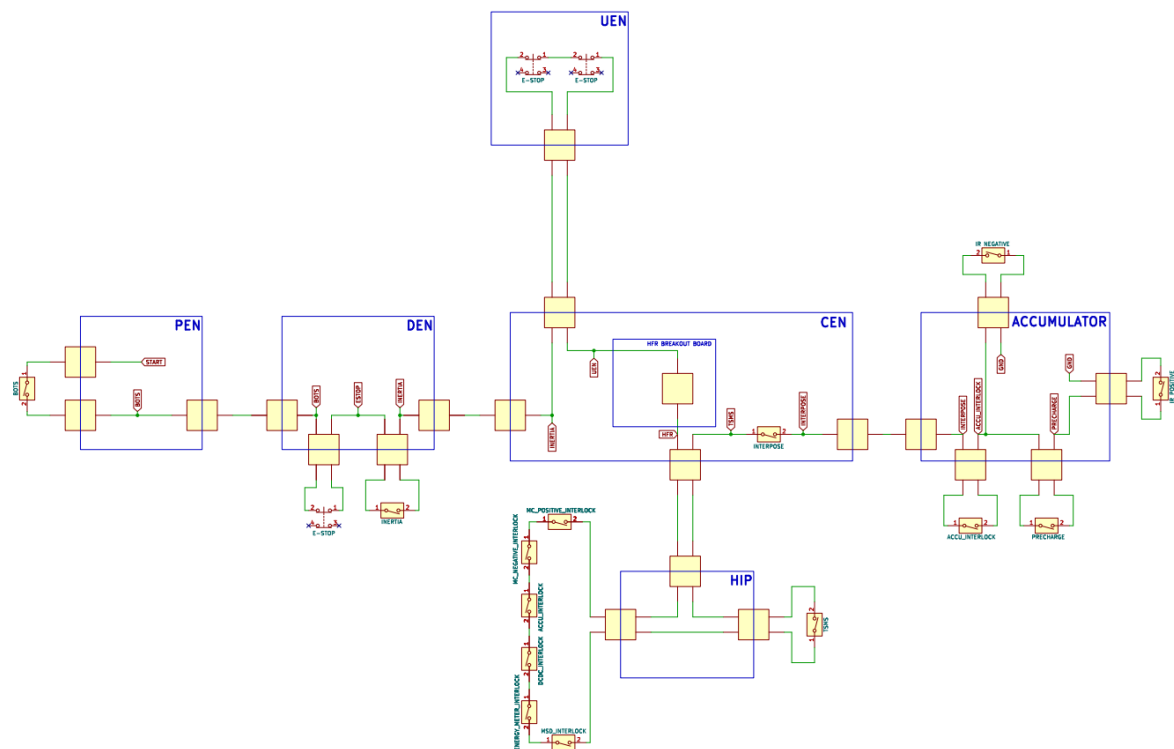


Figure 20. NU25 Shutdown Circuit

4.3.3. Live Telemetry

Due to the high reliability and robustness of NU24's live telemetry system, NU25 saw no changes to it.

The only drawback to the current solution is that there is a large learning curve to using the proprietary software required. Thus, a Zero-To-Hero style document was created by the Author, with contributions from Jayson Mills, to aid in learning the software. It can be found in SharePoint called 'MoTeC Zero-To-Hero', as well as in the Appendix of this report (Appendix H – MoTeC Zero-to-Hero). At the time of writing it has yet to be finished, readers who are interested are encouraged to contribute to the Zero-To-Hero.

5. Project Scope and Objectives

5.1. Original Project Scope

After the role began halfway through 2023, the Mechatronic Department Lead Engineer role has had time to develop and evolve into what it is today. Due to this, the original scope set at the beginning of 2025 for the Author did not change much during the year. Just the wording and definitions required to encapsulate the work done in this role were edited and revised during the year.

The scope of work was set out as:

- Working with the constraints set by the Team Leader and Chief Engineer
- Working with the Mechanical Department Lead
- Participating and/or leading Industry Partner Visits
- The delegation and management of mechatronic scopes and projects in collaboration with Team Lead
- The Mechatronic Budget
- All the mechatronic components used in the team
- Competition documents (ESF, ESO/ESA)
- The mechatronic aspects of the Cost Report, Design Event and Technical Inspection
- Regular Track Day Attendance
- Mechatronic Testing
- Management of NU Racing stock count
- Mechatronic Performance at Competition
- Mechatronic Safety at Competition
- Additions to scope of work
- Technical knowledge of the Mechatronics team
- Maintaining regular engagement with all Mechatronic team members, both on technical matters and team wellbeing
- Vibes of the Mechatronics team

5.2. Project Objectives and Success Criteria

During creation of a ENGG4200 - NU-Teams-Project-Scope-Form, the following objectives and success criteria were identified:

- Team passes Accumulator, EV Static, EV Functional and Rain Test Inspections
- All new MCHA system designs validated
- Efficient knowledge transfer of processes and systems
- Well-structured information regarding systems on the competition platform, for current and future team member's access
- Documented tasks completed by Mechatronics team
- Training and advice provided to Mechatronics team
- Suggested areas the Mechatronics team can innovate/improve in

5.3. Alignment with Team-Level Performance Goals

Team-level goals for 2025 included:

- Evolution not revolution
- Validation
- Reliability
- Serviceability
- Self-documenting systems
- Sustainability
- Positive team culture

The following decisions and system changes were made to align with these goals. The guiding philosophy for technical change was to implement changes only when supported by data demonstrating a clear need. This approach reduced the risk of introducing changes for their own sake, a consideration that strongly informed the Author's decision-making process. Limiting modifications also enabled increased track testing time, as the changes implemented were minor and involved well-understood systems.

5.3.1. 'Evolution not revolution'

- PEN: The main changes were to the layout and noise suppression on sensor inputs, all other circuits had been tested in previous years, remaining in line with an evolution.
- CEN: All circuit topologies remained the same, with the addition of the interpose circuit that was previously used and tested on the LVD in 2024. The main change for the board was that it was moved to opposite the HIP, which the Author believes to be an evolution instead of a revolution, as all circuit topologies had remained the same, only the location of the board on the car. With the idea that as many aspects as possible were kept the same.
- HIP: Changing the HV connectors to ones with dedicated interlock lines was seen as a potentially risky change and a revolution instead of an evolution, due to using connectors the team had not previously used. But it was decided to go with them since the alternative to remain compliant with the rules was to use external 3D printed enclosures to cover the detaching clips on the connectors, which was a time consuming and demanding task to do whenever the connectors were to be taken off.

5.3.2. Validation

- Validation was implemented at every step of the PCB creation process, from LTspice simulations, bench testing circuits before PCBs were ordered, and extensive bench testing of PCBs once they arrived and were manufactured. This significantly reduced the number of unknown issues arising during integration of new Mechatronic systems.
- The Expansion Board saw substantial success in 2025, with work completed by Oliver Bunny, in-depth information can be found in his FYP report [16], available on the NU Teams SharePoint. A summary of the project includes; The board elegantly implemented wheel speed sensors, brake temperature sensors, wheel temperature sensors, and shock potentiometer sensors on all four wheels, steering angle sensor, radiator temperature sensors and strain gauges all into the car's existing CAN Bus system, allowing it to be read and logged on i2 Pro. Each sensor circuit was extensively bench tested before integration with the car and allowed for many other FYP students to get critical data to show both improvements and gain validation of their decisions with their projects. One example was obtaining brake temperature data from all four wheels, which allowed Jethro Raj to conclude that the brake pads that the team uses on the front wheels are insufficient for how temperature range of the front wheels get. More information can be found in Jethro Raj's report [17] on the NU Teams SharePoint.

5.3.3. Reliability

- CEN: The new position meant that it was above shielded HV wires, and out of the way of noise from the inverter and DCDC, resulting in no issues being caused from noise on the CEN during the testing of NU25. With appropriate fuse sizing, the system became more reliable as fuses would no longer blow frequently during normal operation. The only downside to this change was the increased distance that the analogue output from the current sensor in the HIP would have to travel to get to the BSPD. This downside was avoided by using separate shielded cable for this signal within the CEN to HIP loom.

5.3.4. Serviceability

- CEN: The new position and enclosure design made it far quicker and easier to get to when issues arose, being four bolts to lift off the lid and access all components, as opposed to six bolts and two hard-to-get-to zip ties.
- LVD: All surface mount and through-hole components (apart from the DT connectors) were placed on the opposite side to the DT connectors, allowing for quick fault finding and probing when issues arose during testing.

5.3.5. Self-documenting systems

- Self-documenting systems refers to a design aspect for a system. It is the idea of explaining as much as possible to a user through physical aspects about the design.
- Examples include having name plates on all PCBs with the name of the creator, year and month, version number, and car that it was used on. Or having a detailed schematic for a PCB that is as simple as possible, with minimal spaghetti traces and all signal names labelled with easy-to-understand names. Or within code sets for PCBs, having simple variable names, separating long lengths of code into their own functions, commenting (shock horror) out blocks of potentially confusing code. If ever in doubt, ask another team member to check over your work and see if they can understand it without you having to explain it to them. If there is anything they get confused about, it is on you as the 'transmitter' to provide more explanation in the design.
- NU Racing successfully implemented this goal in 2023, with improvements in 2024. Thus in 2025, not much had to be changed for this, only to maintain what had already been set out.

5.3.6. Sustainability

- Being a goal that is not directly related to systems on the car, the Author interpreted this goal as writing down and creating as much knowledge transfer documents as possible during the year. The documents came in the form of Zero-To-Hero style documents aimed at individuals who are familiar with NU Racing already. Alongside these documents, an in-depth KiCad schematic detailing the Mechatronic system layout of the car, with all associated signals and connections, called the 'Electrical System Overview' was created. These documents are talked about in detail in the Knowledge Transfer and Documentation Outputs section of this report.
- On-boarding and upskilling the next generation of NU Racing leaders as early as possible significantly helps with knowledge transfer. Being an overarching team goal, the Author contributed to this by upskilling Jayson Mills, the next Mechatronics Department Lead Engineer subtly during the year, then once he was officially announced for the role late October 2025, giving more in-depth advice and including him in decisions that were made towards the end of 2025. It is recommended that in future years a similar process is followed, if not officially chosen as the next lead within the year, at least give directed advice to an individual who enjoys the team and shows initiative in learning the Mechatronics systems of the car.

5.3.7. Positive team culture

- Being another non-technical goal, as well as a potentially hard to measure for success goal, the Author approached this by creating bonds with team members, learning about their lives outside of NU Racing. As well as attempting to try to break down social barriers between members.
- From the point of view of well-being management, checking in with members in a non-technical sense to see if they needed breaks in workload. If larger issues came up, that were out of the scope of help that the Author could provide, forwarding the information up the organisational chart.
- The Author found this easiest to do in short five-minute casual chats, then if anything concerning came up in those chats, scheduling time later to talk it out and provide more avenues for help if required.

6. Project Context and Constraints

6.1. Team Structure and Leadership Interfaces

6.1.1. Team Structure

NU Racing operates under a hierarchical team structure, depicted in Figure 21. NU Racing Team Structure. The structure allows efficient flow of communication during both slow and busy periods of work. It also displays the structure in a well-organised format for new members joining the team.

While effective, the Author believes that it is not a hard and fast solution to the team's structure. This became evident at times during the year where information needed to be relayed to the team, and instead of informing the Team Leader first, then allowing a team post to be sent, it was most efficient to simply inform the individuals present at the time, then allow a team post to be sent later.

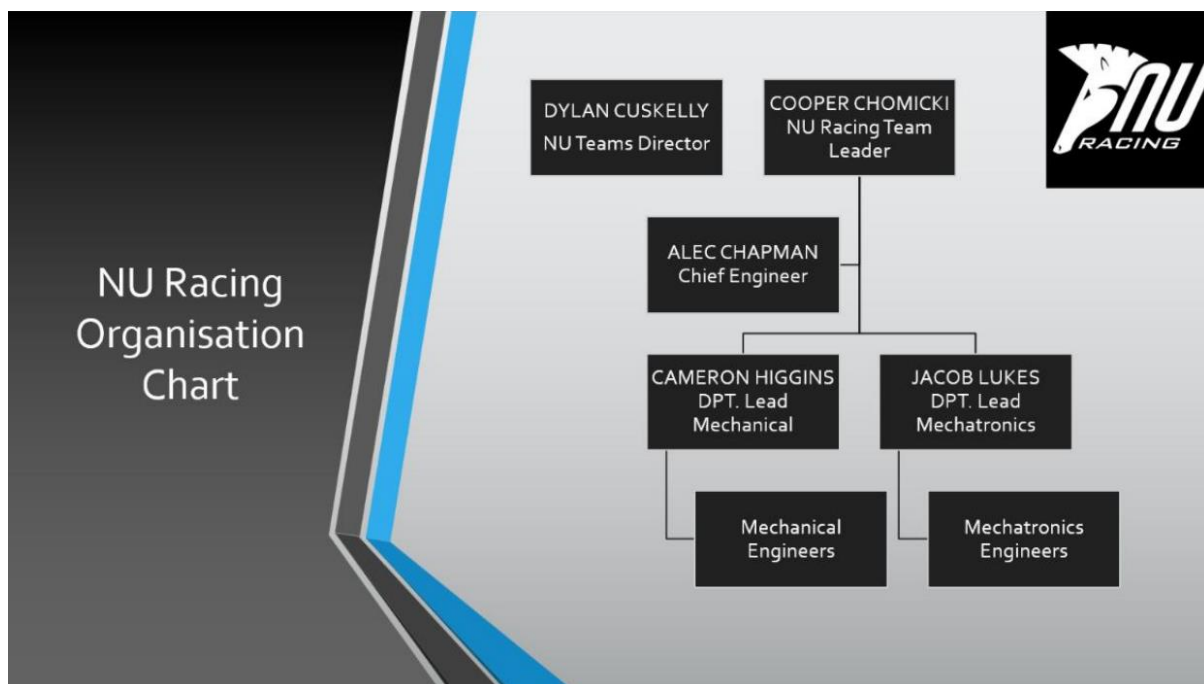


Figure 21. NU Racing Team Structure

6.1.2. Leadership Interfaces

The Author's communication with the Mechatronic Engineers often consisted of Discord posts in the 'mcha' channel, this was used to give quick updates to the team during the year. Example type of messages sent in 'mcha' discord channel shown in Figure 22. Example of Leadership Interfaces. In addition to this the Author found it valuable to continually check in on members in the Mechatronic team in a casual format. This was found to be an effective way of gaining updates on where individuals were at in their projects, allowing bonds to slowly build over time, and as to not pull them away from their work for extended periods of time to have one on one meetings.

This medium of communication for the Mechatronics team had its flaws, as at times there would be periods where individuals would not be present in the NU Teams building and would fall behind in work output. When this occurred, it was the role of the Author to regain communication with the individual and check-in with their technical struggles and well-being, providing further assistance where required.

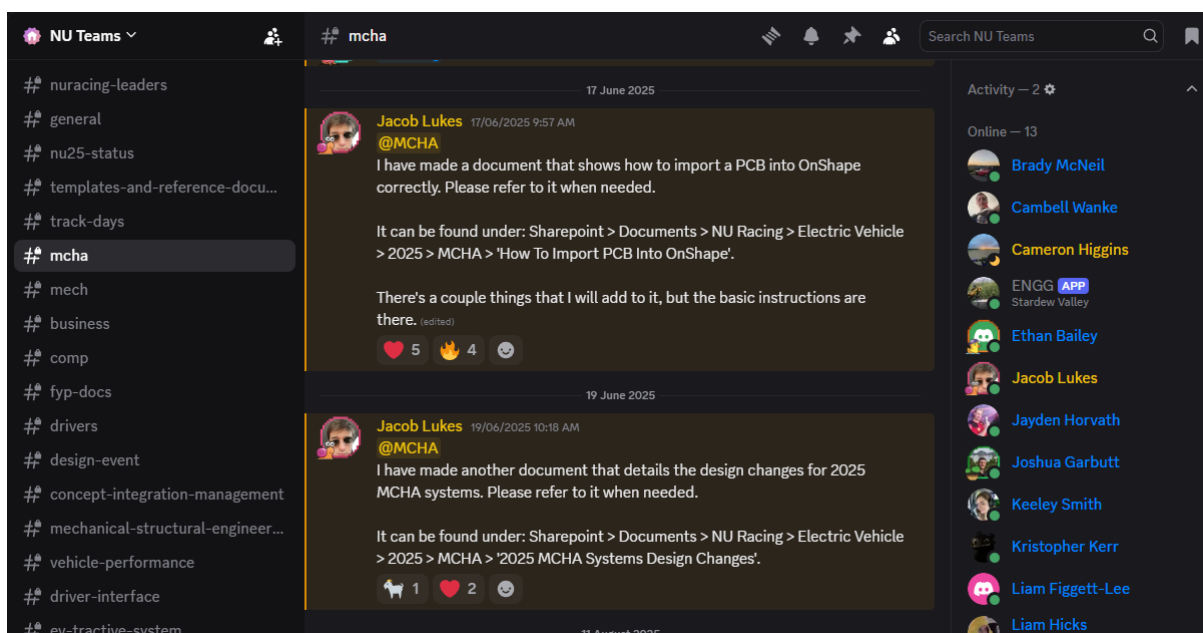


Figure 22. Example of Leadership Interfaces

For projects that had more unknowns about them than others, the Author saw it valuable to have weekly 30 minute to 1-hour meetings with individuals about the project, to help structure thoughts and remain on top of progress. The project where this most helped was the Dynamometer project, initially with just Samuel Connett handling the electronics and control side, then later with the addition of Victoria Daniela to design the mechanical aspects. The weekly meetings along with a ‘Dyno Progress Meeting’ Excel document helped to keep both individuals on track with their projects throughout the semesters. Agenda for these meetings were often going through what the individual had filled out in the Excel document, of what they had done in the week, what they will do the following week, any challenges they had, and any areas where they needed help. Checking over these often brought up points of discussion, that were then written down in a discussions section after the meeting. The ‘Dyno Progress Meeting’ Excel can be found in the NU Teams SharePoint, with an example snippet shown in Figure 23. Dyno Progress Meeting Excel Snippet.

Date	Progress This Week	Aim for Next Week	Challenges, Risks and Opportunities	Help Needed	Discussions	Additional Discussions
14 18/9/2025	Resolver manual done, but gonna make some improvements, designed CEN and HIP supports but they need to be fixed (to seat the nut), also worked on HIP enclosure which we delayed	lots of UNI, soz	Timelines, but chatting with Cam it seems we are on track		We worked out solutions to the cooling connections to the pumps, and the order shall be made tomorrow hopefully and we just need to pester Malcolm about it. Motor mount has passed FEA and Victoria is finding quotes before working out if it'll be made in house or by a company. Driveshaft connectors shouldn't be too difficult to get manufactured.	
15 26/09/2025	lots of Uni, but also working on making sure everything MCHA is ready. Including making a breadboard circuit for the TSMS and GLV buttons/switches, and the torque pot	work experience, but also ensure nothing else is forgotten, find the hip bus bars, and make DYN0 CEN code				
16 10/09/2025	mounted transformers, got new E-stop for E-stop box.	CEN code, Bread PCB Perf board, the looms required for the first test,	we also have to make a safety doc before the first test, some of the bolts in the dyno room are imperial and weird, and the workshop is busy		Malcolm meeting was epic Notes below: * can move Dyno back * Need Malcolm for first test * Need a safety doc before that. * Can make long looms for PAU to YourDyno but Malcolm will install them. * New EStop can EStop 1-3 lines at once. Need to transfer EStop box line from old to new EStop. * Need more bolts for mounting PAU. See other note. * Malcolm will chat with Michael about computer * Also bolts in Dyno room are imperial.	
17 22/10/2025	Perf Board almost done, looms still to complete. I've done a lot of Uni. Put together the motor mount and looked at how to move the dyno back with Victoria.	finish Perf today, CEN code tonight, looms next week	when are these parts arriving, and when to book Malcolm for first test,		We discussed timeline, what we're waiting on, what TS and Motor to use, when we will test, and a HIP to Power Supply loom and CEN to Power Supply loom,	Looms: CEN to Power Supply, HIP to Power Supply, INV madness, Rear (w/ Relay) to HIP, bump switch to not let the motor/dyno spin when the guard shield is open, wear masks cause a lot of old dust will be kicked up, get a key to lock the door, people can see in from outside the building, so E-stop and tell them to leave, flush out heat exchanger with pressure and velocity, leaks of very hot water are not good
18 28/10/2025	perf board done, met with Malcolm about RA and SOPs	KiCad models for looms, make looms, CEN code			discussed Sam and Victoria's scopes. discussed that we cant trust the heat exchanger specs, we dont need TS or ACC mounts, get more readily available fittings, assume fittings will fail and the HIP enclosure must be water proof, silicone it closed,	

Figure 23. Dyno Progress Meeting Excel Snippet

6.2. Regulatory and Competition Constraints (FSAE-A Rules)

The engineering problem that FSAE-A gives to teams are as follows “Develop and build a rules compliant single seat race car that could be manufactured by a hypothetical engineering firm to sell at a profit to a weekend autocross racer.”

The overarching FSAE-A ruleset is the same as the FSAE International ruleset, it aims to give detailed constraints in some areas of design, such as safety and areas where there has been previous incorrect interpretation of rules. But overall attempts to allow for a wide range of solutions to the problem that it gives to teams. In addition to the International ruleset, there are also ‘FSAE-A Local Rules Addendums’ that are released periodically throughout the year. They give further clarification on certain rules from the International ruleset, to both help teams and outline the differences between the Australasia competition and the International one. At times these rulesets are painfully vague with their wording, and it is strongly encouraged that further NU Racing students contact the FSAE-A committee asking for rule clarifications, and store any replies received, as they serve as evidence against scrutineers for the chosen design during Technical Inspection.

The rules and addendums can be found on the SAE Australasia website, under ‘Formula SAE-A > Current-Year Competition > Documents & Templates’, screenshot provided in Figure 24. Instructions to Find FSAE-A Rules. It is strongly recommended to become familiar with this website early in the year, as it is messy and all over the place, and it is their main form of communication with teams, outside of direct email communication with the Team Leader.

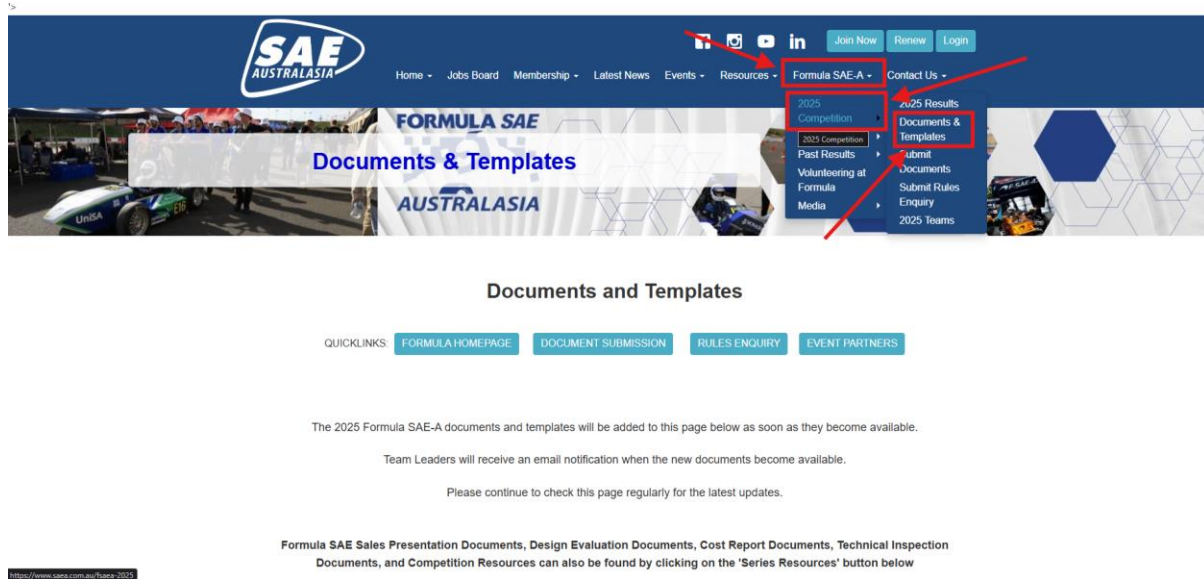


Figure 24. Instructions to Find FSAE-A Rules

6.3. Time, Resource, and Budget Constraints

6.3.1. Resource Constraints

At the beginning of the year, the leadership team was tasked with creating all the scopes of work for soon to be NU Racing members of the team. It was a juggling act of ensuring enough technical development was achieved for the year, while remaining within budget, ensuring all individuals had enough work to satisfy either FYP or Directed Reading course requirements and that members of the leadership team would not become overwhelmed with members to manage.

The Author had assumed to receive a small number of students who studied the Mechatronics Engineering degree but was met with none. The team was instead met with many Mechanical students. To remedy this the leadership team picked those who were most interested in Mechatronic roles, as well as who would excel best in them, based off limited interaction with the individuals. This principle was followed because if a project is given to an individual with little interest in it, then they will not put much care or create much output during the year.

6.3.2. Budget Constraints

2025 saw a 23% decrease in budget from 2024, from \$90,000 down to \$69,000. This significantly impacted the scope of projects that the leadership team could produce. In 2024, the University of Newcastle had a system where NU Teams would be injected with a boost of money halfway through the year, but in 2025, this had been removed, causing further budget constraints and concerns from the leadership team.

The leadership team was tasked with creating a budget, summarising where money was to be spent during the year. It was encouraged to make three versions of this budget, each version promptly named 'Splurge', 'Realistic' and 'Skeleton'. Within each budget giving an initial cost for the item, then with a confidence percentage on top, to show the NU Teams Director that the team understood where there would be unknowns and possible money sinks in a project, and where there were not. Splurge was made to see how much money the team could spend, if given no constraints, this ended up being around \$166,000. Realistic, being what the team would ideally receive, of \$70,000. Skeleton, what was the smallest amount the team could survive on and still make it to competition, this was found to be \$53,000.

The leadership team decided to go with an extremely in-depth, itemised budget, providing as much detail as possible for each purchase, with the idea of if the number of unknowns were reduced, the teams initial budget of \$70,000 would be more likely to be granted. Each budget was split into General, Mechanical and Mechatronic department purchases. The budget was successfully granted, and the leadership team was given praise for the work that went into the budget's creation.

The creation of the budget was closely linked to the number of projects that the Mechatronics team could undertake. Both the scoping document and budget were essentially created together, as one could not be finished without the other. It is encouraged that this workflow is followed in subsequent years, as it gives the team the most likely chance of receiving their desired budget, as it shows NU Teams Directors that the team is acutely aware of any money sinks that could be encountered during the year.

The Mechatronic budget was managed exclusively by the Author, and this approach proved effective, with NU Racing as a whole finishing the year approximately 4% under budget.

6.3.3. Timeline Constraints

The Author also created a Gantt chart detailing project timelines up to June of 2025. This helped to see overlaps in project deliverables and potential clashes with integration of projects. While helpful, the Author also recommends that future Mechatronics Department Leads should have a document detailing all remaining manufacture of a project. To help further detail what still needs to be done before a PCB, for example, can be put on to the car (e.g. enclosure manufactured).

The final version of the Gantt the Author created can be found on the NU Teams SharePoint, as well as Appendix E – MCHA Gantt Chart.

6.4. Project Dependencies and External Interfaces

Being the Lead of the electrical systems on the car, this project was heavily reliant on both individuals of the Mechatronics and Mechanical team. These team members were critical in completing PCB designs, software and sensor integration, as well as neatly packaging all hardware components onto the car. There was a large need for clean and consistent communication between the Mechatronics Department Lead Engineer and the Mechanical Department Lead Engineer. This was found to be most efficient form of communication between the departments as both individuals were most up to date across their department and were able to see potential clashes in integration early on. This was one of the areas of struggle during 2025, as there were times of inconsistent communication between the Department Leaders, which would lead to delays in new component integrations, and ultimately less track testing.



External interfaces for this project consisted of liaising with companies for component purchasing, as well as PCB manufacture. It is recommended that future Mechatronic Department Lead Engineers and other NU Racing students become comfortable with sending emails to companies asking for product updates, as well as potential sponsorships, to save the team money where possible.

When liaising with external companies, it is recommended to use a NU Teams/NU Racing email signature at the end of all emails. This helps with showing the company the team's professionalism, helping yourself become familiar with using email signatures, as well as helping the company learning who you are. An example signature has been attached in the Appendix, Appendix A - Email Signature; the Author encourages readers to use it and edit it to be their own.

7. Responsibilities and Management Approach

7.1. Technical Responsibilities of the Mechatronic Department Lead

Technical Responsibilities for the 2025 Mechatronic Department Lead were as follows:

- Maintain consistent and reliable validation of systems
- Overview data collection from onboard vehicle data logger
- System optimisation
- Compliance
- Innovation
- Provide technical knowledge to Mechatronics team
- Filter information to Chief Engineer and Team Leader
- Work with Mechanical Department Lead Engineer
- Aid in troubleshooting of Mechatronic systems
- Resource allocation
- Timeline management
- Budget overview
- Competition Technical Inspection and preparation
 - Electrical Safety Officer while at competition
- Help with Design event and preparation
- Help with Cost event and preparation
- Aid with preparing the car for each dynamic event at Competition

7.2. Safety, Compliance, and Quality Assurance Responsibilities

7.2.1. Safety-Critical System Oversight

Safety of team members was always the number one priority. The Author aided with this in the following ways: ensuring all team members working on the car had correct PPE and informing and training members of the risks they were exposed to when working on electrical systems on the car. Ensuring all safety-critical systems were functioning correctly before the car was driven, as an example conducting the Accelerator Pedal Plausibility Sensor (APPS) check as well as testing the dash shutdown button and precharge/discharge circuit after work had halted on the car before a track day. Other ways include preventing unsafe and untested designs from reaching the car, enforcing validation and unit testing as much as possible to reduce the risk of a new component reaching the car and harming nearby systems or potentially the driver.

7.2.2. Competition Compliance

Compliance with FSAE-A regulations was upheld during the design process of new systems. During integration it was acknowledged that compliance could not always be achieved due to components having issues and safe work arounds being put in place to continue testing. An example would be that the DCDC on NU25 had to be manually started between power cycles due to a baud rate clash between systems. This was not compliant by FSAE-A rules but was fixed later by having all systems be on the same baud rate again.

7.2.3. Quality Assurance and Design Review Process

During the Author's previous experience in NU Racing years, it was acknowledged the importance of quality assurance. Previous year teams had created Design Decision Reviews (DDRs) to aid with the PCB creation process, minimising the chance of unreliable and dodgy designs making their way onto the car. The Author upheld the DDR standard in 2025, incorporating previous lessons learnt into the DDR, such as a point for ensuring fuses to be added to traces that would power external systems to the PCB.

Other areas include holding a documentation standard for any test procedures or validation conducted throughout the year, maintaining consistent version control between PCB iterations, and training individuals on manufacturing looms. Crimping of pins (DT and Molex), where often the most common issue that caused wildly random issues to stem from.

7.2.4. Validation Gates and Readiness Criteria

Being an overarching team goal of 2025 to increase conducting and documenting validation, the Author enforced this within the Mechatronics department by requiring new circuits to be LTspice tested (simulated), proving their functionality, then adding another step to the design process of conducting sub and full integration test, reducing the number of unknowns at each step. Each stage was to be checked off by the Author before the individual could proceed.

This extended strategy to the design process proved successful by the fact that the team completed 842 km of testing, with six new PCBs being introduced to the car throughout the year.

7.3. Budget Management and Resource Allocation

7.3.1. Budget Management

Throughout 2025, the leadership team struggled with a confusing layout for the budget. It had many revisions over the year and eventually ended up with something that we were happy with. The final approach was to have a summary positioned at the top of the document, of current money spent, the delta from the approved budget of \$69,000, and other important aspects that the Team Leader and Chief Engineer requested to see briefly, example shown in Figure 25. Example of Budget Summary. Then within each section (General, Mechanical, Mechatronic), there were a ‘Spending Delta’ column for each item and their cost. This was used to easily track where overspending had occurred. This allowed decisions on subsequent purchases to be made easily. Example shown in Figure 26. Example of Budget Spending Delta.

The 2025 NU Racing Budget can be found in the NU Racing 2025 section of the NU Teams SharePoint.

Approved Total 2025 Budget	Excel Currently Spent	Asana Currently Spent	Asana Delta	Overall Budget Delta	Predicted Total 2025 Budget
\$69,000.00	\$57,051.59	\$ 60,028.02	\$2,976.43	\$2,219.19	\$ 66,780.81

Figure 25. Example of Budget Summary

[illegible]

Figure 26. Example of Budget Spending Delta

7.3.2. Resource Allocation

The leadership team began this process by reviewing the year prior, and searching for bottlenecks in the system and areas that either needed or could do with improvement. Once this list was gathered, the Author created multiple scoping documents for varying sizes of FYP and Directed Reading members that would join the Mechatronics team, from zero FYPs and four Directed Readings/Extracurricular up to five FYPs and two Directed Readings/Extracurricular. Multiple versions of any document should be avoided if possible and instead combined all into a single document. This helps with file management as well as ensuring information does not get lost in old documents.

The Mechatronics FYP team was comprised of two Mechanical students and one Aerospace Systems student. These individuals were chosen as they expressed interest in the respective roles. It turned out to be an extremely switched on and dedicated group of FYPs, which the Author will be forever grateful for.

Aside from FYP students, the Mechatronics team relied heavily on Directed Reading and Extracurricular students during the entire year. During semester one the non-FYP Mechatronics team comprised of two Directed Reading students, Zak Lobko and Jayson Mills, and three Extracurricular students, Mia Cosentino, Oliver Dowzer and Hugh O'Brien. Though Oliver Dowzer and Hugh O'Brien's projects were cut short due to lack of communication with the FSAE-A committee at the beginning of the year. In semester two, there was one Directed Reading student, Mia Cosentino, and two Extracurricular students, Fletcher Cogoi and Jayson Mills. These individuals were given specific projects, but there were many other individuals from the Mechanical and ENGG2200 cohorts who were given many small but incredibly important jobs.

7.4. Communication and Coordination with Other Departments

Communication is by far the most important aspect of a leadership role. There were multiple times during 2025 where a breakdown in communication occurred between the leadership team, and it caused a noticeable negative impact on the rest of the team.

For the Author, this occurred most with the Mechanical department lead. It is believed that it was a combination of neither of us attempting to dedicate time to ensuring communication was constant at the start of the year. Then when projects began to get busy, we had quickly forgotten and did not realise we were not communicating. The first consequence of this was during the integration of the new HIP PCB (V1.3). The enclosure had not been completed, and the Author was about to cut and re-crimp critical HV cables to interface with new connectors that had been chosen as a part of the new design. The Chief Engineer luckily had picked up on this, and halted work to ensure the upcoming track day was able to go ahead.

After this incident, the Author and Mechanical department lead decided to dedicate an hour every Friday morning to go over where each department was up to in manufacture or integration. This helped significantly in highlighting areas of potential integration conflict.

Communication with the Team Leader and Chief Engineer was consistent through the weekly Leadership Meetings. Through the year though, the Author realised that summarising all critical information about each project was difficult to do on the spot. To remedy this, the Author ensured to take a quick five-minute break before a Leadership Meeting to gather thoughts and note any areas of concern, as well as areas of big wins.

In Leadership Meetings, it is recommended to take a cynical approach to conveying information about projects. This attempts to reduce the likelihood of forgetting to convey important information to the rest of the leaders during the meetings. After all the 'fires' had been talked about, then big wins can be talked about. The 2025 leadership team arrived at this approach towards the end of 2025 when there was little time to have leaders' meetings, and issues were the main thing that had to be talked about. Otherwise, there was a risk of wasting time on topics that did not matter.

8. Engineering Decision-Making and Technical Oversight

8.1. Systems-Level Engineering Perspective

An FSAE EV race car is comprised of many systems and sub-systems. All that are interlinked in one way or another, like one massive spiderweb. Being able to see the links between the systems is a difficult task but must be done to be able to make informed decisions when it comes to making technical changes to the car.

All design decisions in 2025 for the Mechatronic systems had this philosophy in mind. For example, mounting of the Brake Over Travel Switch (BOTS) had to be positioned correctly in the Pedal Box to both ensure the brake pedal will press it if the brake master cylinders fail, but also not press it if they don't fail, while also ensuring that the switch was rotated correctly so that the switch would be in the correct orientation when the car enters HV.

Vital aspects to consider when considering changes for systems are:

- Safety
- Compliance
- Reliability
- Inspect-ability (easy to show during Technical Inspection)
- Integration effort (how much needs to change to implement the design)

8.2. Design Review and Validation Strategy

8.2.1. Design Review

All new Mechatronic systems that went on NU25 in 2025 underwent a thorough design process. The process is outlined below:

- Design
- Simulate
- Manufacture
- Unit test
- Integration test
- Commission
- Validate

The stages of Design, Simulate, Manufacture, Unit Test, Integration test and Commission are all touched on in the Design Decision Review (DDR) document that is created for each PCB that is made. There was a template DDR version made in a Template File Structure folder in the NU25 GitHub Repository, which was copied and edited to each PCB made.

The individual making the PCB must not commence to the next stage of design until the stage they are up to in the DDR has been checked off by the Mechatronic Department Lead Engineer. This ensures that mistakes are not carried through the whole design process, then upon review at the end a small mistake is picked up on and costs many hours of reworking the design.

Each year this document is added to with lessons learnt within the year. The Author recommends that the document and process is continued in future years. Due to both the number of times it has been tested and proven valuable, and due to it being a medium for easily carrying knowledge between the years.

After the DDRs creation in 2023, it is estimated to have been used to create over 32 PCBs, an estimated six in 2023, nine in 2024 (multiple iterations of each main PCB on NU24), and 17 in 2025 (also multiple iterations of each main PCB on NU25).

8.2.2. Validation Strategy

Being an overarching team goal for 2025, the team saw the first dedicated push to have every new system on NU25 validated, showing that a new design made a positive or negative impact to the systems around it and why.

Being the first year to push for it, there was not much infrastructure or previous knowledge on tested methods of validating Mechatronic system changes.

Most changes would show positive impact through data, being specific to the change. For example, once regenerative braking had been turned on in 2024, and math channels created in i2 Pro to calculate power regenerated through an endurance run, it was clear that regen had a positive impact and resulted in the Accumulator having far more capacity at the end of a run. More information can be found in Joshua Haywards FYP report [5], available on the NU Team's SharePoint.

Smaller changes, such as changing a connector to a different type, would show improvements in ease-of-life, serviceability and service life of a system, but there were no hard numbers showing the change had a positive impact.

One of the failings of the Author in 2025 was not creating a structure on how to validate Mechatronic systems particularly for smaller design changes.

8.3. Key Engineering Decisions and Trade-Off Analyses

The main design change for NU25 was adding in a Differential into the drivetrain. In 2024, the housing for the drivetrain, named Powerbox, had issues with bending under high torque loads from the EMRAX 228 Motor, more information can be found in Ty Doubikin's 2025 FYP report [18], as well as Lachlan Fisher's 2024 FYP report [19]. Both can be found in the NU Team's SharePoint. Due to these two reasons, it was imperative that many kilometres of driving would have to occur to ensure the implementation of the Differential was safe and reliable.

As a result of, Mechatronic design decisions were centred around reliable operation of the car to maximise testing time and minimise the likelihood of electrical or control-related failures masking and potentially compounding drivetrain issues. The trade-off for this decision was the deferral of performance related improvements, in favour of stable, repeatable system behaviour throughout testing and competition.

Another area that the Author was always concerned about was the Accumulator, since 2022, there have been many issues that were carried into competition, often costing the team many points lost in the competition due to the issues. Thus, it was decided to be the focus system to validate and ensure reliability. Only compliance level changes, this limited the scope of developmental experimentation but significantly reduced the technical and compliance risk.

In 2024, an issue with the CM200DZ inverter, it was sent off for repair, and a new one was bought, which also had integration issues. These issues accumulated to 3 months of wasted testing time between the end of July and late October. The broken inverter was received at the beginning of 2025, and was tested at a track day in May, where it showed no signs of being broken. This demonstrated a need for a conservative approach to validation and configuration for NU25's tractive system.

The effectiveness of this method was validated at the 2025 competition as there was not a single issue caused by the Accumulator or Powerbox, a massive improvement from previous years and reflecting the substantial development and validation effort undertaken by Oscar Rudman, Ryan Bird and Ty Doubikin during the 2025 season.

8.4. Risk-Based Decision Making

Identifying engineering risk is critical. It appears in all aspects, from safety, compliance, reliability, and performance. In 2025 compliance, reliability and performance risks were not formally identified and documented, instead they were continually discussed and managed during Leaders Meetings. It is the Authors recommendation to formalise this process earlier on in the year, so that if anything the leadership team are made aware of areas to be cautious.

The only formal structure of risk awareness was within the budgeting stage of the year, where the leadership team was tasked with applying 'Factor of Importance', an estimated percentage of how unknown the cost of an item would be, ultimately referring to the risk involved in the design of the system.

An example tolerance structure of how risks in safety, compliance, reliability, and performance could be hierarchised with the NU Racing mindset is shown below:

- Safety risks: zero tolerance
- Compliance risks: very low tolerance
- Reliability risks: zero tolerance
- Performance risks: moderated/managed tolerance

Through discussions of risk in leadership meetings, the leadership team were able to identify designs that required more in-depth unit testing and validation before the design was deemed reliable. This was most prominent in the Powertrain project, though another area where this philosophy was used was in the Expansion Board project, where the Author saw Oliver Bunny's scope as major risk since the team had not undertaken a PCB that integrated the number of sensors that was desired. As a result of this, the Author ensured to dedicate large amounts of time to design reviews of the PCB and ensure that there was in-depth testing procedures created to prove the PCBs functionality. This decision was validated with the success of the PCB, having all sensors tested during 2025, allowing significant Vehicle Dynamic research, powertrain cooling research, informed and efficient driver training, and informed brake pad selection to be conducted.

The discussions within leadership meetings significantly reduced the amount of risks the team faced, though where residual risk remained, acceptance was contingent on mitigation strategies, and ultimately an assessment that further reduction was not feasible within the available timeframe. Residual risk was most pronounced in the inverter issue that the team faced after Acceleration event testing had been conducted with the Differential Powerbox. After researching the fault, the team acknowledged that the solution could not be implemented in the available time left before competition. Instead, a temporary fix of limiting the torque during Acceleration runs was implemented. More information can be found in Samuel Connett's FYP report [10].

9. Mechatronic System Development and Integration

9.1. Overview of Mechatronic Systems on the Competition Vehicle

An overview of the Mechatronic systems of NU25 are depicted in NU25 Mechatronic System Architecture. A more detailed overview of the electrical connections and system topology can be found in Appendix C - NU25 Electrical System Overview.

9.2. Design Evolution and Change Management

NU Racing follows a ‘build first, design second’ philosophy. In 2024, most designs had been decided for NU25. In 2025, the Mechatronic systems were required to change as little as possible, as discussed in the Key Engineering Decisions and Trade-Off Analyses section of this report, only minor revolutions to designs were allowed. This resulted in the Author creating a detailed list of changes to each node, a process that has its cons and are discussed in the Onboarding, Training, and Mentorship section of this report.

Designs were evaluated with the help of consultation with the Chief Engineer, Malcolm Sidney and Dylan Cuskelly. Along with Decision Matrices, that allowed for un-biased decisions to be made in a quantitative way.

During the year, the Author found it difficult to keep track of changes to the race car (NU24 and NU25) between each track day. Thus a ‘Changes to NUxx’ list was written up on a whiteboard in the TA workshop. This was found to be a quick and efficient way of capsulating changes to the car. The Author thought it was beneficial to ensure that the method of documenting the changes were quick, because of the reduced time both the Author and Mechanical department lead had to document the changes, a whiteboard in the workshop seemed most fitting. After the changes were finalised for the track day, the Chief Engineer would then formally document the changes in the ‘NU Racing Track Day Testing Log’. A process that the leadership team found to be efficient. An example whiteboard summary is shown in Figure 27. Whiteboard Change Management Solution.

This whiteboard summary also helped to fault find during track days. Since it was usually an integration issue that was causing a fault, being able to quickly see what the changes were for the track day allowed the Mechatronics team to efficiently find and fix issues. Resulting in more efficient use of track days and increasing driving time.

CHANGES TO NU25 BEFORE		date: 5/12
CHECK	\$	REFIT
MECH		MCHA
- Rear suspension - Diff/sprockets (Diff oil replaced) - All fasteners replaced - Steering system - Pedal box - Wheel bearing pre-load		• hot glued tees & PEN & BSD pots • Sealed DEN, CEN, HIP enclosures • removed all expansion board stuff (still have) • Dotted HIP-CEN (beacon) (LV) (CEN-ACCU (LV))
CHANGES	\$	UPGRADES
MECH		MCHA
Segment housings		• CANvaMon 1 • Kapton tape all LV & HV wires in accu

Figure 27. Whiteboard Change Management Solution

Other forms of change management included the creation of a document called '2025 MCHA System Design Changes', that outlined all the changes to PCBs made in 2025. Available on the NU Team's SharePoint as well as in Appendix G – 2025 MCHA Systems Design Changes. Design Decision Reviews (DDR) were also used as change management, at the beginning of the DDR, there is a section that gets the individual to detail the changes to the new PCB version compared to the last version. DDRs are further discussed in the Quality Assurance and Design Review Process and Design Review and sections of this report.

9.3. Integration with Mechanical and Vehicle-Level Systems

- PCB and enclosure issues (CEN and HIP mounting)
- Sensors on wheel assembly (uprights)
- TS stack (DCDC, inverter, lv battery)

9.4. Issues Encountered During System Integration

- Implement as many steps as possible to integration to make sure that everything that can be tested is tested before it is all put together.
- Do a dedicated test day of new Mechatronic systems on previous year's competition car. Makes so that (ideally) all changes to Mechatronic system is tested before new competition car is running.

10. Testing, Validation and Compliance

10.1. Validation Strategy and Test Philosophy

Validation is the process of ensuring that a product, service, or system meets the operational needs and intended use of the end user [20]. It is a critical step in the engineering process, as it is the direct way to know if a change to a design has made a positive impact or not. It ensures safety and aids to decrease the risk of a new design.

With improvements in 2025 with validation in NU Racing, the ideal strategy employed consists of testing a design in a controlled environment, one where unknown and uncontrollable variables are kept to a minimum, or if possible reduced to zero.

New designs should only be allowed on the competition platform once controlled validation has taken place, and unit tests have been successfully passed. This method reduces the number of unknowns and late-stage failures of a new design as much as possible. Though note should be taken that one cannot remove all possibility of an element being missed.

In 2024, it was demonstrated that insufficient or mislead diagnostic clarity and methods could lead to misidentification of hardware faults, further reinforcing the need for a staged, multi-step validation and unit testing process.

10.2. Simulation and Unit Testing

The Author insured to push for simulated circuits before PCBs could be ordered in 2025. This proved to help with individuals understanding of the circuits on their PCBs, as well as easily highlight circuit designs that would not work.

The chosen software in 2025 was LTspice, due to it being taught in the University courses, as well as its wide range of components and its large online community. Other simulators included Falstad, an online quick to use real-time simulator. Though not as in-depth and good as LTspice, it was used to quickly sketch a circuit and test its functionality with a visualisation of current flow.

Simulations of circuits are recommended to be conducted in future years.

Unit tests were an extremely valuable tool throughout 2025 in the creation of PCBs; it helped to find issues once a PCB had been manufactured but not yet put on the car. An in-depth unit test plan and check list can save many hours troubleshooting on the car. A detailed example unit test checklist written by Mia Cosentino is provided in Figure 28. HIP V1.4 Unit Test Checklist.

Stage	Function	Description	Verification Method	Steps
10. Before Assembly	QUALITY CONTROL	PCB Quality Control	Inspection	Check the PCB is has been manufactured to our expected quality. Eg: - check continuity across zones (12V RAW, 60A, PREGLV) - check continuity across all other (meant to be) connected sections/traces Check each component has been manufactured to expected quality. - ensure all components are as expected - correct resistances - ensure regulator works as intended (breadboard) - fusing intact
11. Before Assembly	QUALITY CONTROL	Component Quality Control	Inspection	Use a multimeter to read that the resistance between power and ground is an open loop. (multimeter will display OL (open loop) basically -> not connected)
12. Before Assembly	POWER DISTRIBUTION	Power/Ground Isolation	Test	Use a multimeter to read that the resistance between TS+ and TS- is an open loop.
13. Before Assembly	TS+/- DISTRIBUTION	TS+/TS- Isolation	Test	Use a multimeter to read that the resistance between anything/everything in HV and anything/everything in LV is an open loop.
14. Before Assembly	HV/LV DISTRIBUTION	HV/LV Isolation	Test	1. Solder on all components from Voltage Regulation and Protection Block (solder a fuse holder and then insert correct fuse, do not solder fuse)(ensure leds are blue). 2. Use a multimeter to read that the resistance between power and ground is an open loop.
15. During Assembly - Unpowered	POWER DISTRIBUTION	Power and Ground are isolated	Test	Ensure correct connections and operations. 1. Connect screw terminals to 12V DC power supply. 2. Ensure LED's light up as intended (blue). 3. Using multimeter ensuring that 12V and 5V are only where expected (dont forget current sensor should receive 5V, LED's, big and small testpoints) 4. Ensure fusing is working as intended (continuity) 5. Use diode function on multimeter to ensure reverse polarity protection. can you check that a diode lights up by placing power supply on either side or not bother? yes basically, do this.
16. During Assembly - Powered	POWER INSPECTION	Regulator Quality Control	Test	1. Solder on LED's (red) and resistors from Masterswitch and Interlocks blocks (3 LED's/resistors total) 2. Use a multimeter to check continuity where expected (ensure solder joints are correct)
17. During Assembly - Unpowered	QUALITY CONTROL	LED/Resistor Quality Control	Test	1. Solder on 15k 500VDC Body Protection Resistors from TS Voltage Block (in HV) (2 resistors total). 2. Use a multimeter to check resistance and continuity is as expected (ensure solder joints are correct).
18. During Assembly - Unpowered	QUALITY CONTROL	Body Protection Resistor Quality Cn Test	Test	1. Solder on Fuses (fuse holders) from Test points block and insert correct fuses (1A) (2 fuses total). 2. Use a multimeter to check continuity is as expected (ensure solder joints are correct).
19. During Assembly - Unpowered	QUALITY CONTROL	Banana Plugs Fusing Quality Contrc Test	Test	1. Solder on Fuses (fuse holders) from I/O block (DCDC) and insert correct fuses (4A 500VDC) (1 fuses total). 2. Use a multimeter to check continuity is as expected (ensure solder joints are correct).
20. During Assembly - Unpowered	QUALITY CONTROL	HV Fusing Quality Control	Test	1. Solder on remaining connectors - banana plugs in Test points block (12VRaw, PREGLV, TS+, TS- -> red)(GND -> black)(S total) - Current sensor (Molex Micro-fit 2x02 3.00mm Vertical) - Arc Interlock (Molex Micro-Fit2x02 3.00mm Vertical)

Figure 28. HIP V1.4 Unit Test Checklist

Another area where unit tests for Mechatronic systems occur was in loom creation, by continuity checking looms to ensure pin outs were correct. In 2025, there was no way of unit testing parts of the Tractive System, such as the motor. Once the Dynamometer is functional, it will fill this gap.

Further discussion on the positives of simulation and unit testing can be found in sections Validation, Validation Gates and Readiness Criteria and Design Review of this report.

10.3. Integrated Vehicle Testing

During integration testing in 2025, many issues were found and fixed, below is a summary of some of the issues encountered. Further information about each issue can be found in NU Racing's Lessons Learnt document, available on the NU Team's SharePoint.

- Removing motor controller interlock connectors inside hip before comp (from track day testing they would rattle and disconnect)
- Inverter overcurrent faults (more information found in Samuel Connett's FYP report)
- Inverter DC overvoltage faults (more information found in Samuel Connett's FYP report)
- CEN code upload issues
- Updating the cars DBC without updating code on nodes where changed signal IDs came from
- Beacon issues
- DCDC baud rates
- AIL popping capacitors during power cycle
- BSPD tuning
- BOTS issues with shutdown circuit not turning on interpose relay (and the car)

10.3.1. Orion 2 BMS Data Viewer Software

Battery Management System (BMS) faults were frequent at track days. When the BMS does fault, it stores a snapshot of parameters within the BMS at the time of the fault. It creates a CSV file that can be viewed and saved to disk from the Orion 2 BMS software. With significant help from the AI Claude, the Author made a Python script that takes the CSV files as input, and outputs graphs of cell resistances, as well as open and instantaneous cell voltages.

The Author enforced the need to save the CSV files to disk after a fault had occurred, as in previous years the fault would simply be cleared after viewing the CSV, and the CSV would be lost. The Author found this useful to quickly see what cells were causing issues and that may need replacing. This script can be found in the BMS folder in the Racing-NU25 GitHub Repository.

10.4. Technical Inspection Preparation and Outcomes

Technical Inspection at the FSAE-A competition is extremely important. If one of the eight Inspections are not passed then the car is not able to compete in any of the dynamic events at the competition, were 67.5% of the available competition points lie. If done efficiently, they significantly reduce the stress felt by team members during the competition, which is incredibly valuable.

Though a struggle, the team performed extremely well in Technical Inspection at the 2025 competition, the team Mechatronic team passed EV Static Technical Inspection first try, Accumulator Technical Inspection second try, and EV Functional Technical Inspection first try.

The hard work was rewarded, competition recognised this by awarding NU Racing with the Technical Inspection specialty award, shown in Figure 29. 2025 Technical Inspection Award. This award could not have been achieved without the endless hours of dedication from the Mechatronic and Mechanical team members who were present and helped with Technical Inspection.



Figure 29. 2025 Technical Inspection Award

10.4.1. Inspection Strategy and Preparation

10.4.1.1. *Practice Technical Inspections*

Mock Technical Inspections are extremely important to conduct throughout the year. They help team members involved in Technical Inspection to learn and practice for the Technical Inspections at competition. In 2025 these began in late July, with the Technical Inspection sheet for the 2025 competition released October 24th, weekly Technical Inspections occurred from late November. It is recommended that these occur more frequently throughout the year, as it was a push and down to the wire to get the car compliant in 2025. It is important to get in as many Alumni for this process as possible, as it is an easy way to get multiple informed perspectives on the team's preparedness, as well as the compliance of each system.

10.4.1.2. *2024 & 2025 Philosophy and strategy to Technical Inspection*

In 2024, the Mechatronic Department Lead Engineer, Alec Chapman, created a 'MCHA Bible' for Technical Inspections. Which were the Technical Inspection printed out with brief descriptions of how to answer each point. The Author saw how effective these Bibles were and decided to continue their use in 2025. They are available in the Technical Inspection section of the SharePoint, as well as in Appendix D - 2025 Technical Inspection MCHA Bible.

Also carried from 2024 was to have a dedicated document handler present at each Technical Inspection. The role of this person is purely to handout and organise all documents that are required for the given Technical Inspection. This helped with the fast-paced environment that Technical Inspection becomes. Technical Inspection is ultimately a battle of interpretations of the competition rules and technical inspection points. It is vital to remain level-headed and calm during competition Technical Inspection. It is likely they will pick up on points or interpret a section differently to how you have. You must be prepared to argue your interpretation, as well as provide evidence and reason with the Scrutineer. Do not fold immediately when they pick up something, since at times they have not fully read the Technical Inspection point, or have read it wrong.

It is important to note that not all the Scrutineers have experience in the topic they are inspecting, be prepared to have to explain or let them figure out the Technical Inspection points on the spot and even guide them to your understanding of the point, to help pass it quicker.

10.4.1.3. Technical Inspection at Competition

Then there are the Scrutineers who have been doing it for years and enjoy pulling teams up on small insignificant points and interpretations. They will stick to their guns and will not change their mind without a calm counterpoint from yourself. If they really are extremely not happy with a design, the Technical Inspection Captain will be called upon to give a final decision. In 2024 and 2025, this person was easy going and waived most concerns.

An aspect that can be used to your advantage is the chaos of the situation, in 2025 there were three Scrutineers in EV Static, two in four in Accumulator Technical Inspection, and one in EV Functional. It is very situational, but the Scrutineer may skip points or brush over stuff in the essence of time. If this occurs, again very situational, but it is possible to leave it and hope they don't pull you up again on a potentially non-compliant point.

Each Technical Inspection is given one hour to be completed; this is done since the competition has many teams to get through in the two days given to run static events. Due to this, the team has historically and did the same in 2025, constantly snooped and checked if Technical Inspection bays were free to see if the team could jump the queue. This has been valuable as often teams are not very organised, and it gives extra time if issues occur in the given Technical Inspection.

10.4.1.4. Advice to Future NU Racing Students Undertaking Technical Inspection

- Do as many mock Technical Inspections as reasonably possible. They will start out with extremely poor performance, but over time the team will learn and the car more compliant, it is only repeated attempts at Technical Inspection that create an efficient and compliant car.
- Use and replicate the 'MCHA Bible'.
- Dedicate specific people to each Technical Inspection, with a dedicated document handler.
- Best advice is to prepare as best you can, gather as many documents as possible to prove that the given design is compliant.
- Be prepared to have to argue your interpretations or even explain the Technical Inspection points to the Scrutineers because they read it wrong or not fully.
- Use the chaos of the situation to your advantage.
- Continually check to see if other Universities are dropping the ball and not showing up to their Technical Inspections and jump the queue if the chance arises.
- Be wary of both in-experienced and experienced Scrutineers either not reading points correctly or having an out-dated understanding of how a point works. As well as experienced Scrutineers looking to bite you on the smallest of things. Like grounding of hose clamps....
- Be confident, be prepared and argue your interpretation of rules.

10.4.2. Accumulator Technical Inspection

In 2025, the Accumulator Technical Inspection was split into three sections, safety tools check, charger check, and the actual Accumulator check. Due to the one-hour timeslot allocation, the Author split the team into two groups, charger and Accumulator. The Accumulator team would conduct both the tools check and the actual Accumulator check itself. There was a dedicated document handler for the charger and Accumulator sections.

Once the Accumulator Technical Inspection is passed, the Scrutineers hand the team a thermal strip, to place on the most negative terminal of the Accumulator. In 2025, it was placed in the location shown in Figure 30. Thermal strip viewing hatch. This was inspected later in Parc fermé, to show that the Accumulator did not exceed temperature restrictions defined in the Accumulator Technical Inspection sheet. Further information on Parc fermé can be found in the Parc fermé section of this report.

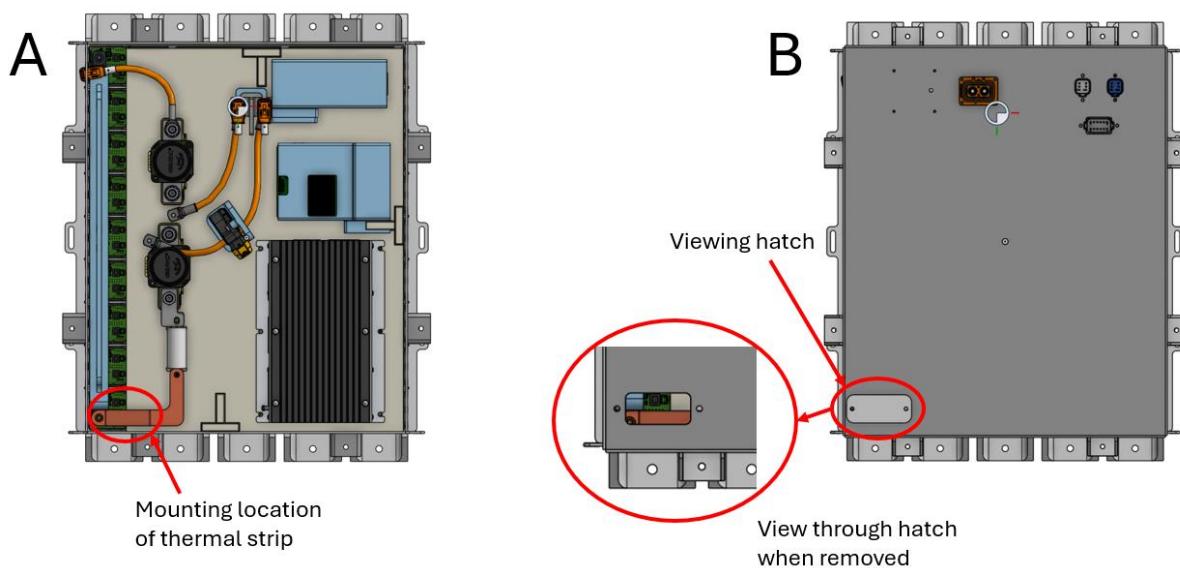


Figure 30. Thermal strip viewing hatch

At competition the team was able to do the charger and tools sections early. The tools section passed first try, but the charger section did not.

10.4.2.1. *Charger*

The charger section consisted of both an Accumulator trolley inspection as well as the charger inspection. The Scrutineers picked up a very small wire located at the back of the charger that was un-terminated. The Author is still shocked that they managed to see the tiny wire in the depths of the charger, but they were right. They also picked up on that the handle of the charger trolley was not correctly grounded when the charger was earthed to the trolley. After checking with the Technical Inspection Captain, the charger was brought back to the pits and disassembled to remove the un-terminated wire. To resolve the trolley grounding issue, an earth wire was simply run from the trolley ground connection point to the trolley handle. Once the changes were complete, the team brought the charger and trolley back to the Technical Inspection area and both sections passed.

10.4.2.2. *Accumulator Inspection*

The Accumulator Technical Inspection failed on its first attempt. The Scrutineers failed the team on exposed copper connection on the Radlok pin that the quick disconnect to the lid attached to, it could be touched with a 6 mm diameter, 100 mm length probe, which is not a point in the Accumulator section of the Accumulator Technical Inspection, but it was a valid concern. The other point where they failed the team on was the nuts holding the AIL to the lid of the container, they were nylon nuts on nylon bolts, which the Author initially decided to go with so that there was not a metal bolt extruding from the inside of the Accumulator, but forgot that it had to be positively locked.

10.4.2.3. Fixing Accumulator Inspection Fails

These two fails meant that the first Technical Inspection had to be stopped, and the Accumulator brought to a clean room to disassemble to make the make the changes. They luckily were quick fixes. Both issues are shown in part A and fixes shown in part B in Figure 31. Accumulator Technical Inspection Fail Point 1 and Figure 32. Accumulator Technical Inspection Fail Point 2.

Once they were fixed, the team requested that a Scrutineer came over to the clean room to check the changes done. They were happy to do this, and it saved the team having to reseal the Accumulator and take it back to the Technical Inspection Bay. The Scrutineer was happy with the work conducted, though he noted that we were using rubber like tape as insulation in the case of Figure 31. Accumulator Technical Inspection Fail Point 1, which would violate AI.5.24, but was happy to pass it at the time. Care should be taken in future years around this point, as well as be aware that they could check for things that are not explicitly stated in the given Technical Inspection but instead show up in other parts of Technical Inspection. For example, the touch protection point they caught us on is a point in the charger section of Accumulator Technical Inspection (AI.4.18) and a section of EV Static (SI.5.2).

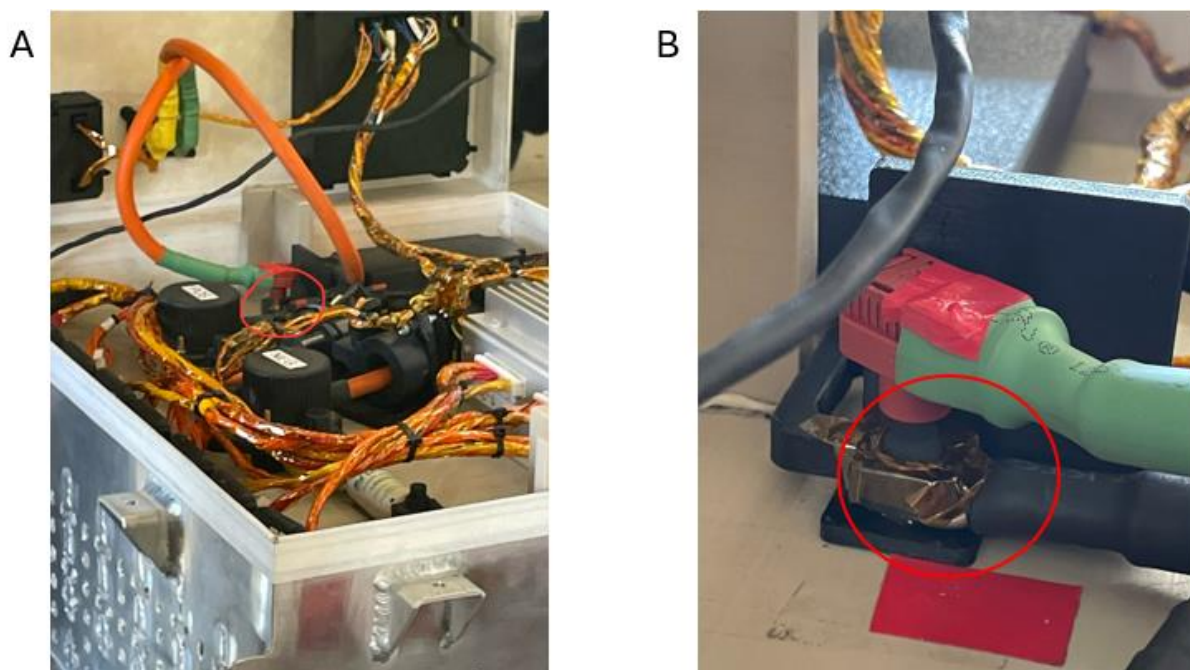


Figure 31. Accumulator Technical Inspection Fail Point 1

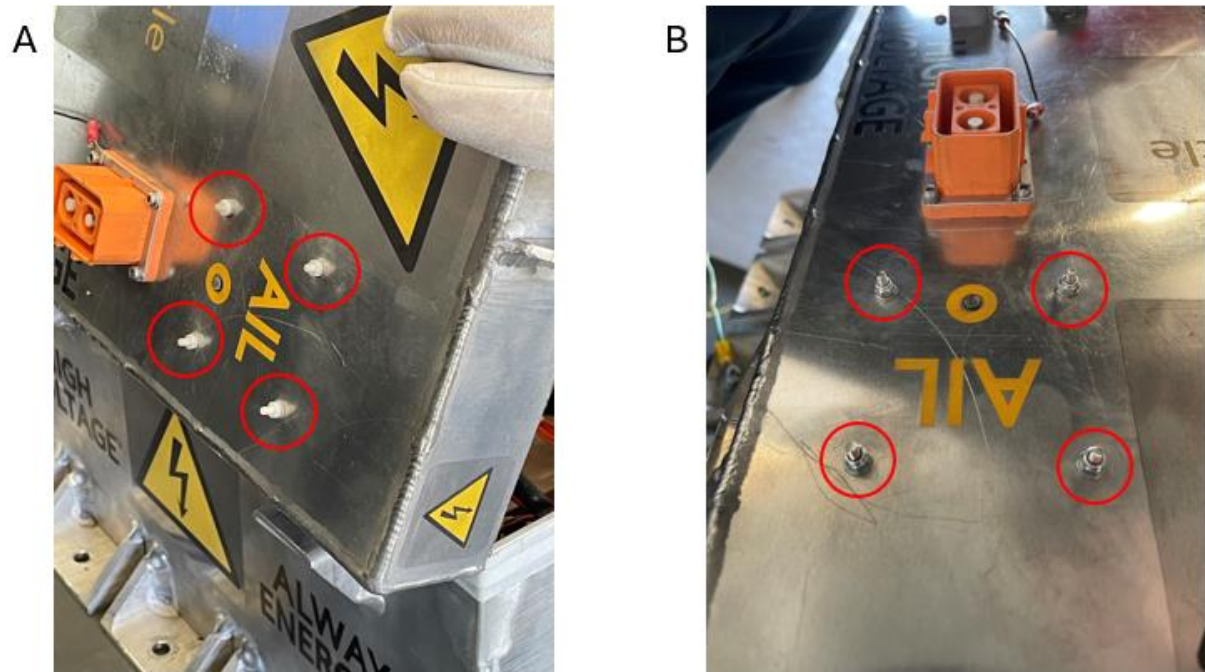


Figure 32. Accumulator Technical Inspection Fail Point 2

10.4.3. EV Static, EV Functional and Rain Test

10.4.3.1. EV Static

EV static passed first try. Though there were points that the Scrutineers picked up on, and after some debate and small argument, they were happy to pass. These points include:

- SI.8.2, the phase cables that entered the motor were not sufficiently secured. The point made was that the crimp was being strained, which is valid. It is recommended that this is fixed for 2026, possibly with the use of cable glands. Depicted in Figure 33. Visualisation of where phase cable securing is needed.

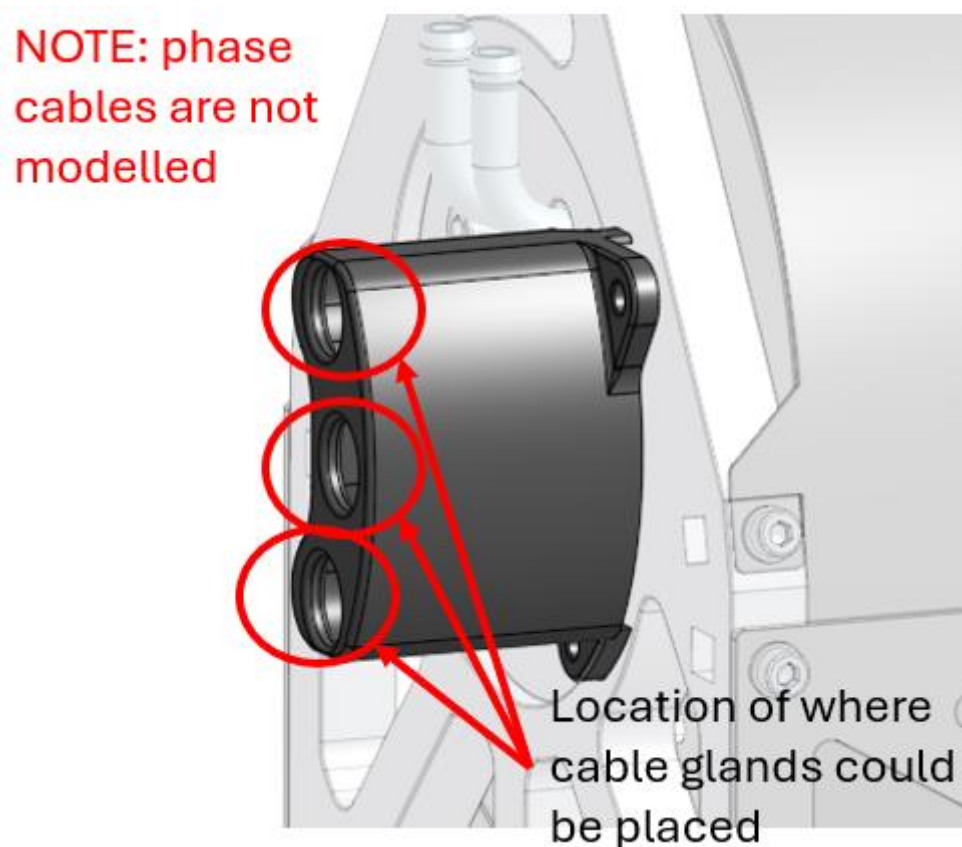


Figure 33. Visualisation of where phase cable securing is needed

- Not a point in EV Static, but the external bolts that hold the MSD in place on the HIP must be positively locked. They briefly mentioned it at the start of EV Static, but forgot to recheck it, because it is not a point. One solution that should be implemented in 2026 would be to lock-wire the two bolts together. Illustrated in Figure 34. Location of MSD bolts that should be positively locked.

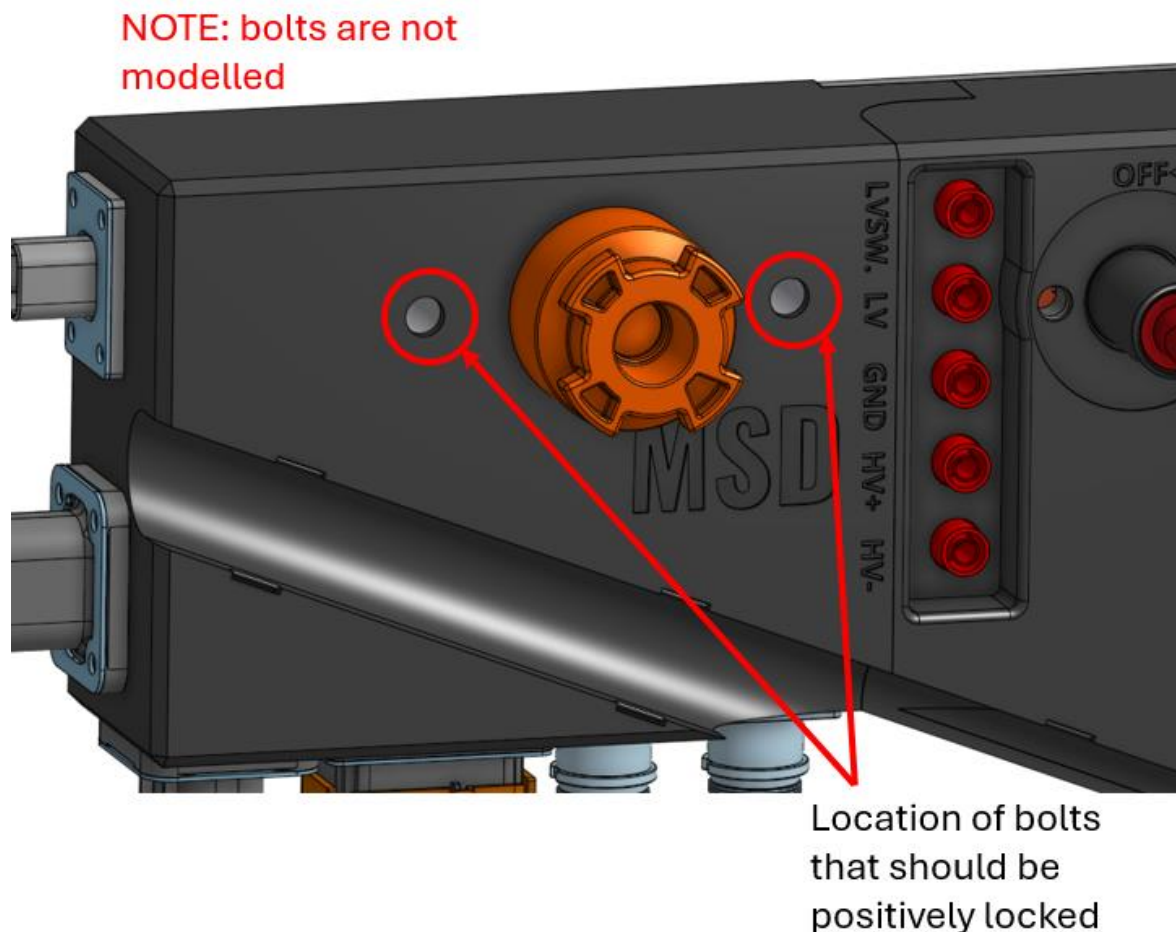


Figure 34. Location of MSD bolts that should be positively locked

- SI.5.2, when the MSD is removed, it could be argued that the contacts inside the MSD enclosure that are exposed can be touched with fingers. But the 6 mm diameter probe cannot touch the contacts. It is recommended to 3D print a cover for this in future years, just to cover your ass in case they get annoyed by it. Figure 34. Location of MSD bolts that should be positively locked, shows item of discussion, except that the MSD is in place in the figure.

- SI.18.16, they were not happy that the seat belt harnesses were not grounded. It is specified that items only need grounding if they are within 100 mm of HV components. This Scrutineer did not seem to care about this and pushed very hard for this point. The Author and Jayson Mills were sitting there sanding the heck out of the chassis to make the connection for the seatbelt harness holders (shown in Figure 36. Seatbelt harness holder locations), but they were not making a good connection. The Scrutineer ended up folding and waived it as not necessary. Though the point of components not needing to pass the 0.3 Ohm grounding requirement (shown in Figure 35. Showing 100 mm proximity grounding requirement in EV Static), the Scrutineer continued to check the grounding of items such as the steering and pedal box assemblies. From the perspective of the Technical Inspection point, the Author was very confused why they were checking these components. But from the perspective of safety, it is very understandable that they would check this, as if the chassis was to become live with HV in the event of an accident, bad connections to the steering and pedal box assemblies could become hot if current was pulled through them, but then again there is very little chance of that ever happening. This is an example of the Scrutineers checking random things that are not on the Technical Inspection sheet, either because it used to be a point in previous years, they get a kick out of it.

EV CLASS - EV STATIC INSPECTION		
VEHICLE GROUNDING		
It is expected that all conductive components within 100mm of the Tractive system will be appropriately grounded (electrically bonded to the GLV system ground).		
1. By equipotential bonding the conductive parts within the vehicle, the hazard posed by touching a voltage gradient is reduced, and		
2. By ensuring a conductive path back to the chassis ground common point, the likelihood of the IMD correctly detecting the fault and shutting off the vehicle is increased, reducing the period of time for which the hazard can be present.		
Scrutineers will look for any conductive components which are in close proximity (within 100mm) of any tractive system component (HV) element, which could become live in the event of a failure of a component, breach of insulation or similar either from normal operation, or in the event of a collision.		
Any wire, braid or similar used for vehicle grounding must be appropriately terminated, robustly mounted, and of sufficient size and mechanical strength for its purpose.		
This test involves using a Fluke 289 Multimeter on the lo-ohm setting to ensure there is a low resistance connection from the item being tested in reference to the chassis ground test point. It is acceptable to subtract the resistance of the test leads if necessary. Teams may remove paint, resin or other coverings to expose a bare conductive surface if required.		
Teams also have the option of measuring resistance via method of voltage drop using a 1 amp current source if necessary.		
Measurements to the conductive component should be taken within 100mm of the tractive system or as close as practical. Teams may remove body panels and covers to demonstrate compliance if required.		
The pass criteria are:		
a. Electrically conductive parts 300 mOhms Examples: parts made of steel, (anodized) aluminum, any other metal parts		
Criteria	Reinspect	Passed

Figure 35. Showing 100 mm proximity grounding requirement in EV Static

**NOTE: the chassis
is not powder
coated in this**

Location seatbelt
harness holders'
locations



Figure 36. Seatbelt harness holder locations

10.4.3.2. EV Functional

The team passed EV Functional first try. A massive attribute to all members from the Mechatronics team that were involved with both preparing for this event and who took part in the event.

During preparation for this Inspection, every mock Technical Inspection for EV Functional had something go wrong. Often it was the Brake Over Travel Switch (BOTS) mechanically failing during the shutdown circuit section. Shown in part A of Figure 37. A) shows location of BOTS in Pedal Box. B) describes reason for continual breaks, the switch was failing due to the frequent servicing of the brake lines in the pedal box. When serviced, brake fluid would get everywhere, and overtime caused the plastic of the switch to deteriorate. This combined with a poor mounting solution that created a moment on the thread of the switch, caused it to fail when operated. This was fixed by constant replacing of the switch, especially close to competition once brake line servicing stopped, as well as a proper mounting solution of supporting the switch housing by the base. Described in part B of Figure 37. A) shows location of BOTS in Pedal Box. B) describes reason for continual breaks

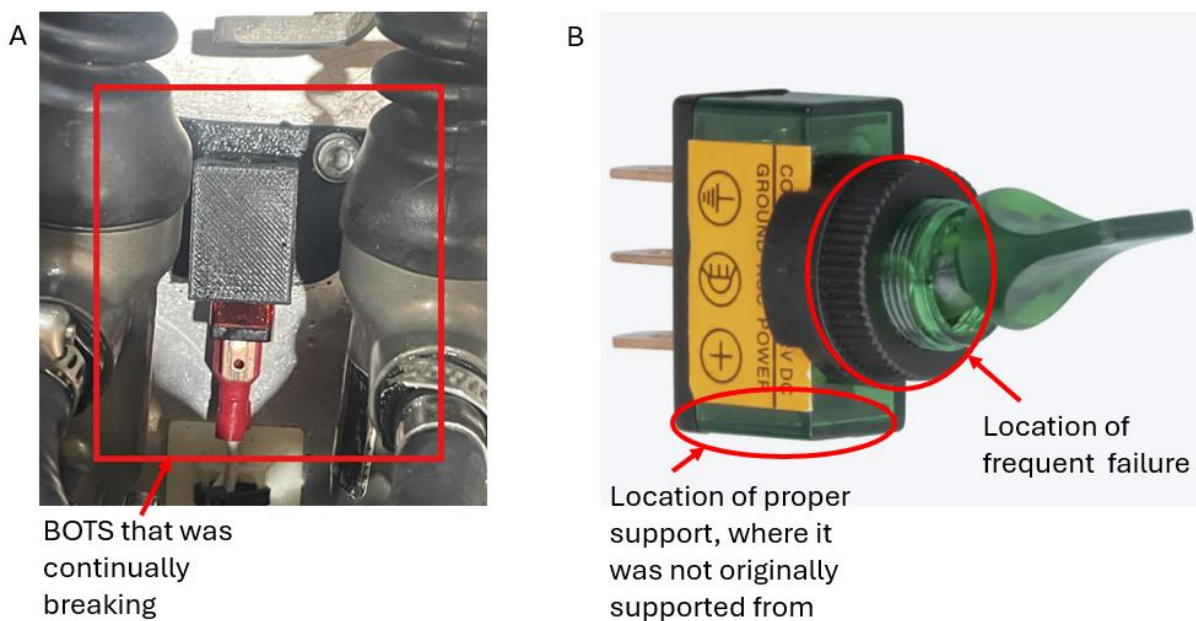


Figure 37. A) shows location of BOTS in Pedal Box. B) describes reason for continual breaks

There were also two instances of the fuse on the output of the interpose circuit on the Central Electric Node (CEN) would blow due to the frequency of switching on and off the Tractive System. This was occurring due to an incorrectly specified fuse, it was set to 1 A, when it should be a minimum of 2 A due to the transient current draw of the MegaVac Isolation Relays used in the Accumulator. This fault was occurring very close to competition, so the Author decided to put a 10 A fuse in the holder just in case. Fuse location shown in Figure 38. CEN fuse that was blowing during EV Functional Mock Technical Inspections.

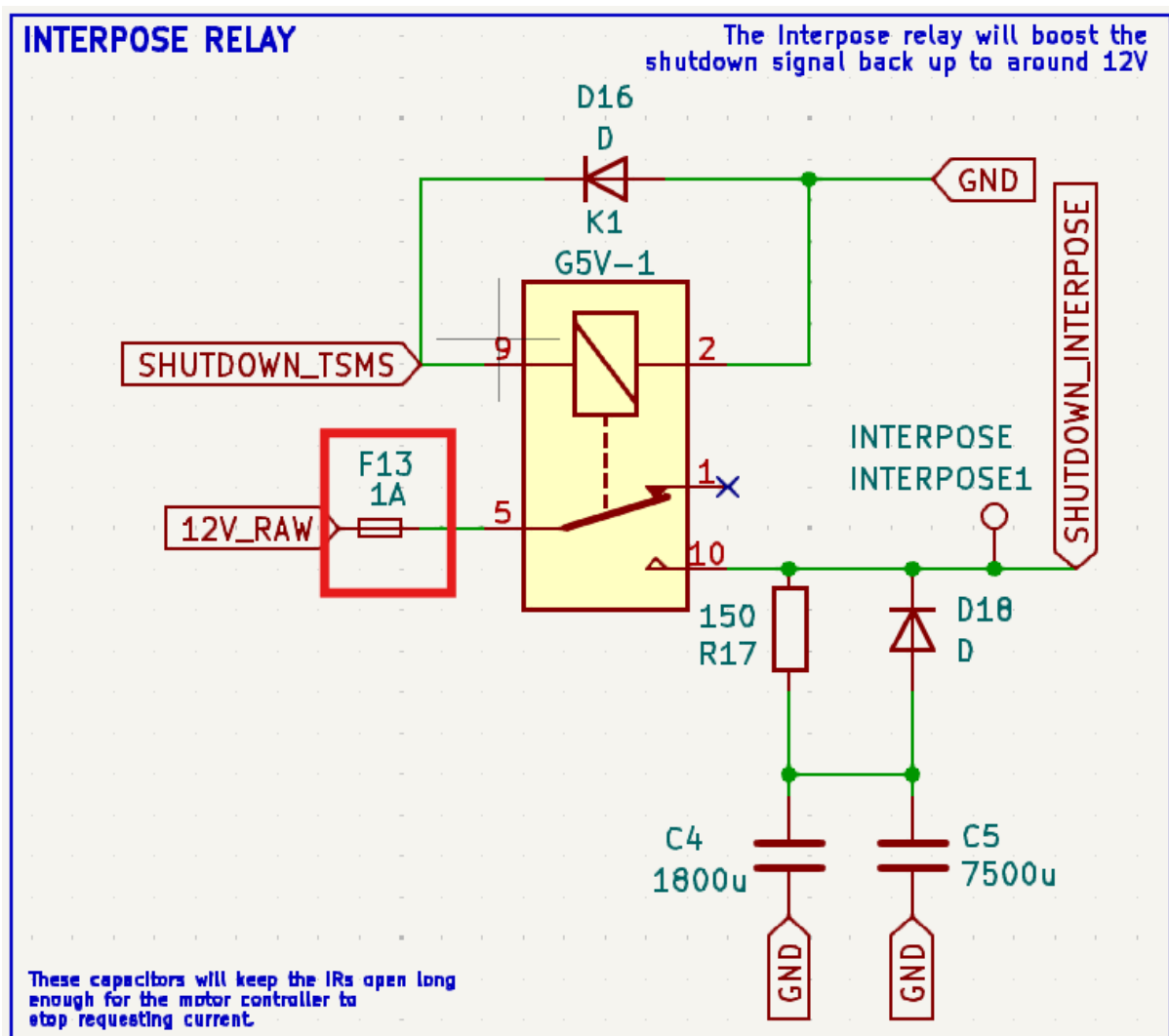


Figure 38. CEN fuse that was blowing during EV Functional Mock Technical Inspections

During the last EV Functional mock Technical Inspection, an Alumni found a weak point in the compliance of the teams Insulation Monitoring Device (IMD) test resistor box. The Technical Inspection point in question was FT.5.2, shown in Figure 39. FT.5.2 Explanation.

FT.5.2	IMD	The team must provide a test resistor manufactured with insulated banana plugs with an appropriate resistor in place. The resistor must have an appropriate voltage and power rating and be encased in a hard shell enclosure. The resistor should have appropriate power rating and be installed in a hard shell enclosure (heat shrink/tape insulation is not acceptable). Check the resistance of the test lead with the Fluke 289 Multimeter. The resistance should be a minimum of $R_{Test} = (\text{maximum TS voltage} * 250\Omega/V) - (2 * BPR)$ Measured Resistance: Should be $(453.6V * 250\Omega/V) - (2 * 15k\Omega) = 83.4k\Omega$ *BPR = the value of the Body Protection Resistors in Ohms BPRs are installed on HTP, present schematic if asked	Inspect	p.g.	
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Figure 39. FT.5.2 Explanation

Which is already a confusing point but can be easily interpreted as the test resistor must be rated for the correct power and voltage. Which the Author had not prepared material to prove that it was. Because of the Author explained that there was a single resistor with a power rating of 1 W, rated to 500 V inside the enclosure, the Alumni asked if that was enough and said to prove it, the Author did an on the spot calculation and it showed that it was not rated for the expected power of 1.8 W. Due to the Author's lack of confidence in answering the question, the Alumni had asked for the enclosure to be opened, which showed that there was two resistors in series, meaning that the dissipated power between the two was 0.9 W, very close to the 1 W rating the Author had told the Alumni.

It also showed additional screw terminals in the box that was not rated for 500 V, as well as a dodgy solution. The box is shown in Figure 40. Non-compliant IMD Resistor Box.

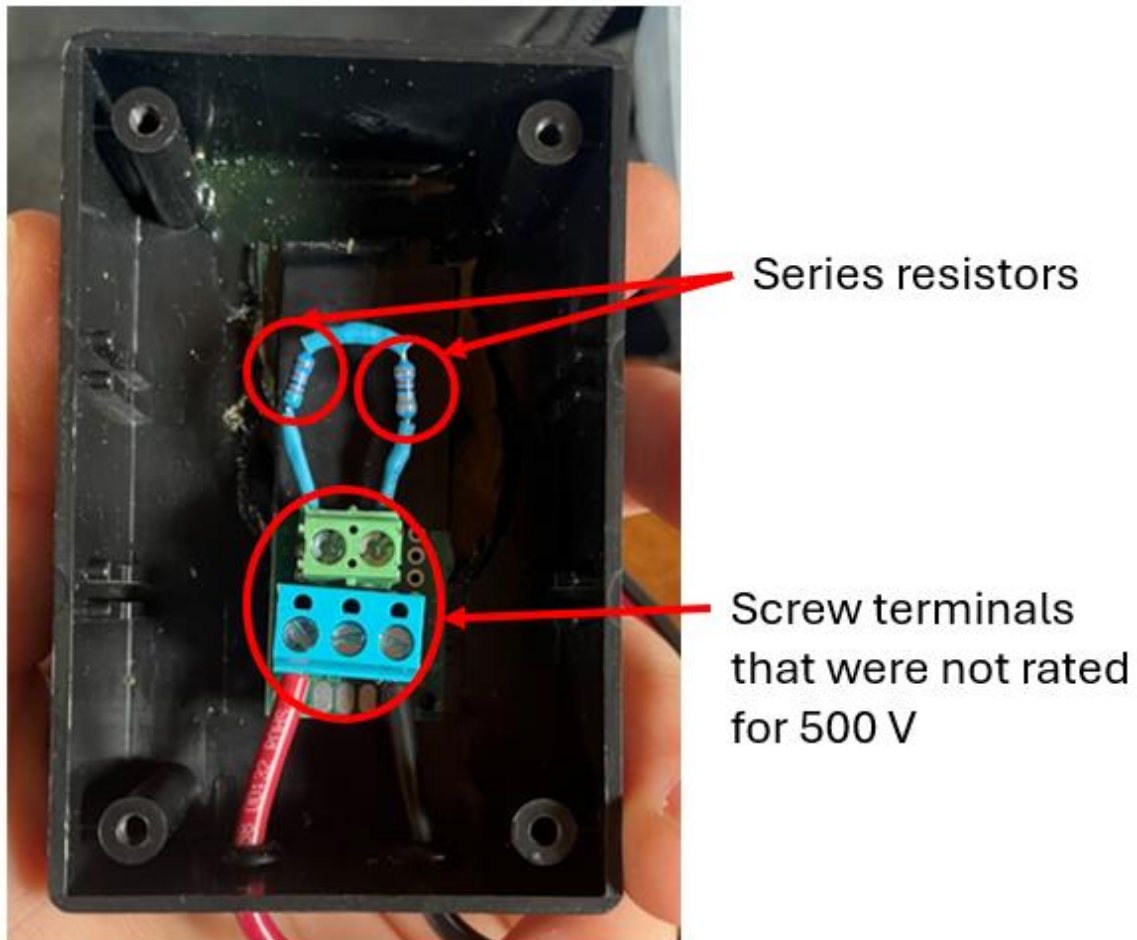


Figure 40. Non-compliant IMD Resistor Box

After the mock Technical Inspection, the Author quickly made this a priority to fix and gather evidence showing that it was compliant.

To start, the Author said that the resistor was instead rated for 2 W, this option is not recommended, but with the Author had three days before the competition EV Functional Technical Inspection. Next, the Author asked Oliver Dowzer to print an enclosure for the new solution. Malcolm Sidney advised that a solution to removing the screw terminals would be to solder wires directly to the resistor legs, then retain the wires through the enclosure with cable glands. The compliant solution is shown in Figure 41. Compliant IMD Test Resistor Box. The Author then wrote up hand calculations showing that the 2 W power rating were a significant safety margin for the actual power that the resistors would dissipate. The hand calculations are shown in Figure 42. Proof of IMD Test Resistor Power Requirement Compliance.

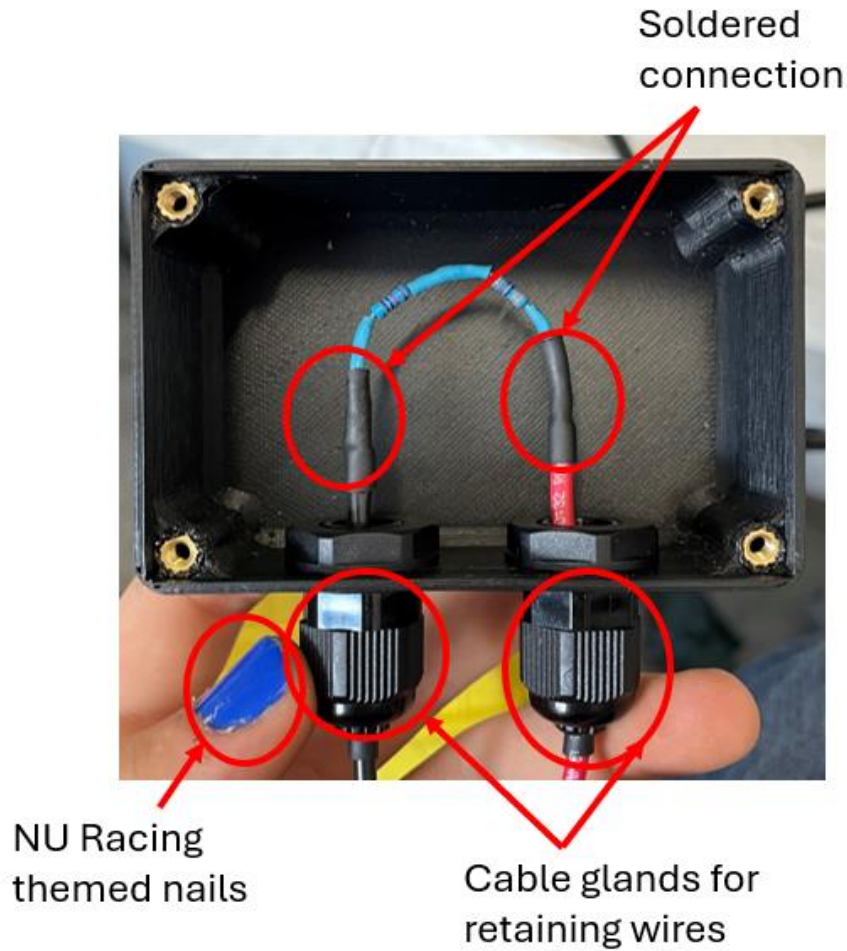
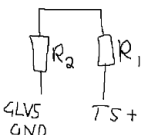


Figure 41. Compliant IMD Test Resistor Box

Circuit



Equation

$$P = \frac{V^2}{R}$$

$$P_{R_{1,2}} = \frac{(V_{R_{1,2}})^2}{R_{1,2}}$$

$$= \frac{(226.8 V)^2}{41.7 k\Omega}$$

$$= 1.23 W$$

Variables

BPR = Body Protection Resistor
= 15 k Ω

NOTE: R_{test} Equation given in FT.5.2 Tech Inspection point

$$R_{Test} = (V_{TSmax} \cdot 250 \Omega/V) - (2 \cdot BPR)$$

$$= 83.4 k\Omega$$

$$R_{1,2} = \frac{83.4 k}{2} = 41.7 k\Omega$$

$$P_{R_{1,2}rating} = 2 W$$

$$V_{TSmax} = 453.6 V$$

$$V_{R_{1,2}} = \frac{V_{TSmax}}{2} = 226.8 V$$

Conclusion

$\rightarrow P_{R_{1,2}} < P_{R_{1,2}rating} \rightarrow \text{Compliant}$

Figure 42. Proof of IMD Test Resistor Power Requirement Compliance

10.4.4. Parc fermé

At the end of the Endurance event, the car enters Parc fermé, regardless of if it successfully finishes the event or not.

In 2025 the structure of Parc fermé was:

- Push bar the car into the Parc fermé bay.
- All members of the team had to leave the bay and wait approximately 15 minutes.
- Once 15 minutes had passed, the Electrical Safety Officer (ESO) (the Author), was the only one allowed in the bay.
- The Scrutineers would inspect:
 - IMD resistor test
 - APPS plausibility test
 - Remove Energy Meter from car

Once the above had been completed, the car could then be removed from the bay and be officially done with competition and laid to rest.

The Scrutineers commented on the team's efficiency during Parc fermé.

In 2024, the Scrutineers also asked to see the thermal strip in the Accumulator, for this the Accumulator had to be removed from the car and the viewing hatch removed (shown in Figure 30. Thermal strip viewing hatch). In 2025, they did not ask to see the thermal strip. The Author asked if they wanted to, and they said no. The Author then later overheard Scrutineers saying that they had forgotten to ask to see team's thermal strips.

11. Team Leadership and Knowledge Transfer

11.1. 2025 Mechatronic Team Structure and Member Roles

The structure of the Mechatronics team is largely dependent the number of Final Year Project (FYP) students that the team receives at the start of the year. After the FYP roles are determined, Directed Reading and Extra Curricular roles can be established.

In 2025 the final Mechatronics team was comprised of three (technically four if including parallel projects) FYP, three Directed Reading and one Extra Curricular student. Their roles are detailed below. Mechatronics team shown in Figure 43. From left to right, Zak Lobko (Sem 1 Directed Reading), Samuel Connett (HV Powertrain), Oscar Rudman (Accumulator Electronics), the Author, Jayden Horvath (Autonomous Vehicle Development), Oliver Bunny (Data Engineer), Jayson Mills (Sem 1 Directed Reading), seated in car Mia Cosentino (Sem 2 Directed Reading), and Fletcher Cogoi (not pictured).



Figure 43. From left to right, Zak Lobko (Sem 1 Directed Reading), Samuel Connett (HV Powertrain), Oscar Rudman (Accumulator Electronics), the Author, Jayden Horvath (Autonomous Vehicle Development), Oliver Bunny (Data Engineer), Jayson Mills (Sem 1 Directed Reading), seated in car Mia Cosentino (Sem 2 Directed Reading), and Fletcher Cogoi (not pictured)

11.1.1. Data Engineer

The scope of this role was to increase the mediums of which data can be collected on the competition platform. As well as maintain the DBC of the car's CAN Bus system. Specific tasks that the individual performed in 2025 can be found in Oliver Bunny's report [16].

11.1.2. HV Powertrain Engineer

The scope of this role was to maintain the High Voltage (HV) Powertrain on the competition platform. As well as develop the validation capability of the dynamometer. Specific tasks that the individual performed in 2025 can be found in Samuel Connett's report [10].

11.1.3. Accumulator Electronics Engineer

The scope of this role was to maintain the Accumulator electronics on the competition platform. As well as develop the electronics of the next generation of NU Racing's Accumulator. Specific tasks that the individual performed in 2025 can be found in Oscar Rudman's report [15].

11.1.4. Sem 1 ENGG3200, CEN PCB

Increase serviceability of Central Electronics Node (CEN). Specific tasks that the individual performed in 2025 can be found in Zak Lobko's report [13].

11.1.5. Sem 1 ENGG3200, PEN & Breakout Board PCBs

Increase serviceability of Pedal Electronic Node (PEN), ensure compliance of Breakout Board PCBs, create an introduction to designing PCBs for NU Racing. Specific tasks that the individual performed in 2025 can be found in Jayson Mills' reports [12] [21].

11.1.6. Sem 2 ENGG3200, HIP, BEN, LVD PCBs

Ensure compliance of Human Interface Panel (HIP) on NU25, make desired technical changes to BEN and LVD PCBs on AV.One. Specific tasks that the individual performed in 2025 can be found in Mia Cosentino's reports [14] [22] [23].

11.2. Proposed 2026 Mechatronics Team Structure and Member Roles

While the final number of Mechatronics roles for 2026 is largely dependent on the number of members the team receives, below is a recommended list of roles for the 2026 team to consider.

11.2.1. Data Engineer

Continue development of data collection methods and validate sensors readings implemented in 2025

11.2.2. HV Powertrain Engineer

Continue to maintain HV Powertrain of competition platform, help Dynamometer Engineer implement inverter/motor tuning.

11.2.3. Dynamometer Engineer

Finish implementing the system to control and power the Dynamometer. Validate motor datasheet values, aid in Hub Motor development.

11.2.4. Accumulator Electronics Engineer

Implement Pouch Cell Accumulator into the competition platform. Validate improvements in capacity, power output, and PCB designs developed in 2025.

11.2.5. Directed Roles

The Author does not have any recommendations for Directed Roles in 2026 but is certain that projects and tasks will fall out of the scoping process conducted at the start of the year.

11.3. Onboarding, Training, and Mentorship

Onboarding, training and mentoring of new members into NU Racing is a critical task. It is arguably the largest contributor to an efficient and remarkable team. It can be seen as the sole reason why NU Racing does so well compared to other University FSAE-A teams who have tripled the budget, double the manpower and exponentially more experience and knowledge.

In 2025, the Author learnt many lessons on the onboarding, training and mentoring of individuals within the Mechatronics team. The Author also received valuable mentorship from other NU Racing Leaders, and NU Teams academic and technical staff.

11.3.1. Starter Projects

In 2025 and previous years, new FYP members are tasked with a 3-week starter project to give them a taste of what the role has to offer. It also serves as a way for the leadership team to gauge whether the individual is fit for the role.

In 2025 new FYP members were tasked with reading the last two years of reports from individuals who had done their role in the past. Two years was chosen as it was believed that this would give the individual a wide enough perspective of their role, without bogging them down with report reading. It is recommended that this idea is followed in future years.

Hard set deliverables and due dates play an important role in Starter Projects. They force the individual to work to a 'deadline', which has the benefit of showing the leadership team that the individual can either, get things done on time, or easily fall behind and let things go past their due date. If it is the latter, it can mean that this will be a repeated offence during the year.

The scope of work for a Starter Project should intentionally be fleshed out. So that it is as clear as possible to the individual what is expected of them. It also serves as a point to fall back on if they argue that the project was not well-defined after the due date has passed.

It can be hard to know how much work should be put into the scope. The Author recommends trying your best to put as much work in that you believe is reasonable. This does require previous experience in the team. Otherwise consulting previous leaders, or even the individuals that were in the role, or role most similar in the previous year.

From the Author's perspective, the point of the Starter Projects is to serve three main points:

- Allow leaders to gauge if the individual is fit for the role. As well as if they are fit for the team as whole, regarding project timelines and dependencies.
- Give the individual as much of a summary as possible of what they will be doing within their FYP project. This involves letting the individual get their hands on as many things as possible, in a succinct manner, that makes sense for their project.
- To let the individual learn as many lessons as possible about the role. This is usually best done through failure. This ensures that the individual will be up-skilled as quickly as possible.

The Author believes that by following this strategy, a Starter Project can be efficiently scoped.

11.3.2. FYP Starter Projects in 2025

11.3.2.1. *Data Engineer*

- Starter Project
 - Read Jackson Boustany's FYP report and Alec Chapman's ENGG4200 Report
 - Read Nick Lyall's FYP report for amplifier circuit for wheel speed sensors
 - Read datasheet to see if this circuit is needed at higher revs, as Nick discovered this using a 3d printed mount to imitate a wheel
 - Failing datasheet retrieval, attempt to use Nick's set up with a drill to see if higher revolutions can be achieved.
 - Read relevant rules from 2025 rules
 - Read through rule changes from 2024 to 2025
 - Mount wheel speed sensors (3d printed)
 - Design 3d printed mount to attach to potentially all 4 wheels of NU24
 - So that if project is finished, sensors can be easily tested
 - Do the front two first wheel speed sensors first

- Creation of loom to plug into EXPANSION Board
 - Pinout of OEM connector researched
 - Pinout of new DT connector correct and able to interface with EXPANSION Board
- Wheel speed sensor data can be read from MoTeC
 - Create 4 messages in NU25 DBC file for wheel speed sensors (one for each wheel), with correct scaling, offset, bit length and value range
 - Implement code for wheel speed sensors on the EXPANSION Board
 - With data from wheel speed sensor sent using the message from previous dot point
 - Configure MoTeC to read and display new message
- Deliverable:
 - Front two-wheel speed sensors mounted
 - Working with the EXPANSION Board on CAN King and (hopefully) MoTeC

11.3.2.2. *HV Powertrain Engineer*

- Starter Project
 - Read and understand both Josh Hayward's and Josh Dawson's FYP reports
 - Implement any recommendations from Hayward's report
 - Implement pack current derating at cell/pack temperatures over 50 degrees.
 - PEN code
 - Use either MaxSegTemp or MinCellVolt/MaxCellVolt
 - Derate from the 3 degrees under the max temperature rating of the VTC6
 - Limit current down to ~30A under high temperatures. Shut down all current above 60 degrees
 - Learn how to charge the car
 - Read relevant rules from 2025 rules
 - Read through rule changes from 2024 to 2025
 - Deliverable:
 - Current derate at high temperatures works as intended
 - Test to show it off could include having the max allowable temperature to be set to a low value (eg 30 degrees) and have a track day where the car is driven hard, if it works correctly, then current will be limited and cut out to the driver upon approaching and reaching the set limit
 - Know how to charge the car

11.3.2.3. Accumulator Electronics Engineer

- Starter Project
 - Read Daniel Iveson's and Jedd Reeves' FYP reports
 - Design and commission new AIL, breakout board mounting, implement change from Ivo's report
 - Read relevant rules from 2025 rules
 - Read through rule changes from 2024 to 2025
 - Learn how to charge the car
 - Deliverable:
 - AIL ordered (with LTspice or breadboard test)
 - Know how to charge the car

11.3.3. Lessons Learnt by The Author During the Onboarding, Training and Mentoring Processes

- Do not treat people like trades people. They are bright and keen to learn engineers, who should be treated like engineers, they merely have a lack of institutional knowledge.
- Let and support people make mistakes. It is imperative that they are supported and allowed to make mistakes within their roles. As this is what helps the individual grow both technically and as a person. There is a critical balancing act at play here, due to the tight ship that NU Racing runs financially. Resulting in the leadership team needing to be acutely aware of sunk cost fallacy of projects.
- Be able to take criticism. From people in roles below, adjacent and above you.

11.4. Delegation, Progress Tracking, and Review Processes

11.4.1. Delegation

Delegation is a critically important role of the Mechatronics Department Lead Engineer. It is important to know when a task can be delegated or delt with by Department Lead.

NU Racing operates on a system level basis. With individuals assigned to those systems. Design and implementation are the primary aspects that get delegated to assignees. Tasks or issues that arise with that system also fall to the ownership of the assignees, depending on the situation (such as track day testing) and issue, it quickly requires more than the assignees of the system to fix it. Safety-critical changes, design review and quality control should remain assigned to the Department Lead.

As members evolve in their roles, it is critical that more delegation takes place, and they are trusted to conduct the work to a high standard.

When delegation is done correctly, the Department Lead should have time to ensure aspects such as competition submissions are on time and allow time for knowledge transfer documents and mediums to be developed, to improve internal processes, such as validation strategies and unit test procedures.

Though care should be taken as to not over delegate, as it will come across as laziness and the team will feel like you are purposely putting yourself above the rest of the team, and make them feel like you are undervaluing them.

11.4.2. Progress Tracking

Progress tracking of projects is a critical role of the Mechatronics Department Lead Engineer. As it allows the Department Lead to make informed decisions about timeline conflicts and member well-being.

In 2025, the Author informally tracked progress through the year with short casual but technical catchups with individuals. When it came to high workload and crunch periods, the Author saw it valuable to keep Tasks Lists of what individuals had to do by what date. This helped ensure that all project dependencies would line up correctly and hold individuals accountable if areas fell behind. There were moments in 2025 where keeping a Task List during high workload periods was difficult for the Author, as often the Author would be helping individuals get the work done on time. If possible, this should be done in a reduced capacity so that all aspects of the implemented solution are ready at the same time. To remain “high level” as one could put it.

11.4.3. Review Process

When tasks are delegated, there needs to be a method of review for the Department Lead to ensure the implementation is sound, compliant, and safe.

This can be in the form of manual review by the Department Lead, follow a chain of command through a peer review process, or requiring the individual implementing the change to create evidence that it is sound, compliant, and safe.

11.5. Knowledge Transfer and Documentation Outputs

The Author used the few downtime moments to create knowledge transfer documents to help support this report, as well as to create other mediums for transferring the technical knowledge that the Author has had the privilege to gain since second semester 2023.

These knowledge transfer outputs consist of:

- The Electrical System Overview. A summary of all the electrical connections on NU25. To give an overview of the complexity of the system, as well as to help with fault finding, and helping with individuals understanding of the Mechatronics systems of NU25. The Author believes it is valuable to update and change the Electrical System Overview as the competition car changes year to year. Found on the NU Teams SharePoint or the Racing-NU25 GitHub Repository, as well as Appendix C - NU25 Electrical System Overview.
- MoTeC Zero to Hero. A summary of all the information about the wireless telemetry on the competition platform. This document was not completed in 2025, as the Author ran out of time to complete it. It is highly recommended that when you feel as though you have learnt enough about the wireless telemetry (C125 programming, updating variables on the screen of the C125, how to set up the wireless telemetry at a track day, how to use either Dash Manager or Dash Editor) that you add this information into the Zero to Hero. In 2024 and 2025 it was arguably one of the least documented systems on the competition platform. Found on the NU Teams SharePoint, as well as Appendix H – MoTeC Zero-to-Hero.
- How to Crimp NU25 HV Connectors. Being that the Author decided to change the HV connectors on the NU25 to ones that the team had not used previously, the Author saw it valuable to document the manufacturing process for it. Found on the NU Teams SharePoint, as well as Appendix I – How to Crimp NU25 HV Connectors.
- How to Import PCBs Into OnShape. One of the less explained sections of the PCB creation process, and something that if not done correctly, can waste many hours. The target audience is PCB designers. It can be found on the NU Teams SharePoint, as well as Appendix J – How To Import PCBs Into OnShape.

12. Risk Management and Lessons Learnt

12.1. Technical Risks and Mitigation Strategies

Entering the 2025 competition, while the car was extremely tested and the result of competition was extremely positive. The Author had the following technical concerns about NU25 as the car entered its first dynamic event.

- BMS reliability. The highest concern for the Author the entire 2025 competition. These faults were occurring during 2025 due to the smallest issues of bolts not being tight enough, or something rattling loose. The chance that something had rattled loose during transport between Newcastle and Melbourne, or during driving any of the dynamic events was far too high. Despite this concern, the Author had trust in his Accumulator team. And the BMS remained active for the entire 2025 competition because of the trust and did not fault once. A massive achievement and relief, since from 2022, when the NU Racing first had a competitive EV car, the team has had Accumulator issues that caused the car to stop during AutoX and Endurance events at competition each year. And 2025 was the break of the curse!
 - The most contributing factor to not having the fault was the continual servicing of the Accumulator by the Accumulator Electronics and Mechanical Engineer throughout 2025.

- Inverter overcurrent fault. While it is not exactly known why this fault occurs yet, it is believed to occur due to two reasons. Reason 1 is from a high torque request, on a high grip surface, the Inverter could not hand the requested torque by the motor. Reason 2 is due to incorrect tuning between the motor and Inverter, causing incorrect current supply to the motor during rapid high torque request.
 - The fault occurred during two of the four Acceleration runs at competition. Causing the car to stop on track and the time for the run not being scored. The fault occurred due to a naïve implementation of torque ramping, which would create a step change after the driven axle would surpass a calculated speed of 30 Kph. From one torque to another. More information on the faults can be found in the Acceleration section of this report.
- During 2024, the team used dodgy and relatively cheaply made battery isolator switches for the Grounded Low Voltage Master Switch (GLVMS) and Tractive System Master Switch (TSMS). These switches would frequently have contacting issues, resulting in an unreliable method for turning on NU24's LV and HV. An issue that carried through to the 2024 competition, where the LV switch failed an hour before the car was due to do its first Skid pad run. Luckily the team had a replacement and were able to replace it in time.
 - In 2025, the Author opted to implement higher quality battery isolator switches, which proved to be beneficial as there was not a single failure of these switches during the running of NU25.

- Most of the LV looms on the car has been in use since 2023, most of which the Author is unsure if correct sized wire AWG was used for the expected current passing through the wire. After an extremely high temperature track day, the CEN to Accumulator LV loom had burnt both sides of the loom on the DT connector at two of the DT pin locations. These pins supplied low voltage, high current to the Accumulator cooling fans, which would run at peak loads of approximately 24 A. After the uncountable amount of power cycles and kilometres that loom had endured, it eventually failed. This was swiftly replaced by a new loom the day it was found.
 - This highlights the potential for other looms on the competition platform that has been recycled year to year to randomly overheat and fail. Not all, but the ones carrying high current signals. Care should be taken to regularly inspect looms for damage.
- The E-stop located on the DEN (named the Dash E-stop) has had its mating contacts worn down over time. The Author assumes due to incorrect use of the E-stop (pulling it out instead of twist pulling to release it). It was noticed to be worn down 4 days from competition, and the Author had a handful ordered to the University that arrived after the team had left for competition. Luckily Mr Malcolm Sidney had not yet left, so was able to bring them down with him.
 - The replacement E-stops did not have to be used during the 2025 competition, but it should be noted as a potential point of failure for the shutdown system.
- The Accumulator saw two very late mechanical design implementations due compliance concerns. Namely the container and segment housings were replaced very close to competition. The Author had integration concerns about these two late implementations, as there are usually things missed during design. Luckily the issues that occurred were easily fixable in the timespan. Details on these issues can be found in Ryan Bird's FYP report [11].

12.2. Team and Workload Management Risks

The risks that the Author had identified for mainly himself, but also for other team members of the Mechatronics team are detailed below:

- Burn out
- Overworking
- Timeline management
- FSAE-A rules
- Team Leader/Chief Engineer inconsistent participation in the team
- Constraints from Team Leader/Chief Engineer
- High voltage systems

12.3. Schedule and Delivery Risks

The Author's mindset to schedule and delivery risks and how to minimise them as best as possible was to follow the philosophy of fail early and fail cheaply. Failure is the key building block to a good engineer. Failing early allows time for design changes. Failing cheaply refers to both time and money cost. Due to the team's small budget, failing cheaply is a key constraint.

From this, Mechatronic designs were able to have the most testing time of new systems on the competition platform. It gave the most time to implement changes, resulting in low risk from schedule and delivery timeframes.

12.4. Key Lessons Learnt

Arguably the largest risk that the Author was met with personally was letting NU Racing consume his life. It became all I thought about, which was not healthy. I found it incredibly hard to tell myself to switch off when I needed to. Something I learnt from Dr Dylan Cuskelly during my performance review was that a way of switching off is by physically doing a task to turn your mind to something else. For me, I found this came in the form of ‘documenting’, writing down the concern at the forefront of my mind at the time helped me to not stress about this as much. But other methods include dedicating a specific shirt or item of clothing to a task. For example, wearing the NU Racing shirt only when doing NU Racing work and for nothing else. Or to distract yourself by exercising, playing video games, hanging out with loved ones. Something to physically separate yourself from your usual environment to force your brain to stop thinking.

Talk about things that are bugging you. About people’s behaviour or technical concerns about the car or how the team is running. Again, Dr Dylan Cuskelly is a valuable asset to have, who will drop everything he is doing to make sure that you are heard, and will give advice on the situation, or begin to put a plan in place to change things. It is never a sign of weakness to speak up, and the situation can literally only be better for it.

13. Project Outcomes and Performance Evaluation

13.1. Achievement of Project Objectives

In section Project Objectives and Success Criteria of this report, the following project objectives were detailed. Here, the Author will outline if these were achieved or not.

- Team passes Accumulator, EV Static, EV Functional and Rain Test Inspections
 - Achieved, all Technical Inspections were passed. Efforts in preparation and professionalism were recognised by the competition through the Technical Inspection Specialty Award
- All new MCHA system designs validated
 - Partially achieved:
 - 7/7 PCBs were validated to show improvements
 - 5/7 Expansion Board sensor readings validated, with strain gauges and radiator temperature sensors being the only ones that were not validated
 - 0/2 Motor outputs validated, due to the Dynamometer project not being completed in 2025
- Efficient knowledge transfer of processes and systems
 - Achieved
 - 4 knowledge transfer documents were created on topics that had not been previously covered:
 - Electrical System Overview
 - MCHA System Design Changes
 - MoTeC Zero-to-Hero
 - How to Crimp NU25 HV Connectors

- Well-structured information regarding systems on the competition platform, for current and future team member's access
 - Achieved:
 - Creation of Electrical System Overview, available on Racing-NU25 GitHub Repository, and NU Teams SharePoint
 - Additions to NU Racing's Lesson Learnt document
 - Updated ESF to current competition vehicle
 - All relevant datasheets for competition platform Mechatronic parts added to a structured file system on NU Teams SharePoint
- Documented tasks completed by MCHA team
 - Partially achieved
 - While not consistent during 2025, the beginning and end of the year's tasks completed by the MCHA team were documented. Shown in Appendix F – 2025 Tasks Completed by Mechatronics Team.
- Training and advice provided to MCHA team
 - Achieved, a hard to measure metric, but the Author believes this was shown through how efficiently the Mechatronics team passed Accumulator, EV Static, and EV Functional Technical Inspections.
- Suggested areas the MCHA team can innovate/improve in
 - Achieved, list provided in the Opportunities for Future System Improvement and Innovation section of this report.

13.2. Impact on Team Performance and Competition Readiness

13.2.1. Static Events Performance

13.2.1.1. *Design Event*

An explanation of this event can be found in the Design (150 Points) section of this report.

At the time of writing (11th of January 2026), the feedback from the judges for the Design Event has not been released. Meaning that there can be no comments made on if the Author had a positive impact on the outcome of the event. Overall, the team placed 5th, with 122 out of 150 points.

13.2.1.2. *Cost Event*

An explanation of the Cost Event can be found in the Cost Report, Scenario and Event (100 Points) section of this report.

The Cost Report accuracy rating came back with 100% accuracy, meaning the team had made no mistakes on the pages in the Cost Report, this result felt very rewarding as the amount of hours put towards this report by the Author was extremely high. The Cost Scenario did not do very well; it was marked poorly due to lack of validation of the supplied solution.

The Author did not participate in the Cost Event at competition due to a timetable overlap with Accumulator Technical Inspection.

13.2.2. 2025 Dynamic Events Performance

13.2.2.1. Skid pad

An explanation of the Skid pad event can be found in the Skid Pad (75 Points) section of this report.

No Mechatronics related issues occurred during the Skid pad event. The two drivers did extremely well considering the track conditions. The team placed 7th overall in this event.

13.2.2.2. Acceleration

An explanation of the Acceleration event can be found in the Acceleration (100 Points) section of this report.

Acceleration has historically not been a dynamic event that NU Racing designs for, as NU Racing runs a single rear motor with very limited capability. NU Racing placed 2nd in Acceleration. An impressive result considering track conditions and that other teams run hub motor designs in both two-wheel and four-wheel configurations. Though the team's fastest acceleration time was slower than 2024's, the team showed their preparedness by successfully being on time and participating in the event. That being said, NU25 did have two Inverter faults causing the car to lose power during two of the four allowed Acceleration runs. Details listed below.

A screenshot of the logged MoTeC data can be seen in Figure 44. 2025 Acceleration Faults.

- The first fault occurred in the first driver's second run where torque stepping was not enabled. The driver was instructed to put their foot down as quickly as possible, and the Inverter faulted before the car even moved.
- The second fault occurred in the second driver's second run with a torque step from 160 Nm to 220 Nm, at a calculated speed of 20 km/h. The Author and Chief Engineer were very sceptical of this implementation, but both saw the reward from the high-risk decision. Which turned out to be too high of a risk and the car faulted after the step change occurred.

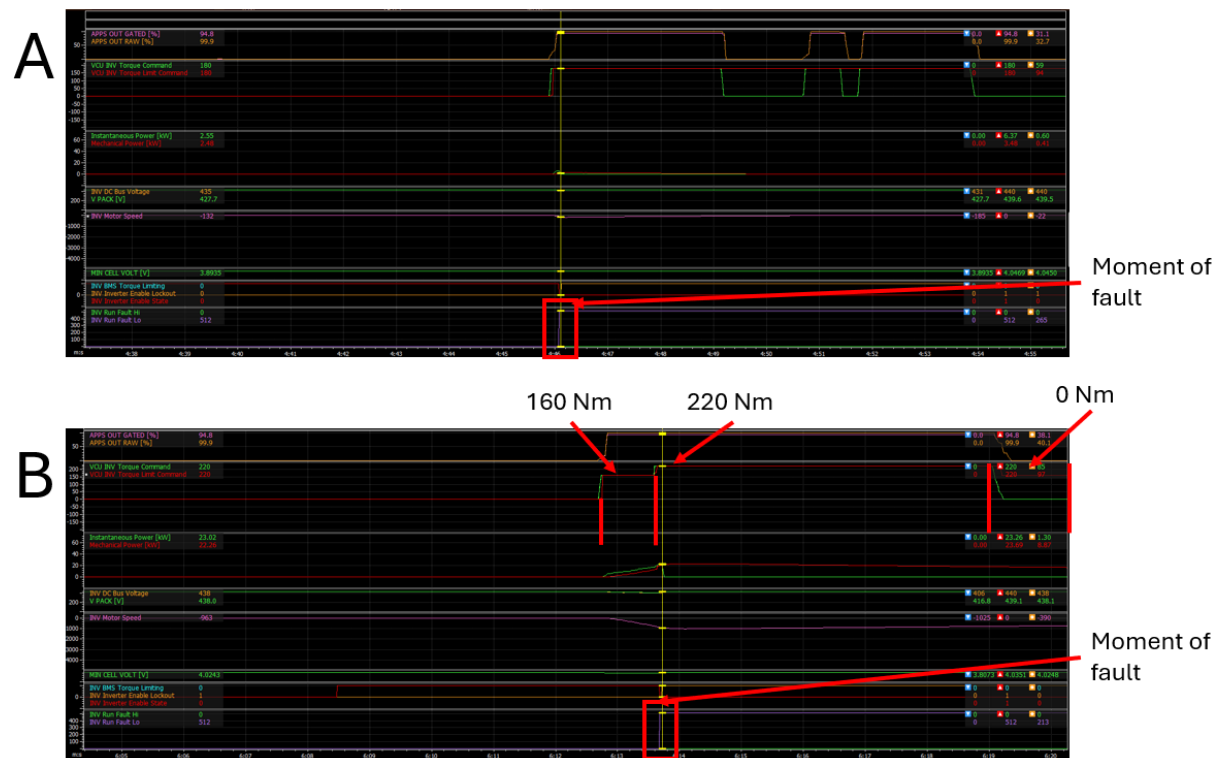


Figure 44. 2025 Acceleration Faults

13.2.2.3. *AutoX*

An explanation of the AutoX event can be found in the Autocross (AutoX) (125 Points) section of this report.

NU Racing has been chasing a first place in the AutoX event since 2023. This year, the team achieved this milestone with their first ever first place in any event at the FSAE-A competition.

There were no Mechatronic related faults during AutoX.

13.2.2.4. *Endurance & Efficiency*

An explanation of the Endurance and Efficiency event can be found in the Endurance (275 Points) and Efficiency (100 Points) sections of this report.

Being the event that halted the team's success in 2023, the Endurance & Efficiency event was one of high concern. As the highest weighted event and the longest that the car must run throughout the entire competition, the team was rightfully concerned and stressed. Despite this, the team successfully finished the 22 km Endurance with no car stopping faults. The team placed 5th overall in Endurance, and 3rd overall in Efficiency, the highest the team has ever placed in this event.

The only issue that arose was that the Accumulator was not fully charged to a pack voltage of 445 V (~95% SOC) before the event, it was left at 430 V after AutoX. The thought behind this being that regenerative braking would kick in earlier during the run. This would allow for a higher efficiency score and let the driver gain a feel for regenerative braking earlier in their run.

This turned out to be a risky decision, as the car ended Endurance with a pack voltage of 340 V (~4.5% SOC). This meant that high current limiting was in place in the last few laps. Resulting in the car completing very slow lap times and impacting the overall performance in the final laps. The Author is unsure if the finishing low pack voltage was due to the Accumulator not being fully charged, or because the competition purposely designed the track to be tighter than 2024. By reducing the number of high-speed sections. High speed sections allow for the Accumulator to regain charge easier.

A comparison of contributing factors can be found in Table 1. 2024 vs 2025 Endurance. For interested readers, the Author suggests further review of MoTeC data should be conducted for a full understanding.

Table 1. 2024 vs 2025 Endurance

	2024	2025
<i>Discharge Torque</i>	180 Nm	180 Nm
<i>Regen Torque</i>	40 Nm	40 Nm
<i>Starting Pack Voltage</i>	442 V	428 V
<i>Ending Pack Voltage</i>	380 V	340 V
<i>Starting SoC</i>	94.5%	85%
<i>Ending SoC</i>	30%	4.5%
<i>Distance Travelled</i>	22,065 m	22,222 m
<i>Average Speed</i>	39 km/h	35 km/h
<i>Power usage</i>	3.72 kW	3.89 kW
<i>Power regenerated</i>	0.92 kW	0.97 kW
<i>Power discharged</i>	4.64 kW	4.87 kW
<i>Endurance Score</i>	250.87/275	176.32/275
<i>Efficiency Score</i>	40.82/100	51.49/100

14. Reflection and Recommendations

In the Mechatronic Department Lead Engineer role, the Author gained invaluable engineering experience. The Author feels grateful to have been considered as a candidate for the role, and hope that he made an ever-lasting impact on NU Racing and the people that he was lucky enough to work with.

An extensive summary of the lessons learnt and recommendations for future Mechatronic Department Lead Engineers, as well as potentially any future NU Racing engineers are listed below. The Author hopes that these learnings can help the reader better understand and prepare for the role.

Technical:

- Phase cable retention in the phase cable cover in the power box.
- Make i2 and T2 telemetry be able to stream on a local network at track days so that multiple laptops can view live data, and access logged data quicker (improves fault finding and driver training).
- Change port that either i2 or T2 accesses so that it doesn't clash with Dash Manager and BMS Orion 2 software.
- Potentially move the BSPD to the HIP, wouldn't need shielded wire on current sense signal, would mean more signals go to HIP, and larger connector.
- Use the Electrical Systems Form (ESF) to help make technical changes to the competition platform, being a calculator, it can quickly pick up on non-compliant design choices. In 2025 it was used to see that the discharge resistor for the Tractive System (TS) circuit was too large, causing too long of a discharge time. In 2024 it was used to see that the Accumulator insulation lining material was not compliant.

Non-technical:

- Be a leader, not a boss
- Communicate with other departments and leaders
- Keep a diary
- Fail early and fail cheaply
 - Test all new PCBs as soon as possible.
 - Things will get missed, which will only be found in integration, be aware of it, but don't be lazy during design.
 - “A piece of advice the Author received from Jye Hollier was that time spent fixing known issues from a prototype is far more valuable than time spent focusing on potential issues considered during design.” Lachlan Fisher [19].
- Put consistent work into the project, there is not one major defining thing that makes someone good at this role, but the continued persistent drive to be better, and not to stay knocked down when something goes wrong.
 - Recommended books that explain this philosophy (both are a very good read):
 - The Slight Edge by Jeff Olson
 - Atomic Habits by James Clear
- Create a document detailing remaining things to do with projects (HIP PCB being ready but not enclosure, almost cut up HV lines without new HIP being ready).
- Scope out next Mechatronics Department Lead Engineer as early as possible, to help with the knowledge transfer process.

- Vital aspects to consider when deciding changes for systems are: (Systems-Level Engineering Perspective)
 - Safety
 - Compliance
 - Reliability
 - Inspect-ability (ensure the design is easy to show during Technical Inspection)
 - Integration effort (how much needs to change to implement the design)
- Create a document or process around how to validate a Mechatronic system
- Create a risk structure analysis, further detailed in the Risk-Based Decision Making section of this report
- Create a leadership onboarding module, to help in the leadership handover at the beginning of the year
 - Which includes a summary of all documents the team uses and why
- Don't change stuff on the car less than 24 hours from a track day
- Be cynical during leaders' meetings, talk about the shit before the good
- Make a 'remaining manufacture' list, keeps track of what's left to do for a design to get on the car
- Keep doing the 'changes to NUxx' before a track day

14.1. Reflection on the Mechatronics Department Lead Role

The role is extremely multi-faceted in many aspects; it combines project management with high technical expertise and coordination of members. For it to be done well, it is a major time commitment, but if willing, it is arguably one of the quickest ways to upskill and learn about oneself technically and mentally at the University.

It is the perfect role for an individual who is dedicated, enjoys learning, hungry for a challenge, and wants to improve themselves.

14.2. Recommendations for Future Mechatronics Leads

Below is a list of recommendations for specifically future Mechatronics Leads. They are both a mix of technical improvements and suggestions to help look after oneself during the whirlwind of a year. These suggestions are based on the Authors 1,531 hours of experience in this role. Effective implementation of these suggestions could make the role more manageable and create even better results for future teams.

- Keep diary
- Use ESF
- Target issues in designs that are easy to fix (low hanging fruit)
- Remain honest with people
- Maintain consistent technical communication with Mechanical Department Lead Engineer
- Have hobbies outside of NU Racing
- Try not let NU Racing consume you
- Make connections, make friends, talk shit, go outside, frequent Bar On The Hill (BOTH)

14.3. Opportunities for Future System Improvement and Innovation

For 2026 and beyond, the Author sees the following areas as the most beneficial technical improvements for the competition platform. They aim to target low hanging fruit and bottlenecks of the Mechatronic systems.

- Pouch cell Accumulator
- Stricter power limiting solution
- Front wheel speed sensor and traction control tuning
- Dynamometer
 - Test bed for future designs
 - PCBs
 - Cooling
 - Motor
 - Inverter
- Inverter and motor tuning
- ESPs used on nodes instead of Teensy 4.0s
- Rear wheel hub motor implementation
- Custom BMS

15. Conclusion

In conclusion, the Author is incredibly lucky to have been given the opportunity to be one of NU Racing's Mechatronic Department Lead Engineers. The Author learnt many valuable lessons about himself that will be used throughout his engineering career.

The Author would like to thank the amazing team of 2025, without them none of the achievements and accomplishments covered in this report would have been possible.

Obtaining 3rd place in the FSAE-A competition is no small feat. It is a direct result of the blood, sweat and tears that NU Racing team members put into the year of 2025. It shows how comradery and the tight bond that NU Racing creates between team members absolutely smashes out the teams with higher budgets, resources and member counts any day of the week.

Recommendations are listed in the Reflection and Recommendations section of this report. The Author hopes that these are read and actioned in 2026 and even years beyond, so as to not make the same mistakes that the Author made in 2025.

The Author believes this report has shown the importance of this role to NU Racing, as well as what consistent and dedicated time put into this Project can do.



Figure 45. NU Racing 2025

16. References

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17. Appendices

17.1. Appendix A - Email Signature

Jacob Lukes

NU Racing - Mechatronic Department Leader

NU Teams is the official on-campus professional student engineering teams at The University of Newcastle

Bachelor of Electrical Engineering

Email: Jacob.Lukes@uon.edu.au

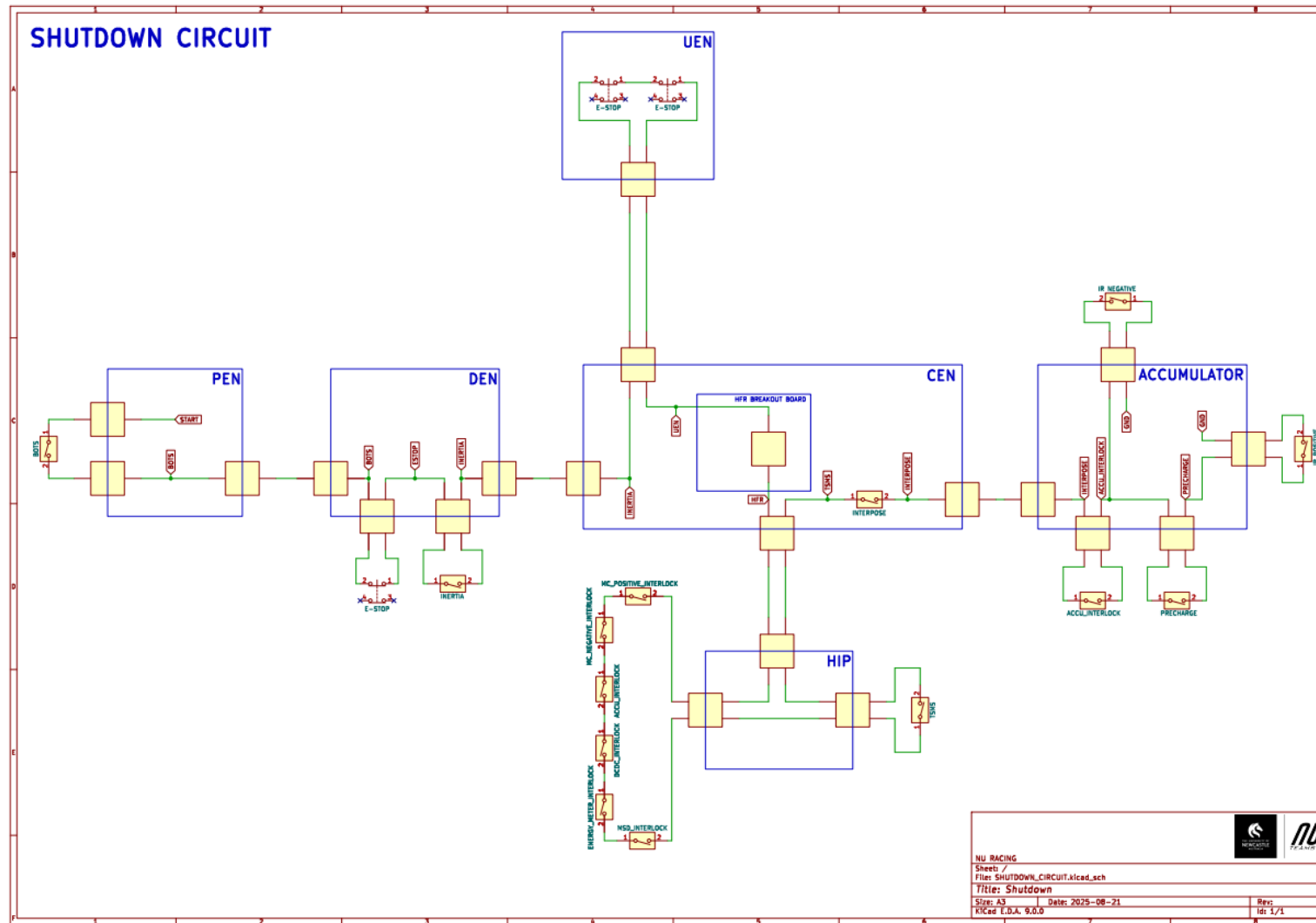
Web www.nuteams.org

The University of Newcastle Callaghan Campus, NSW 2308, Engineering CI



17.2. Appendix B – NU25 Shutdown Circuit

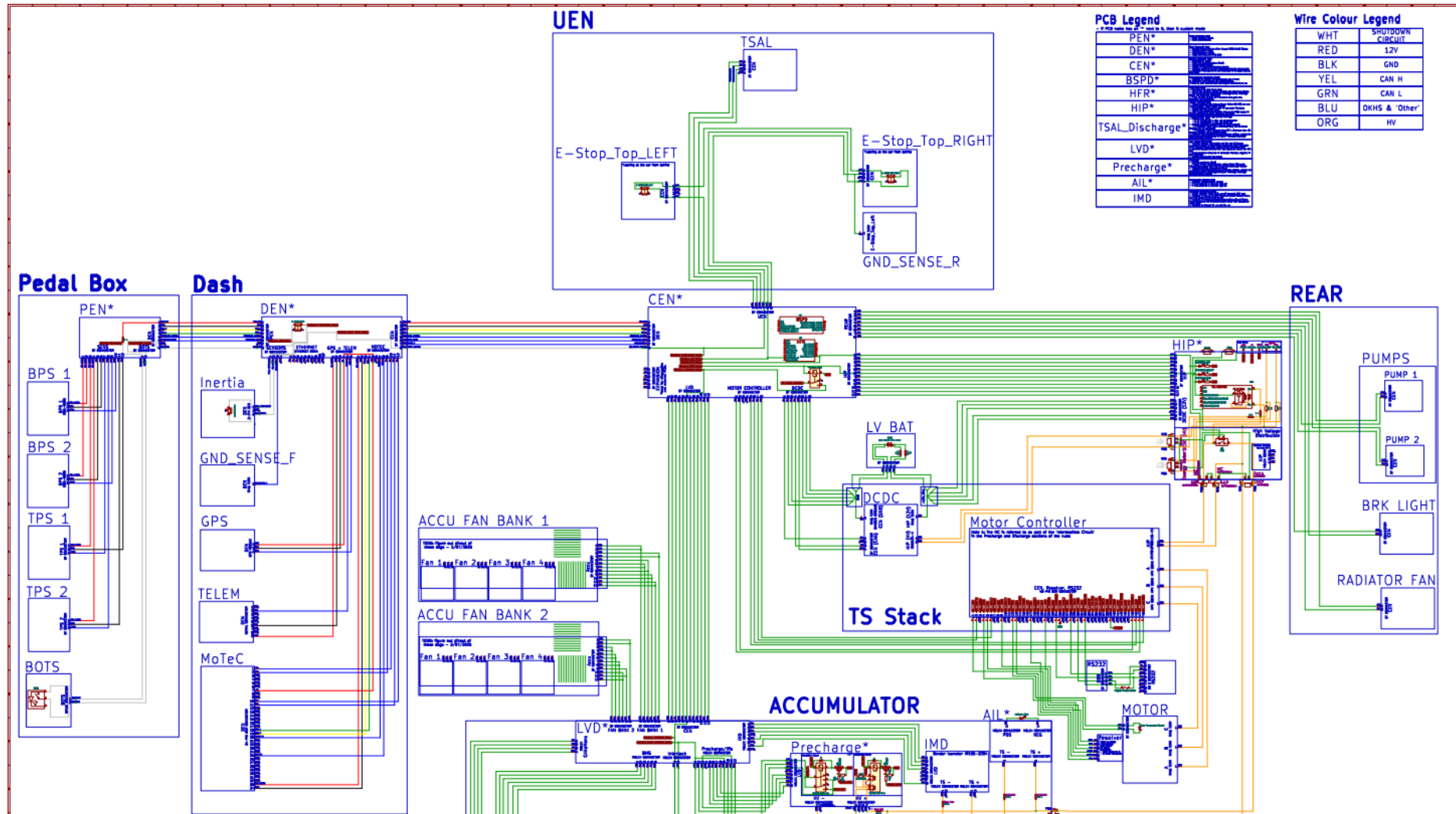
A high-resolution version can be found in the NU Teams SharePoint, or in the Racing-NU25 Github Repository.

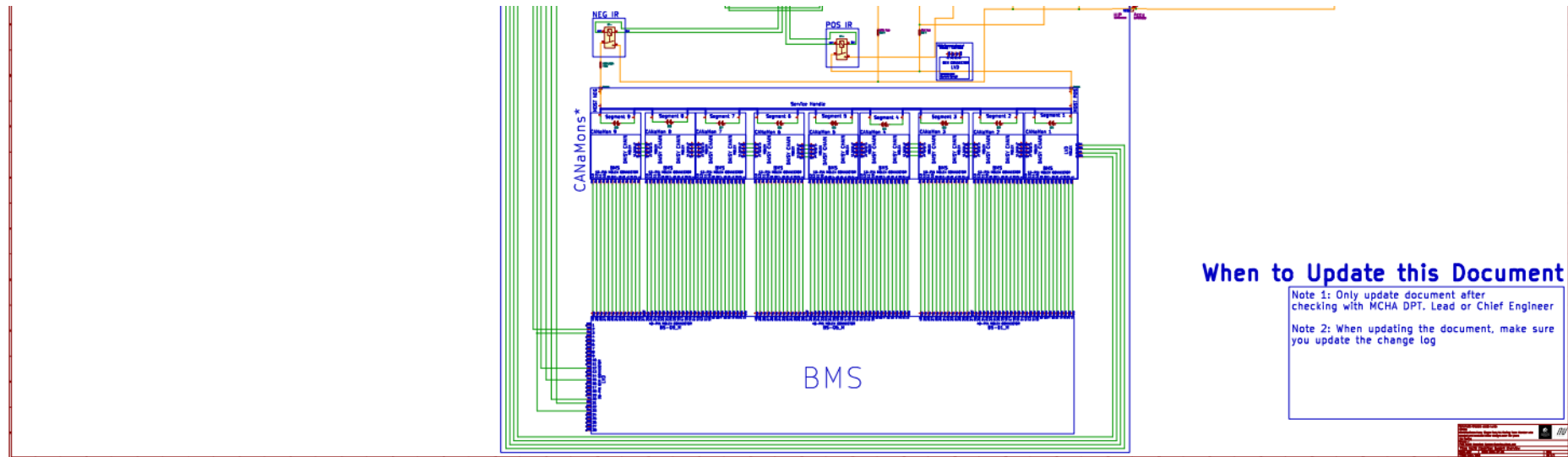




17.3. [Appendix C - NU25 Electrical System Overview](#)

The document is too large to fit on one page. A high-resolution version can be found in the NU Teams SharePoint, or in the Racing-NU25 Github Repository.





17.4. Appendix D - 2025 Technical Inspection MCHA Bible

2025 Formula SAE-A Technical Inspection Sheet ACCUMULATOR INSPECTION							
Initial Presentation Scrutineer Name:		Initial Presentation Lane Number:					
1st Reinspection Scrutineer Name:		1st Reinspection Lane Number:					
2nd Reinspection Scrutineer Name:		2nd Reinspection Lane Number:					
3rd Reinspection Scrutineer Name:		3rd Reinspection Lane Number:					
PERSONNEL				Inspection #			
Teams are required to have an Electrical Safety Officer.				Initial ins.	1st reins.	2nd reins.	3rd reins.
AI.1.1	Identify the Electrical Safety Officer (ESO)	Ask for the ESO					
AI.1.2	Only team members involved in technical inspection can be in the technical inspection bay	Ask for non-essential people to leave the area					
DO NOT PROCEED WITH TECHNICAL INSPECTION IF THE ESO IS NOT PRE SENT							
INSTRUCTIONS FOR SCRUTINEERS This section applies to Electric Vehicles (EV) and Autonomous Vehicles (AV) and Dual Purpose Vehicles. If an item is acceptable, initial the "Passed" column. If an item is unacceptable, initial the "Reinspect" column and inform the team members they have not meet the requirements for that item. Teams are allowed to take their vehicle away from technical inspection and address any issues before representing for reinspection. Scrutineers may continue with the inspection when unacceptable items are found, provided it is safe to do so. Teams are required to pass all items before the vehicle can proceed to the Brake Test or compete in Dynamic Events. Scrutineers reserve the right to refuse to proceed with Technical Inspection if a sufficiently serious issue which might jeopardize the safety of the technical inspection team, event personal or any other persons is found at any time.							

EV CLASS - ACCUMULATOR INSPECTION

- ESO (Lukes), Oscar, Ryan, Jayson and Malcolm (charger), Bunny (document hander), Cam (SES)
- Charger, Accumulator (race ready) bolted to the accumulator trolley
- Accumulator spares box
- Red toolbox, callipers (both plastic and metal)
- Plastic ruler
- Large cable crimps (blue handle)
- Tech Box which samples, documents and tech bibles
- 2 x HV Mats
- 4 x Safety Glasses
- 3 x LV Gloves and 3 x Leather outers
- HV Insulated tools – Socket sets, Spanners, Screwdrivers, cutter, shears, Allen keys
- Fluke Multimeter with probes and banana jack probes

EV CLASS - ACCUMULATOR INSPECTION					
SAFETY EQUIPMENT AND TOOLS					
Teams are expected to have appropriate safety equipment and tools to allow them to work on their vehicle. All safety equipment and tools must be in good usable condition, and comply to Australian industry best practice standards. There must be a sufficient quantity of appropriate safety equipment (PPE) for everyone who needs to work on electrical systems. Insulated tools must be rated and appropriately marked as compliant to DIN/IEC 60900 (VDE 1000V), or an equivalent recognized standard. Improvised or repaired tools using heat shrink, tape or similar are not acceptable. Modified tools are not acceptable.					
				Reinspect	Passed
AI.2.1	Safety Glasses	At least one pair per team member working on electrical systems. Safety Glasses should be non-conductive (non-metallic). Display safety glasses which should be laid out, know where safety rating is	Inspect		
AI.2.2	HV Insulating Gloves	HV Isolating gloves with leather outers. Gloves must comply with AS2225 or ASTM D120 or AS IEC 60903 The inner gloves must have been tested in the last 12 months and be free from damage or punctures. Outer gloves must be in serviceable condition. Gloves must be labelled with a working voltage, higher than the vehicle's maximum battery voltage. At least one pair per team member working on electrical systems. Size of the gloves should be appropriate for those conducting the work. Note: Class 00 gloves are rated for up to 500V, Class 0 gloves are rated for up to 1000V. Lay gloves out separated from the outers	Inspect 	D.g.	
AI.2.3	HV Isolating Blanket	Two HV Isolating Blanket (at least 0.83 square meter) Blanket must be in good condition and free from punctures or tears. Blanket must be labelled with a working voltage, higher than the vehicle's maximum battery voltage. Measure or look at data sheet	Inspect	D.g.	
AI.2.4	HV Insulated Tools	Check the team has an appropriate set of tools to maintain the vehicles electrical systems. These must include as a minimum all of the following: - Insulated cable shear Display - Insulated screwdrivers Display - Insulated pliers Display - Multimeter with protected probe tips (CATIII 600V or better) Display - A pair of quality insulated 5mm banana test leads (600V rated or higher) Display	Inspect	D.g.	

EV CLASS - ACCUMULATOR INSPECTION					
AI.2.5	Additional Insulated Tools	If the accumulator uses bolted connections the team must also present tools appropriate for manipulating the selected fasteners such as: - Insulated spanners Display - Insulated socket set Display - Insulated Allen keys (hex wrenches) Display - Any other special tools required to maintain the accumulator and other electrical systems safely	Inspect	D.g.	
CHARGING TROLLEY				Reinspect	Passed
AI.3.1	TSACHC Wheels	The hand cart must have at least two wheels. Appropriately rate to take the load of the accumulator and cart. Display datasheet, the accumulator is 67 kg, trolley weight is 40 kg	Inspect	D.g.	
AI.3.2	TSACHC Brake	The hand cart must have a brake which acts on at least 2 of the wheels that is always on and only released if someone pushes the handle, or similar. The brake must be capable of safely stopping the fully loaded hand cart. Display by moving the trolley and operating the brake lever	Inspect		
AI.3.3	TSACHC Accumulator Mounting	Accumulator must be securely attached to the hand cart to enable safe transportation. Display trolley with accumulator bolted to the base.	Inspect		
AI.3.4	TSACHC Labels	The label on the TSACHC must include The vehicle number, the university name, and the ESO phone number(s) must be displayed and written in Roman characters of at least 10mm high on the lid or top of each TSACHC. The characters must be clearly visible and placed on a high-contrast background. Display labels on Trolley, Use a ruler to show height of letters	Inspect		
BATTERY CHARGER INSPECTION				Reinspect	Passed
Only chargers presented and sealed at technical inspection are allowed.					
AI.4.1	Battery Charger	Charger needs to be professionally built. All connections of the charger(s) must be insulated and covered. No open connections are allowed. Power-lite HV connector must be covered, Malcolm to be present and to explain that he built it and is a qualified electrician	Inspect		
AI.4.2	Battery Charger Wiring	Battery Charger Power lead must be free of any damage, nicks, cuts, grease or grime. The plug shall be suitably rated for the current load and conforming to EN 60309-2, AS/NZS 3123, Charger shall have a valid test and tag sticker or tag attached. Display Datasheet, display testing tag and conforming sticker	Inspect	D.g.	
AI.4.3		HV wires must be marked with gauge, temperature rating and voltage rating, serial number or norm is also sufficient, if the team shows the datasheet in printed form. Display samples and datasheets	Inspect	D.g.	

EV CLASS - ACCUMULATOR INSPECTION					
AI.4.4		Wire temperature, current and voltage rating must be suitable for use in the charger Display Datasheet	Inspect	D.g.	
AI.4.5		Using only insulating tape or rubber-like paint for insulation is prohibited. We do not use	Inspect		
AI.4.6		The AC power supply to the battery charger and other associated devices must include a residual current device (RCD) with over current protection (fuses or an appropriate circuit breaker) or residual current circuit breaker (RCBO). The RCD or RCBO device must act to disconnect both the active and neutral supplies. The trip sensitivity of the RCD must not exceed 30mA. Where possible 10mA is preferred. Display Datasheet and show on charger. Display schematic	Inspect	D.g.	
AI.4.7		Traction System charging leads must be orange. Display HV Charging Cable	Inspect	D.g.	

EV CLASS - ACCUMULATOR INSPECTION						
BATTERY CHARGER INSPECTION CONTINUED...						
Only chargers presented and sealed at technical inspection are allowed.						
					Reinspect	Passed
AI.4.8	Charging system measuring points	Two charging system voltage measuring points must be provided on the charger output lines. Display Schematic and Test Points	Inspect	D.g.		
AI.4.9		The measuring points must be protected from being touched with the bare hand/fingers once the housing is opened. Display Test points with covers	Inspect	D.g.		
AI.4.10		4mm shrouded banana jacks rated to a voltage higher than the maximum tractive system voltage must be used Display datasheet	Ask the team to provide a datasheet	D.g.		
AI.4.11		The Charger System Measurement Points must be protected by body protection resistors per EV6.8.4 Display schematic, datasheet and be prepared to demonstrate using FLUKE by doing the HV+ test point and outlet and the HV- test point and outlet. They should both be 15kΩ	Inspect	D.g.		
AI.4.12		The Charger System Measuring Points must be marked with HV+ and HV- Display test points	Inspect			
AI.4.13	Battery Charger Emergency Stop	The charger must include a push-type emergency stop button which has a minimum diameter of 25mm and must be clearly labelled. The Pushbutton must be easy to reach with the accumulator in place. Display Estop and the length of the accumulator charging cables. Use callipers to measure button size	Inspect			
AI.4.14		When the E-stop is pressed, the output from the charger must be electrically disconnected from the battery, and the output from the charger must fall to less than 60 V in 5 seconds. The shutdown method must be implemented using a positive switching circuit (Contactor, relay contacts, etc.) which is opened if the charger shutdown button is pressed, or the charger is disconnected from the accumulator. Display video and charger PCB schematic	Inspect			
AI.4.15		The international electrical symbol consisting of a red spark on a white-edged blue triangle must be affixed in close proximity to this switch Display sticker	Inspect			
AI.4.16	Battery Charger Earthing	The battery charger, accumulator and accumulator charging trolley, and any associated metallic components must be equipotentiality bonded and connected to the AC power supply earth such that the RCD function is not impeded. Cables used for earthing must be yellow/green stripped and at least 2.5mm ² cross section. Display photos and earth cable connecting everything, display datasheet	Inspect	D.g.		
AI.4.17	Battery Charger Galvanic Isolation	The charger DC output must be galvanically isolated from the AC input. Interfaces to external devices (such as laptops via serial or USB) must be galvanically isolated from the AC input and from the DC output Display charger PCB and overview schematic, photos of inside, display datasheet of chargers	Inspect Ask the team to provide a datasheet, schematic or similar evidence	D.g.		

EV CLASS - ACCUMULATOR INSPECTION					
AI.4.18	Touch Protection	It must not be possible to touch any live components of the charger with a 100mm long, 6mm diameter insulated test probe when the covers are in place. All output and interface plugs must be touch protected. Ensure test points covers and blank connectors are on. Let the inspector probe the charger, comply to their requests	Inspect		
ACCUMULATOR INSPECTION					
Accumulator technical inspection will inspect the accumulator outside of the vehicle in the designated battery room. Teams are expected to bring the necessary tools for the safe opening of the accumulator for inspection. All accumulators will be physically inspected.					
				Reinspect	Passed
AI.5.1	Accumulator Ready For Inspection	Using appropriate safe work practices, remove covers from accumulator and remove segment links. Comply with inspector requests	Ask team to perform		
AI.5.2	Accumulator Construction	HV Accumulator(s) must be enclosed in container(s). Display	Inspect		
AI.5.3		The bottom of the accumulator must be steel of minimum thickness 1.25mm or Aluminum of minimum thickness 3.2mm. Walls (internal and external) and cover/top, must be Steel of minimum thickness 0.9mm or Aluminum of minimum thickness of 2.3mm. If alternative material used, they should present proof of equivalency (per SES or other) with a sample. Measure the physical container with Vernier Callipers and/or display photos	Inspect	D.g.	
AI.5.4		All components and parts of the accumulator container need to be properly fixed. Internal cables and components must not be able to move around causing chafing, stress to connections, or other damage. Display, wiggle and touch cables	Inspect		
AI.5.5		Fasteners in the tractive system high current path must have a positive locking mechanism. The locking mechanism must be suitable for use at a minimum operating temperature of 90 degrees C. Bolts with nylon patches are acceptable only for blind holes and where used per OEM design. Any bolt which may pose a risk of fire or short circuit if loosened must utilise a positive locking mechanism. Display that only genlocks are used in the accumulator which are rated to 300 degrees C	Inspect		
AI.5.6		The accumulator cells must be secured from moving in the event of an impact. This includes if the accumulator is inverted. Generally friction fitting is not sufficient and some positive retention is required. If the accumulator lid is used as a method to retain the cells, it must be reinforced and robustly attached. Display calc of lid strength and that the top plate is reinforced. Display image of cells under top plate	Inspect		
AI.5.7		Internal vertical walls must be robustly fastened to the container Display welds on the outside of the container and the bottom, show photos	Inspect		

EV CLASS - ACCUMULATOR INSPECTION					
AI.5.8		If the accumulator container is made from an electrically conductive material, the poles of the accumulator stack(s) and/or cells must be insulated against the inner wall of the accumulator container, if the container is made of electrically conductive material. There must be an insulating barrier between the accumulator walls and any electrically live component within. This usually means <u>that all of the internal walls of the accumulator needs</u> to be lined with an appropriate insulating material. <u>Display photos and lining itself and the datasheet for the Nomex paper and tape, the CANaMon PCBs are made from FR4 and protect it from above, as well as the PEEK on the Top Plate</u>	Inspect	<u>D.g.</u>	
AI.5.9		Holes in the accumulator external structure should be of the minimum practical size for cable passages, plugs etc. Holes for ventilation should be circular, no greater than 10mm in diameter, and not obviously weaken the accumulator container external structure. When installed in the vehicle, the holes must not face towards the driver with the firewall removed. <u>Display photos and measure using Vernier Callipers</u>	Inspect	<u>D.g.</u>	
ACCUMULATOR INSPECTION CONTINUED...					
				Reinspect	Passed
AI.5.10	Labelling	Accumulator must be marked with the ISO7010-W012 symbol triangle with black lightning bolt on yellow background) with triangle side length of 100 mm minimum, with text "ALWAYS ENERGIZED" If accumulator voltage is greater than 60VDC must also be Labelled as "HIGH VOLTAGE" Labelling should be on all practical external sides including top, bottom, front, rear, sides. Smaller labels accepted only if surface too small for defined size. <u>Display Labelling on accumulator, measure using ruler</u>	Inspect		
AI.5.11	Internals - cell connection	Contacting / interconnecting the single cells by soldering in the high current path is prohibited. Soldering wires to cells for the voltage monitoring input of the BMS is allowed <u>Display photo, we do not use solder</u>	Inspect	<u>D.g.</u>	
AI.5.12		Connections to cell tabs must be adequately supported to prevent damage to the cells. Cables, busbars etc. must not be supported entirely by the cell tabs. <u>Display photo, bolted to bus bar</u>	Inspect	<u>D.g.</u>	
AI.5.13	Internals – IR/fuse	Every accumulator container must contain at least one main fuse and at least two accumulator insulation relays. There must be no electrically live connection to the accumulator cells outside of the accumulator enclosure when the IRs are open. This includes BMS wires. IRs must be rated for the maximum expected current draw. The IRs and main fuses must be separated from the cell segments by an electrically insulated and <u>non flammable</u> material. <u>Display the datasheets for the fuse and IRs and point out the High current Path. Display that they are separated by the top plate and the PEEK datasheet</u>	Inspect	<u>D.g.</u>	

EV CLASS - ACCUMULATOR INSPECTION					
AI.5.14	Internals – fuses	The fuse(s) must be appropriately rated to protect the cable(s) and other electronic components. Fuses should be HRC type and marked with a DC voltage rating above the maximum expected operating voltage. Except where specifically stated in the rules, all small wiring connecting directly to the cells or tractive system should have appropriate fuses as close as possible to the source. Fusible links are acceptable but require evidence of appropriate voltage and current rating. Teams may justify undersized (current only) fuses based on average current draw and the manufacturer's thermal time curves. Display datasheet for fuses (list below), and CANaMons schematic to show fuses are used for the voltage taps, all fuses use 22 AWG or larger which is rated for 7 A *ONLY IF ASKED*, supply LVD, AIL and Precharge schematics and fuse & fuse holder datasheets AIL: HV fuse is located as close as possible to input to board, connection from HV connector to AIL is maximum 150 mm, and possesses current that is below the ampacity of the upstream Overcurrent Protection (AIL draws ~2 mA, AIL is fused for 500 mA) EV.6.6.6 If conductor ampacity is reduced below the ampacity of the upstream Overcurrent Protection, the reduced conductor longer than 150 mm must have additional Overcurrent Protection. This additional Overcurrent Protection must be: 150 mm or less from the source end of the reduced conductor - On the positive and the negative conductors in the Tractive System - On the positive conductor in the Grounded Low Voltage System Top plate fuses = 0.5 A, 1000 VDC CANaMon fuses = 1 A LVD fuses: LV = 0.5 A, 5 A, 7 A. HV = 0.5 A, 500 VDC AIL fuse = 0.5 A, 500VDC Precharge fuse: LV = 5 A, HV = 0.5, 500 VDC	Inspect	D.g.	
AI.5.15	Internals - maintenance plugs	Maintenance plugs or similar measures have to be taken to allow separating the internal cell stacks in a way, that the separated cell stacks carry a voltage of less than 120VDC and a maximum energy of 6MJ. The separation has to affect both poles of the stack. It must not be possible to incorrectly install the maintenance plugs causing a short circuit. 1 Segment: $64.8 \text{ (Wh)} \times 3600 \text{ (sec)} \times 12 \text{ (num of cells)} = 2.80 \text{ (MJ)}$ $4.2 \text{ (V_max)} \times 12 \text{ (num of cells)} = 50.4 \text{ V maximum}$	Inspect	D.g.	

EV CLASS - ACCUMULATOR INSPECTION					
		Display the service handle and how it is bidirectional and separates the pack into 9 segments. Also display the tap covers (bolt retainers on CANaMons) we use. Display Methode Bud Connector Datasheet			
AI.5.16	Internals – cell stack barriers	Each stack has to be electrically insulated by the use of suitable material towards other stacks in the container and on top of the stack. Air is not considered to be a suitable insulator for this purpose. The team should provide evidence that the material(s) used are appropriate (i.e. material datasheets, manufacturer's specifications, lab test reports etc.). Team manufactured composite materials are not acceptable (carbon fibre, Kevlar, fiberglass etc.). Display datasheets for Ultrafuse ABS FR Black Filament and PEEK Prepare to display Nomex datasheets	Inspect	D.g.	
ACCUMULATOR INSPECTION CONTINUED...				Reinspect	Passed
AI.5.17	Accumulator container connectors	If HV-connectors of the accumulator containers can be removed without the use of tools, a pilot contact/interlock line has to be implemented which breaks the current through the IRs whenever the connector is removed Display interlock loom running from HV connector to LVD Prepare to display LVD and Shutdown circuit schematics (combined onto 1 page)	Inspect	D.g.	
AI.5.18	Indicator light/voltmeter	Each container must have an indicator light or an analogue voltmeter showing that voltages greater than 60V DC are present outside of the container. This includes when the accumulator is being charged. Display on lid and how it is connected to IRs. *Prepare* to display AIL schematic, in the A3 booklet	Inspect		
AI.5.19	Equalizing valve	If the container is completely sealed, it must have an equalizing valve to prevent integral build up of pressure in the event of fire. N/A	Inspect		

EV CLASS - ACCUMULATOR INSPECTION					
AI.5.20	Electrical Connections	Electrical connections must be secure and of high quality. Connections must be made using appropriate hardware, correct sized lugs for selected cable, appropriate crimping methods etc. Electrical connections must not be able to loosen if they become hot. Electrical connections which rely on plastic or similar material being squeezed, which might soften and become loose if overheated are not acceptable. Information about the electrical connections supporting the high current path must be available at Elect Tech Inspection. Display Datasheets for Methode Buds, Radloks, and aPower-Lite HV connector. Display Blue handle crimps and an example of a crimped lug Prepare to display: - molex crimpers and example crimp - DT crimpers and example crimp	Inspect	D.g.	
AI.5.21	Precharge and Discharge Resistors	Precharge and discharge resistors must be installed such that if they overheat, it is unlikely they will cause any of the accumulator cells to overheat. If the team has elected to utilize precharge or discharge or discharge resistors with a continuous operation (not monitored by the PDOC per local addendum), the resistors must be installed according to the manufacturer's recommendations for heatsinking airflow, and such that heat soak cannot cause the accumulator cells to overheat. We do not have continuously operating Precharge or discharge resistors. Show datasheets for the Precharge and discharge resistor, only the Precharge resistor is in the accumulator. Discharge is in the HIP. The Precharge is only on when the accumulator is recharging, which is no more than a couple second and it will not overheat (controlled by teensy activating a Precharge relay)	Inspect	D.g.	
AI.5.22	Accumulator Internal Wiring	All wires inside the accumulator must be marked with gauge, temperature rating and voltage rating, serial number or norm is also sufficient, if the team shows the datasheet in printed form. Display datasheets and samples Wire sizes used: 22 AWG 16 AWG 25 mm2 16 mm2 10 mm2 BMS wires do not have a datasheet, but all requested information is shown on the samples	Inspect		

EV CLASS - ACCUMULATOR INSPECTION					
AI.5.23		Wire temperature and voltage rating must be suitable for use in the accumulator. All Wires (including GLV and BMS) wires within the accumulator container must be rated for the full tractive system voltage. As above	Inspect		
AI.5.24		Using only insulating tape or rubber-like paint for insulation is prohibited. N/A	Inspect		

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ACCUMULATOR INSPECTION Page 12 of 9

EV CLASS - ACCUMULATOR INSPECTION					
ACCUMULATOR INSPECTION CONTINUED...				Reinspect	Passed
AI.6.1	Plastic and 3D printed Components	Any plastic components which are critical to the integrity of the accumulator (structural elements, bolt locking devices, insulation, barriers etc.) must be manufactured from materials which have a glass transition temperature higher than the expected operation temperature. The operating temperature is considered to be the calculated maximum cable or busbar temperature, maximum battery temperature or pre-charge or discharge resistor operating temperature, or 90 degrees C whichever is higher. 90 degrees C is the highest temperature as our cables and bus bars are over spec'd, maximum battery temperature is 60 degrees C and the Precharge resistor is only in operation during Precharge. Display Polymax datasheet and operating temperatures for bus bars and cable	Inspect	D.G.	
AI.6.2	Spare accumulator(s)	Spare Accumulators must be of the same type (construction, voltage, capacity, weight, fixing etc.) Only applicable if spare accumulators are used. N/A	Inspect		
AI.6.3	Temperature monitoring equipment	Teams will be provided with a thermally sensitive sticker to install into their accumulator. The thermally sensitive sticker must be mounted such that it is in direct thermal contact with the negative terminal of a cell of an accumulator segment such that the thermal sticker will reasonably represent the temperature of the cells. The sticker should be visible from outside the accumulator without needing disassembly of the accumulator. (2025 only - refer inaccessible stickers to Technical Inspection Captain for review) During the fitment of the thermal strip, take photos to show and ensure that it is visible through the hatch in the accumulator	Inspect		

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2025 Formula SAE-A Technical Inspection Sheet

EV STATIC INSPECTION

Initial Presentation Scrutineer Name:		Initial Presentation Lane Number:	
1st Reinspection Scrutineer Name:		1st Reinspection Lane Number:	
2nd Reinspection Scrutineer Name:		2nd Reinspection Lane Number:	
3rd Reinspection Scrutineer Name:		3rd Reinspection Lane Number:	

PERSONNEL		Inspection #			
Teams are required to have an Electrical Safety Officer.		Initial ins.	1st reins.	2nd reins.	3rd reins.
SI.1.1	Identify the Electrical Safety Officer (ESO)	Ask for the ESO			
SI.1.2	Only team members involved in technical inspection can be in the technical inspection bay	Ask for non-essential people to leave the area			

DO NOT PROCEED WITH TECHNICAL INSPECTION IF THE ESO IS NOT PRESENT

INERT VEHICLE		Inspection #			
The Vehicle must be in a condition which prevents unexpected energization of the HV electrical systems or movement of the tractive system		Initial ins.	1st reins.	2nd reins.	3rd reins.
SI.1.3	Disable the HV Systems - Manual Service Disconnect (MSD)	Check the MSD is disconnected			
SI.1.4	Disable the HV Systems - Tractive System Master Switch (TSMS) Lock	Check the TSMS Lock is fitted and effective			
SI.1.5	(AV and Dual Purpose only) Disable the ASMS - Autonomous System Master Switch (ASMS) Lock	Check the ASMS Lock is fitted and effective			

DO NOT PROCEED WITH TECHNICAL INSPECTION IF MSD IS CONNECTED OR THE TSMS LOCK IS MISSING

INSTRUCTIONS FOR SCRUTINEERS

This section applies to Electric Vehicles (EV) and Autonomous Vehicles (AV) and Dual Purpose Vehicles.

If an item is acceptable, initial the "Passed" column.

If an item is unacceptable, initial the "Reinspect" column and inform the team members they have not meet the requirements for that item.

EV CLASS - EV STATIC INSPECTION

Teams are
allowed to

take their vehicle away from technical inspection and address any issues before representing for reinspection.

Scrutineers may continue with the inspection when unacceptable items are found, provided it is safe to do so.

Teams are required to pass all items before the vehicle can proceed to the Brake Test or compete in Dynamic Events.

Scrutineers reserve the right to refuse to proceed with Technical Inspection if a sufficiently serious issue which might jeopardize the safety of the technical inspection team, event personal or any other persons is found at any time.

- ESO (Lukes), Jayson, Sam, Bunny (document hander) and Cam (SES)
- MSD Disconnected (In ESO's possession)
- TSMS Lock Installed (Keys in ESO's possession)
- Car in Race Ready Condition (accumulator installed, everything connected, firewall and body kit on)
- Push Bar with multimeter (with caps for probes), insulating gloves, leather outers, Hi Vis vests and fire extinguisher
- High Stands x 2, with body kit mounts
- Red SCA Jack Stands x 4
- Red toolbox, callipers, spare zipties, tape measure
- DT crimps, DT crimpers and large cable crimps (blue handle)
- Accumulator spares box (with M8 bolt used for bolting Accumulator to car)
- Tech Box which samples, documents and tech bibles
- Fluke Multimeter with probes and banana jack probes
- Go Pro mounted to car
- Multiple hands to lift the car and remove the body kit
- Cam ready with SES open on LAPTOP

Roll car into bay and immediately ask if the car can be put on the stands, leave body kit on

After MSD section of inspection, take body kit off and place onto blue stands

EV CLASS - EV STATIC INSPECTION

VEHICLE VISUAL INSPECTION

The primary purpose of the Vehicle Visual Inspection is to find and highlight any items which might cause a safety hazard for the drivers, team members, track marshals and anyone else who might interact with the vehicle.

The hazards most associated with EV tractive systems include fire, arc flash, electric shock or loss of control. The Visual Inspection is intended to check that the vehicle has been designed and constructed to an acceptable standard such that the likelihood of one of these hazards causing injury is minimized. Scrutineers are not generally concerned with the reliability of the vehicle unless the failure of a component or system might pose a safety hazard.

Scrutineers may also look for rules compliance issues. Rules non-compliances should be referred to the EV Technical Inspection Captain.

Because of the bespoke nature of the vehicles, Scrutineers reserve the right to reject vehicles for any reasonable safety concern based on their judgement, even if not specifically covered in this document.

				Reinspect	Passed
SI.2.1	Race Ready Condition	The vehicle should be fully assembled with the Accumulator in place and generally in race ready condition when presented to technical inspection. Inspection of incomplete vehicles may proceed with permission from the Technical Inspection Captain. <u>Ensure Accumulator is in, and everything is connected (without HV switch and MSD). Body kit and top fire wall must be fitted</u>	Inspect		
SI.2.2	Technical Inspection Access	The vehicle must be lifted/jacked onto stands such that <u>all of the vehicles</u> wheels are at least 100 mm and no more than 300mm above the ground. It must be possible to freely rotate the vehicle's wheels without the vehicle moving. Stands must be robust and the vehicle must be secure. <u>Use red SCA car jack stands on the first height setting to support the car in the air. Use a tape measure to display its height off the ground from the wheels. Body Kit Stays on</u>	Inspect		
SI.3.1	Push Bar	A pair of HV insulating gloves with leather outers, a Multimeter and fire extinguisher must be attached. If a tool is needed to open the MSD this must also be attached to the push bar. <u>Unlock box on the push bar to display the HV gloves with leather outers, a Multimeter and a fire extinguisher. The HV inners should be tagged from Accumulator Tech and the Fire Extinguisher must be in date</u>	Inspect		
SI.4.1	Manual Service Disconnect	The MSD must be clearly marked with "MSD" in large, high contrast lettering and should be clearly visible to a Marshal as they approach the vehicle. The MSD label must be easily distinguishable from the <u>vehicles</u> other markings. <u>Display labels on Body Kit and HIP</u>	Inspect		
SI.4.2		It must be possible to disconnect the MSD without removing any bodywork <u>Display MSD location</u>	Inspect		
SI.4.3		If opening the MSD is possible without the use of tools, a pilot contact/interlock line <u>has to be implemented</u> which breaks the current through the AIRs whenever the connector is removed <u>Display schematic for the Shutdown Circuit and HIP, Display physical MSD and the difference between the interlock and HV connections</u>	Inspect	<u>p.g. 1</u>	

EV CLASS - EV STATIC INSPECTION					
SI.4.4		In ready to race condition it must be possible to disconnect the MSD within 10 seconds. Ask the scrutineer if it is okay to fit the MSD to demonstrate. Fit the MSD and remove it once the inspector is ready. Take body kit and top firewall off after this is done	Ask the team to demonstrate		
SI.5.1	Tractive system protection	All tractive system connections outside of an enclosure must be appropriately insulated. Display HV Cables from the accumulator, the HIP and the MC, Display the Phase Cable Cover	Inspect		
SI.5.2		It must not be possible to touch any tractive system connections with a 100mm long, 6mm diameter insulated test probe when the tractive system enclosures are in place Ensure test points covers are on. Let the inspector probe the car, comply to their requests	Inspect		
SI.5.3		Tractive System components and containers must be protected from moisture in the form of rain or puddles. Use of tape alone is not acceptable. Heatshrink must be appropriately rated for the TS voltage and must be provided with additional mechanical protection. Display Heat Shrink datasheet and samples and say that is it only used on the phase cable cover and in the accumulator	Inspect	p.g. 2	

EV CLASS - EV STATIC INSPECTION						
VEHICLE VISUAL INSPECTION CONTINUED...				Reinspect	Passed	
SI.6.1	Master Switches	The Tractive System Master Switch (TSMS), Grounded Low Voltage Master Switch (GLVMS) and (AV and Dual Purpose only) Autonomous System Master Switch (ASMS) on the right side of the vehicle. At the height of the drivers' shoulders. Rotary type, ON position with key horizontal. ON and OFF position must be clearly marked. Located right side of the vehicle at approx. driver's shoulder height, just behind main roll hoop. Not attached to removable bodywork. Must be easily actuated from outside the vehicle. Display the Master Switch location and all the above details	Inspect			
SI.6.2		All master switches must be a rotary type with a red removable key/handle Display keys and switches	Inspect			
SI.6.3		GLVMS (LV Master switch) must be completely surrounded by an RED circle of min 50mm diameter Marked with international symbol (triangle with red lightning bolt on blue background). Display with callipers	Inspect 			
SI.6.4		TSMS (TS Master switch) must be completely surrounded by an ORANGE circle of min 50mm diameter Accompanied by the ISO 7010-W012 (triangle with black lightning bolt on yellow background). Display with callipers	Inspect 			
SI.6.5		The Tractive System Master Switch must have a lock or similar fitted which prevents insertion of the master key. The lock should be of the type commonly used in industry for "LOTO isolation". Use of a LOTO hasp or similar is acceptable if <u>necessary</u> provided it is secure. The key for the lock and the master keys should be in the custody of the ESO. Display lock on TSMS and the keys in the possession of the ESO	Inspect			
SI.7.1	HV warning stickers	Each housing/enclosure containing HV parts must be marked with the ISO7010-W012 symbol (triangle with black lightning bolt on yellow background) If accumulator voltage is greater than 60VDC must also be Labelled as "HIGH VOLTAGE" Labelling should be on all practical external sides including top, bottom, front, rear, sides. Smaller labels accepted only if surface too small for defined size.	Inspect Check top, sides and bottom of enclosures			

EV CLASS - EV STATIC INSPECTION					
		Display Stickers on HIP, DCDC, Accumulator, Phase cable cover and Inverter (inverter will be hard to see)			
VEHICLE VISUAL INSPECTION CONTINUED...				Reinspect	Passed
SI.8.1	HV Wiring	All tractive system wiring outside electrical enclosures must be enclosed in separate orange nonconductive conduit or use an orange shielded cable Display that the 25 mm² ACC to HIP and the 12 AWG HIP to DCDC are enclosed in conduit and the HIP to MC and MC to Motor cables are 50 mm² shielded cable. Show datasheets for 50 mm², 25mm² and 12 AWG wire Be prepared to show samples	Inspect	p.g. 3, 4 & 5	
SI.8.2		The conduit or shielded cable must be securely anchored (at least) at each end so that it can withstand a force of 200 N without straining the cable and <u>crimp and</u> must be located out of the way of possible snagging or damage. Take note of connections to motors, plugs carrying HV, and any entrances to accumulators, motor controllers, the TSMPs and any other HV enclosures Display boots connecting the conduct to the connector on the ACC to HIP and HIP to DCDC, and that the HV wires are routing in a way that does not show possible snagging or damage.	Inspect		
SI.8.3		Tractive system wiring must be shielded against damage by rotating and/or moving parts. All wiring must be sufficiently restrained such that it cannot come into contact with rotating and/or moving parts. Display, comply with requests	Inspect		
SI.8.4		TS wires and GLVS wires are clearly separated/do not run directly next to each other/bounded together by cable rods or in the same cable channel (ALLOWED ONLY FOR PILOT CONTACTS OR INTERLOCK SIGNALS) Display that no HV wires run together with LV wires	Inspect		
SI.8.5		Using only insulating tape or rubber-like paint for insulation is prohibited. N/A	Inspect		
SI.8.6		Heat-shrink type insulation used on tractive system components and cables may only be used inside rigid enclosures and must be labelled with voltage rating. Serial number or norm is also sufficient, if the team shows the datasheet in printed form. Display heat shrink datasheet and <u>heatshrink</u> samples	Inspect	p.g. 2	

EV CLASS - EV STATIC INSPECTION					
SI.8.7		No soldering in high current path including cable lugs and connectors. <u>We do not use solder in the high current path, everything is exclusively bolted, crimped or connected using a connector.</u>	Inspect		
SI.8.8		Electrical connections must be secure and of high quality. Connections must be made using appropriate hardware, correct sized lugs for selected cable, appropriate crimping methods etc. Electrical connections must not be able to loosen if they become hot. Electrical connections which rely on plastic or similar material being squeezed, which might soften and become loose if overheated are not acceptable. Information about the electrical connections supporting the high current path must be available at Elect Tech Inspection. <u>Display Datasheets for HPK (MC side of HIP TO MC) (p.g. 6)</u> <u>ePower-Lite (HIP TO ACCU) (p.g. 7)</u> <u>Surlok EMI (HIP side of HIP to MC) (p.g. 8)</u> <u>HVA 280 (DCDC and EM) (p.g. 9 & 10)</u> <u>If asked, display datasheets for DT and AT series connectors (p.g. 11 & 12)</u> <u>Display Blue handle crimps and an example of a crimped lug (HIP to ACCU), display HVA 280 crimp, HPK come precrimped.</u>	Inspect	<u>p.g. 6, 7, 8, 9, 10, 11 & 12</u>	
SI.8.9		HV wiring must be entirely within the chassis (side impact structure, rear impact zone and roll envelope) except for cables to hub motors <u>Display Phase cable guide and that the rest of the HV wires are within the chassis</u>	Inspect		
VEHICLE VISUAL INSPECTION CONTINUED...				Reinspect	Passed
SI.9.1	Tractive system measuring points	Two tractive system voltage measuring points and a GLVS ground point must be installed next to the master switches, right side of the vehicle, at shoulder height of the driver. <u>Display Test Points</u>	Inspect		
SI.9.2		The measuring points must be protected by a non-conductive waterproof housing which can be opened without tools. The use of simple plastic/rubber sealing plugs inserted into the measuring points is not acceptable as a waterproofing method. <u>Display banana jack covers and their datasheet, showing that they are IP65 rated</u>	Inspect	<u>p.g. 13</u>	
SI.9.3		The measuring points <u>must</u> be protected from being touched with the bare hand/fingers once the housing is opened. <u>Take of banana jack covers and display</u>	Inspect		
SI.9.4		4mm shrouded banana jacks rated to a voltage higher than the maximum tractive system voltage must be used <u>Display Datasheet</u>	Ask the team to provide a datasheet	<u>p.g. 13</u>	
SI.9.5		The TSMPs must be marked with HV+ and HV- <u>Display markings next to test points</u>	Inspect		

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EV CLASS - EV STATIC INSPECTION					
SI.10.1	GLV Ground Measuring Point	Must be positioned next to the TSMPs and must be marked with GND Display GLVMP and GND label	Inspect		
SI.10.2	GLV Voltage	Measure GLVS voltage between GLVS battery positive Must be less than 60V DC Measure using Fluke, measuring between GND and 12V	Ask team to demonstrate		
SI.10.3	Tractive System Voltage	Measure TS voltage at measurement points Must be less than 60V DC Measure using Fluke, using TSMPs	Ask team to demonstrate		
SI.10.4	Discharge System	The discharge circuit has to be wired in a way that it is always active whenever the shutdown circuit is open. If a discharge circuit is used a low resistance can be measured between HV+ and HV- whenever the tractive system is de-activated. Measure resistance between HV+ and HV- with multi-meter. Must be 2*BPR+ Dis- Charge Resistor (GLVS must be off) Use the Fluke to measure the resistance between HV+ and HV- and show the schematic for the HIP and TSAL Discharge, explain that SHUTDOWN_TSMS activates the coil of the discharge relay (datasheet attached) and disconnects the discharge resistor (datasheet also attached) when HV is on. This number must be $2 \times 15K\Omega + 3.3k\Omega = 33.3k\Omega$ or higher Refer alternative discharge methods to the EV Technical Inspection Captain	Ask team to demonstrate	p.g. 14	
SI.11.1	Insulation Measurement Test	The insulation resistance between the GLV Ground Measuring Point and TSMPs HV+ must be greater than 500 Ohms/Volt, calculated for the maximum TS voltage. The available measurement voltages are 250 V and 500 V. All vehicles with a maximum nominal operation voltage below 500 V will be measured with the next available voltage level. All teams with a system voltage of 500 V or more will be measured with 500 V. Use the Fluke to measure the resistance between HV+ and GND. This number must be more than Accu Voltage: $500\Omega/V \times 453.6V = 226.8k\Omega$ Their Voltage range: $500\Omega/V \times 500V = 250k\Omega$	Measure insulation resistance between GLV Ground and HV+		

EV CLASS - EV STATIC INSPECTION					
SI.11.1		The insulation resistance between the GLV Ground Measuring Point and TSMPs HV- must be greater than 500 Ohms/Volt, calculated for the maximum TS voltage. The available measurement voltages are 250 V and 500 V. All vehicles with a maximum nominal operation voltage below 500 V will be measured with the next available voltage level. All teams with a system voltage of 500 V or more will be measured with 500 V. Use the Fluke to measure the resistance between HV- and GND. This number must be more than Accu Voltage: $500\Omega/V \cdot 453.6V = 226.8k\Omega$ Their Voltage range: $500\Omega/V \cdot 500V = 250k\Omega$	Measure insulation resistance between GLV Ground and HV+		

EV CLASS - EV STATIC INSPECTION						
VEHICLE VISUAL INSPECTION CONTINUED...				Reinspect	Passed	
SI.12.1	Shutdown Buttons	One shutdown button, push-pull or push-rotate-pull on each side behind the <u>drivers</u> compartment (height approx. driver's head), one in the cockpit and easily accessible by the driver in any steering wheel position while wearing gloves and wrist restraints Display photo and positions of estops	Inspect	p.g. 15		
SI.12.2		Minimum diameter of shutdown buttons on the side is 40mm. Minimum diameter of shutdown button in the cockpit is 24mm Measure using callipers	Inspect			
SI.12.3		The shutdown buttons are not allowed to be easily removable, e.g. mounted onto a removable body work Display	Inspect			
SI.12.4		All shutdown buttons marked with international symbol Display	Inspect 			
SI.12.5	Cockpit Shutdown Button	Cockpit master switch <u>Pull-ON, Push-OFF, or Twist-Pull-ON, Push-OFF</u> alongside & unobstructed by steering wheel, easily reached by driver. Minimum 24mm diameter. Marked with international symbol Display	Inspect	p.g.15		
SI.12.6	Brake Over Travel Switch	Brake over travel switch must be secured and positioned behind the brake pedal such that it will be tripped before the master cylinders bottom out or any other stops are reached. The brake over travel switch must be robustly mounted and not be bent or damaged if an over-travel occurs. Display BOTS and then photo and video of it operating	Inspect If necessary, ask team to demonstrate by bleeding the brake circuit	p.g.16		
SI.12.7	Inertia switch	The device must be mechanically attached to the vehicle. It must be possible to demount the device so that its functionality can be tested by shaking it. Mounts including double sided tape, Velcro or similar are not acceptable. Display inertia switch, be prepared to demount using ratchet and <u>allen</u> key to demonstrate its functionality, Remount with bolts.	Inspect			

EV CLASS - EV STATIC INSPECTION						
VEHICLE VISUAL INSPECTION CONTINUED...				Reinspect	Passed	
SI.13.1	Accelerator Pedal Position Sensor (APPS)	Accelerator Pedal must promptly return to original position, if not actuated Display by pressing accelerator pedal with hand	Inspect			
SI.13.2		At least two sensors must be fitted as APPS and must have separate supply, ground and signal lines from the plausibility detection device Display the connectors going to the pedal and how they lines are separate, display the schematic of the PEN	Inspect	p.g.17		
SI.13.3		The accelerator pedal must have a positive stop to prevent sensors from being mechanically overstressed Display Positive stop on pedal	Inspect			
SI.13.4		Two springs must be used to return the throttle pedal to the off position and each spring must work with the other disconnected Display spare accelerator pedal, show the separate springs	Inspect			
SI.13.5	Brake System Encoder	A brake pedal position sensor or brake pressure switch must be fitted to check for plausibility Display brake pressure sensors and the schematic of the PEN	Inspect	p.g.17		
SI.13.6	Brake Master Cylinders	The brake system master cylinder must be actuated directly or by a mechanical connection. Use of Bowden cables or push-pull Bowden cables is not allowed. The first 90% of the brake pedal travel may be used to regenerate brake energy without actuating the hydraulic brake system. The remaining brake pedal travel must directly actuate the hydraulic brake system, but brake energy regeneration may remain active. Display that the master cylinders are directly actuated for 100% of the brake pedal travel. We do use regen but only between 0% and 10% of the accelerator pedal	Inspect			
SI.14.1	Firewalls (EV specific requirements)	A firewall must separate the driver compartment from components of tractive system (including HV wiring). Brake parts and chassis as conductive paths are allowed to protrude through firewall. See T.1.9.3 Outboard wheel motors and associated cables and other components of the suspension members are excluded from this requirement. Display that only the rear brake line and LV wires pass through the firewall, reference rule: T.1.8.6 d. 'Grommets must be used to seal any pass through for wiring, cables, etc	Inspect	p.g.18		
SI.14.2		The side of the firewall facing the tractive system must be Aluminum with a minimum thickness of 0.5mm and must be deliberately earthed to the chassis ground point. Display CAD image and then display ground strap from Firewall to TS rail mount	Inspect	p.g.19		
SI.14.4		The firewall's integrity must not be compromised by forces from the driver. I.e. the driver must not sit directly against the firewall and the driver's seat must not rely on the structural integrity of the firewall unless it is adequately reinforced. Display images from CAD	Inspect	p.g.20		

EV CLASS - EV STATIC INSPECTION																									
VEHICLE VISUAL INSPECTION CONTINUED...					Reinspect	Passed																			
SI.15.1	Team built PCBs	Where tractive system and GLV are on the same circuit board, teams must be prepared to demonstrate appropriate spacing. Separation space on the circuit boards must be clearly marked. The minimum allowable clearances are: <table><tr><th>Voltage</th><th>Over Surface</th><th>Thru Air (cut in board)</th><th>Under Conformal Coating</th></tr><tr><td>0-50 V DC</td><td>1.6 mm</td><td>1.6 mm</td><td>1 mm</td></tr><tr><td>50-150 V DC</td><td>6.4 mm</td><td>3.2 mm</td><td>2 mm</td></tr><tr><td>150-300 V DC</td><td>9.5 mm</td><td>6.4 mm</td><td>3 mm</td></tr><tr><td>300-600 V DC</td><td>12.7 mm</td><td>9.5 mm</td><td>4 mm</td></tr></table> If necessary, teams may provide manufacturer's datasheets for components demonstrating appropriate voltage ratings where the components require smaller than allowed clearances. Display the datasheet for the Conformal Coating samples of the PCBs (without components) and the Conformal coating. Have callipers ready to measure separation. Show LV/HV Separation Book	Voltage	Over Surface	Thru Air (cut in board)	Under Conformal Coating	0-50 V DC	1.6 mm	1.6 mm	1 mm	50-150 V DC	6.4 mm	3.2 mm	2 mm	150-300 V DC	9.5 mm	6.4 mm	3 mm	300-600 V DC	12.7 mm	9.5 mm	4 mm	Inspect	p.g.21	
Voltage	Over Surface	Thru Air (cut in board)	Under Conformal Coating																						
0-50 V DC	1.6 mm	1.6 mm	1 mm																						
50-150 V DC	6.4 mm	3.2 mm	2 mm																						
150-300 V DC	9.5 mm	6.4 mm	3 mm																						
300-600 V DC	12.7 mm	9.5 mm	4 mm																						
SI.16.1	Accumulator Construction and Attachment	If containers are monocoque, they must be mounted with Steel backing plates of at least 2mm thickness. N/A	Inspect																						
SI.16.2		The Accumulator Container must also have a minimum number of attachment points, dependent on weight and use minimum Grade 8.8 metric 8mm bolt at each 20 kg = 4 attachments; 20-30 kg = 6 attachments; 30-40 kg = 8 attachments; >40 kg = 10 attachments); Our Accumulator is greater than 40kg (67 kg) sample bolt and photo of accumulator attachment points.	Inspect	p.g.22																					
SI.16.3		Ensure 25mm clearance from Side Impact Structure. Ask to see SES. Display CAD image and SES Show p.g.24 if they aren't happy with p.g.23	Inspect	p.g.23																					
SI.16.4	Accumulator Crush Zone Clearance	If rear mounted: At least 25mm clearance from Rear Impact Structure or any non-crushable items. N/A	Inspect																						

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EV CLASS - EV STATIC INSPECTION					
SI.16.5		Ensure 25mm clearance of surface from the firewall. Display CAD image	Inspect	p.g.23	
SI.16.6		Non-crushable items behind Rear Impact Structure cannot pass through the structure. Display that the radiator and pumps are too large to pass through the rear impact structure	Inspect		
SI.16.7		Rear Impact protection extends to or past the upper height of the SIS and is supported to the SIS. Display CAD image	Inspect	p.g.23	
ENERGY METER				Reinspect	Passed
SI.17.1	Energy Meter	Energy Meter fitted and secured to the chassis in a location that does not interfere with the mechanical or electrical operation of the vehicle, in accordance with the instructions. Energy Meter must be mounted in protected and readily accessible position, mounted securely and identifiable, so that the operation light can be visually inspected and the unit can be quickly removed in Parc Ferme. Display energy meter placement in reference to the instruction document, explain how we will remove it and that it will be easily inspected	Inspect	p.g.24	
SI.17.2		Confirm retaining ring on all plugs on energy meter is fully engaged (ring must be clicked over detent). Display	Inspect	p.g.24	
SI.17.3		Check the battery voltage cable is installed with the correct polarity (positive is white, negative is black) and the current sensor is orientated correctly (arrow shows direction of conventional current flow i.e. away from battery if installed on positive lead). Display that we have placed the energy meter is on the positive line and meets the requirements	Ask team to demonstrate	p.g.24	

EV CLASS - EV STATIC INSPECTION

VEHICLE GROUNDING

It is expected that all conductive components within 100mm of the Tractive system will be appropriately grounded (electrically bonded to the GLV system ground).

1. By equipotential bonding the conductive parts within the vehicle, the hazard posed by touching a voltage gradient is reduced, and
2. By ensuring a conductive path back to the chassis ground common point, the likelihood of the IMD correctly detecting the fault and shutting off the vehicle is increased, reducing the period of time for which the hazard can be present.

Scrutineers will look for any conductive components which are in close proximity (within 100mm) of any tractive system component (HV) element, which could become live in the event of a failure of a component, breach of insulation or similar either from normal operation, or in the event of a collision.

Any wire, braid or similar used for vehicle grounding must be appropriately terminated, robustly mounted, and of sufficient size and mechanical strength for its purpose.

This test involves using a Fluke 289 Multimeter on the lo-ohm setting to ensure there is a low resistance connection from the item being tested in reference to the chassis ground test point. It is acceptable to subtract the resistance of the test leads if necessary. Teams may remove paint, resin or other coverings to expose a bare conductive surface if required.

Teams also have the option of measuring resistance via method of voltage drop using a 1 amp current source if necessary.

Measurements to the conductive component should be taken within 100mm of the tractive system or as close as practical. Teams may remove body panels and covers to demonstrate compliance if required.

The pass criteria are:

- a. Electrically conductive parts 300 mOhms Examples: parts made of steel, (anodized) aluminum, any other metal parts

			Criteria	Reinspect	Passed
SI.18.1	Equipotential Bonding	Firewall(s) Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm		
SI.18.2		Accumulator Container(s) Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm		
SI.18.3		Components in proximity of Accumulator Container(s) Measure with Multimeter using GND TP and a probe, this would be the TS Stack, HIP and Motor	300 mOhm 5000 mOhm		
SI.18.4		Motor Controller Enclosure(s) Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm		
SI.18.5		Components in proximity of Motor Controller(s) Measure with Multimeter using GND TP and a probe, this will be whatever hasn't been checked	300 mOhm 5000 mOhm		
SI.18.6		Motor Casings, Mounts and Scatter Shields Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm		
SI.18.7		Components in proximity of Motor(s) Measure with Multimeter using GND TP and a probe, this includes the left and possibly right rear suspension, and radiator	300 mOhm 5000 mOhm		

EV CLASS - EV STATIC INSPECTION						
VEHICLE VISUAL INSPECTION CONTINUED...			Criteria		Reinspect	Passed
SI.18.8	Equipotential Bonding	Other HV enclosure(s) Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.9		Components in proximity of other HV enclosures Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.10		Conductive HV cable channels, cable Armor and cable shielding Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.11		Components in proximity of HV Wires and Cables Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.12		Components in proximity of MSD Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.13		Components in proximity of the Tractive System Measurement Points Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.14		Components in proximity of the Energy Meter Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.15		Cooling system components including heatsinks, radiators, pumps and reservoirs Measure with Multimeter using GND TP and a probe, and hose clamps :D	300 mOhm 5000 mOhm			
SI.18.16		Seat Mounts, Seatbelt Mounts, Driver Controls, Pedals or items in the Drivers Area if in proximity to HV components N/A, they are further than 100 mm from HV, display with ruler or the 'test probe' pen	300 mOhm 5000 mOhm			
SI.18.17		Other items:	300 mOhm 5000 mOhm			
SI.18.18		Other items:	300 mOhm 5000 mOhm			
SI.18.19		Other items:	300 mOhm 5000 mOhm			
SI.18.20	Other items:	300 mOhm 5000 mOhm				
SI.18.21	Other items:	300 mOhm 5000 mOhm				

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2025 Formula SAE-A Technical Inspection Sheet EV FUNCTIONAL TEST

Initial Presentation Scrutineer Name:		Initial Presentation Lane Number:	
1st Reinspection Scrutineer Name:		1st Reinspection Lane Number:	
2nd Reinspection Scrutineer Name:		2nd Reinspection Lane Number:	
3rd Reinspection Scrutineer Name:		3rd Reinspection Lane Number:	

PERSONNEL			Inspection #			
Teams are required to have an Electrical Safety Officer.			Initial ins.	1st reins.	2nd reins.	3rd reins.
FT.1.1	Identify the Electrical Safety Officer (ESO)	Ask for the ESO				
FT.1.2	Only team members involved in technical inspection can be in the technical inspection bay	Ask for non-essential people to leave the area				
DO NOT PROCEED WITH TECHNICAL INSPECTION IF THE ESO IS NOT PRESENT						

INERT VEHICLE (EV & AV CLASSES ONLY)			Inspection #			
The Vehicle must be in a condition which prevents unexpected energization of the HV electrical systems or movement of the tractive system			Initial ins.	1st reins.	2nd reins.	3rd reins.
FT.1.3	Disable the HV Systems - Manual Service Disconnect (MSD)	Check the MSD is disconnected				
FT.1.4	Disable the HV Systems - Tractive System Master Switch (TSMS) Lock	Check the TSMS Lock is fitted and effective				
FT.1.5	(AV and Dual Purpose only) Disable the ASMS - Autonomous System Master Switch (ASMS) Lock	Check the ASMS Lock is fitted and effective				
DO NOT PROCEED WITH TECHNICAL INSPECTION IF MSD IS CONNECTED OR THE TSMS LOCK IS MISSING						

INSTRUCTIONS FOR SCRUTINEERS
 This section applies to Electric Vehicles (EV) and Autonomous Vehicles (AV) and Dual Purpose Vehicles.

EV CLASS - FUNCTIONAL TEST

- ESO (Lukes), Jayson, Sam, Bunny (document handler)
- MSD Disconnected (In ESO's possession)
- TSMS Lock Installed (Keys in ESO's possession)
- Car in Race Ready Condition (accumulator installed, everything connected, top firewall, body kit, Go Pro mounted)
- Push Bar with multimeter (with caps for probes), insulating gloves, leather outers, Hi Vis vests and fire extinguisher
- High Stands x 2
- Red SCA Jack Stands x 4
- Red toolbox, callipers, spare zipties, tape measure
- large cable crimps (blue handle)
- Spare Zip Ties
- Tech Box which samples, documents and tech bibles
- Fluke Multimeter with probes and banana jack probes
- Multiple hands to lift the car and remove the body kit
- Charged Chungus Laptop
 - CANdapter (with CANdapter DB9 to 6 DT pin adapter)
 - Kvaser (with Kvaser DB9 to 6 pin DT adapter)
 - CAN KING and BMS software open
- Impact Gun with Battery and 21mm Deep Impact Socket
- Test Resistor for IMD
- Test loom for BSPD
- Power supply, Extension cable

If they are happy with us to, ask if we can then remove the body kit and top firewall. Place the body kit onto the stand

Ask if Car can be put on the small red stands immediately after entering the bay and then ask if we are able to take the rear wheels off the car as it is rear wheel drive.

Ask if Inertia Switch can be demounted to save time

EV CLASS - FUNCTIONAL TEST

EV FUNCTIONAL TEST PREREQUISITES

This section of the Technical Inspection process is intended to ensure that the vehicle's mandated safety systems function as required by the rules.

The order of testing in this section is important as the safety of some of the later tests relies on the correct functioning of systems already tested. As such there is a hold point after the testing of the safety shutdown systems. If the safety shutdown systems do not function as intended, it will be necessary to stop testing and for teams to correct any issues.

Because this section requires the vehicle to be powered on, and will require the wheels to turn under power, the vehicle must have successfully passed all previous steps before proceeding. The Scrutineer must check that the vehicle has passed all required steps before proceeding.

The Technical Inspectors will supply a Fluke 289 **multimeter** and insulated **banana** test leads.

				Reinspect	Passed
FT.2.1	Vehicle ready for Functional Safety Test	The vehicle must have successfully passed the following inspection stages, with all reinspect items addressed and passed. - Accumulator Inspection - EV Electrical Inspection - Mechanical Inspection - Driver Egress	Check page 2 of this document and the stickers on the vehicle's nosecone		
FT.2.2	Race Ready Condition	The vehicle must be fully assembled with the Accumulator in place and generally in race ready condition when presented to technical inspection. Present car in Race Ready Condition	Inspect		
FT.2.3	Technical Inspection Access	The vehicle must be lifted/jacked onto stands such that all of the vehicles wheels are at least 100 mm and no more than 300mm above the ground. The driven wheels must be removed - It must be possible to freely rotate the vehicle's wheels/hubs without the vehicle moving. Stands must be robust and the vehicle must be secure. Use red SCA car jack stands on the first height setting to support the car in the air. Use a tape measure to display its height off the ground from the wheels. Body Kit Stays on	Inspect		
FT.2.4	Line of Fire	The minimum possible number of team members and Scrutineers should be in the area when testing is being conducted. Keep people away from rotating wheels, hubs or other moving parts. Check tires for debris that could be thrown or remove wheels from vehicle. Use the impact gun with a 21mm impact deep socket to remove the wheel nuts on the rear two wheels and remove the wheels.	Inspect		

DO NOT PROCEED WITH THE FUNCTIONAL SAFETY TEST IF ANY OF THE ABOVE ARE NOT COMPLETE

EV CLASS - FUNCTIONAL TEST					
FT.2.5	Driver Egress	If any of the following tests requires a driver to be in the vehicle, the driver must be in full race gear, and it must be possible for the driver to egress the vehicle safely. e.g. The vehicle must not be on high stands. <u>We do not need a driver in the vehicle, we will be actuating the pedals from outside the car</u>	Inspect		

BMS FUNCTION TEST				Reinspect	Passed
FT.3.1	Battery Management System Function	BMS must monitor the cell voltage of each cell *Ask if body kit can be removed to access expansion port on CEN and if GLVMS can be turned on* Display using the <u>CANDanaler</u> and BMS software on the <u>Chungus</u> the voltage of all 108 cells in the live data tab. Disconnect <u>CANDanaler</u> from CEN and <u>Chungus</u> when finished	Activate BMS system and show measurement data of the BMS for each cell (e.g., via laptop)		
FT.3.2		BMS must monitor the temperature of at least 20% of cells in pack Using the <u>Kvaser</u> and <u>CAN KING</u> on the <u>Chungus</u> , display the max, min and average temps of the cells in the segments. Display a <u>CANaMONS</u> PCB and explain how the board connects to the thermistors on each <u>Energous</u> Module, converts the analogue reading into a CAN message and sends the messages to the LVD on an isolated CAN network. The LVD then sends the messages on CAN 1 which is connected to the rest of the car and the BMS		p.g.1	
FT.3.3		A red indicator light marked "BMS" must be installed in the cockpit that lights up, if the BMS shuts down the car Point out the BMS light on the <u>MoTeC</u> , which lights up when the BMS OK High Signal goes bad. Offer to turn the light on for them by turning the car off, disconnecting the accumulator LV loom and turning the car back on. Switch the car off to reconnect.	Inspect		

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EV CLASS - FUNCTIONAL TEST					
TRACTIVE SYSTEM ENERGISATION TEST				Reinspect	Passed
ASK THE TEAM TO CONNECT THE MSD AND UNLOCK THE TSMS LOCKOUT. THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES					
FT.4.1	GLVMS Function	TS only allowed to be powered up, when GLVS is powered up. Connect the Fluke 289 Multimeter to TSMPs and set to measure DC Voltage (600V range). Try to switch on Tractive System with GLVS Master switch in Off-Position. No voltage above 60VDC allowed at measurement points. Demonstrate whilst explaining what you are doing.	Ask team to demonstrate		
FT.4.2		TS only allowed to be powered up, when GLVS is powered up. Switch on Tractive System and then switch off GLVS Master switch. Tractive system must switch off as well - Watch measured voltage on scrutineer's Multimeter fall to below 60V DC. Demonstrate whilst explaining what you are doing.	Ask team to demonstrate		
FT.4.3	Pre-Charge Circuit	A circuit that is able to pre-charge the intermediate circuit to at least 90% of the current accumulator voltage before closing the second AIR has to be implemented. Use the graphing function of the Fluke 289 Multimeter to plot the voltage at the TSMPs during power up of the tractive system. Show that 90% voltage is reached before final AIR closes. Demonstrate whilst explaining what you are doing. Pack will be charged to 50% soc (400ish V) and listen for the second AIR to close	Ask team to demonstrate		
FT.4.4	Tractive System Voltage	The maximum tractive system voltage must be less than or equal to 600VDC at all times as measured by the Fluke 289 Multimeter. Measured Voltage: Demonstrate whilst explaining what you are doing. Should be around 400 V	Ask team to demonstrate		
FT.4.5	Accumulator Active Indicator	Accumulator Indicator Light or analogue voltmeter has to show if voltage above 60VDC is present outside of the accumulator container. Switch off the GLVMS. The Accumulator active indicator must continue to function even if the GLVMS is switched off or if the accumulator is removed from the vehicle. Demonstrate whilst explaining what you are doing. The AIL will remain on until the circuit has discharged below 60 VDC	Ask team to demonstrate		

EV CLASS - FUNCTIONAL TEST					
TRACTIVE SYSTEM ENERGISATION TEST				Reinspect	Passed
THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES					
FT.4.6	TSAL	The TSAL must be installed immediately under the highest point of the roll hoop. The TSAL must not be on top of the roll hoop. The TSAL must not touch the <u>drivers</u> helmet in any seating position. <u>Display</u>	Inspect		
FT.4.7		The TSAL must be continuously illuminated green in colour when the vehicle GLVMS is switched on but the voltage outside the accumulator is less than 60VDC <u>Display, may need to use the fluke to measure TS voltage to prove it is green under 60 VDC</u>	Inspect		
FT.4.8		Switch on the Tractive System. The TSAL must flash red in colour at between 2hz and 5Hz when the voltage outside the accumulator is more than 60VDC. The TSAL may stop functioning if the GLVMS is switched off. <u>Display the frequency of the TSAL flash using a stop watch (phone or something)</u>	Inspect		
FT.4.9		The TSAL must be clearly visible in bright sunlight to a person standing anywhere less than 3m from the vehicle with an eye height of 1.6m. The TSAL must not be blocked by wings or other bodywork. Minor obstruction from the roll hoop is acceptable. <u>Display</u>	Inspect		
FT.4.10		TSAL must be non-programmable. No part of the TSAL function must depend on a programmable device. <u>Display TSAL Discharge schematic, prepare to demonstrate the LV/HV Separation if asked.</u>	Ask Team to demonstrate	<u>D.g.</u>	
FT.4.11	Discharge Circuit	A circuit that <u>is able to</u> discharge the tractive system components outside of the accumulator container to less than 60V DC within 5 seconds of a shutdown must be fitted. Use the graphing function of the Fluke 289 <u>Multimeter</u> to plot the voltage at the TSMPs during power up of the tractive system. Show that the voltage falls to less than 60V DC in less than 5 seconds after the TSMS is switched off. <u>Demonstrate whilst explaining what you are doing, have a timer to time the 5 seconds. If they ask to see a schematic, show TSAL Discharge schematic (with discharge circuits)</u>	Ask team to demonstrate	<u>D.g.</u>	
FT.4.12	PDOC	A red indicator light marked "PDOC" must be installed in the cockpit that lights up, if the PDOC system shuts down the car. The indicator light must be visible to the driver in bright sunlight. The PDOC may be omitted if the pre-charge and discharge circuit is designed for continuous operation in a faulted state and will not adversely affect nearby devices <u>Point out light on MoTeC that turns on when the PDOC OK High Signal goes bad as our PDOC is not continuous. Offer to turn the light on for them by turning the car off, disconnecting the accumulator LV loom and turning the car back on. Switch the car off to reconnect. Can show HFR schematic if asked</u>	Inspect	<u>D.g.</u>	

EV CLASS - FUNCTIONAL TEST						
INSULATION MONITORING DEVICE TEST						
The IMD is one of the primary methods for monitoring the health of the accumulator and tractive system. The IMD works by continuously measuring the insulation resistance between the tractive system and the chassis. If it detects a low resistance path between the tractive system and the chassis, it will shut down the vehicle. The IMD is intended to form a detection system, providing early warning to the team that they have an insulation breakdown before a short circuit can form. The correct function of the IMD is critical to the operational safety of the vehicle and any misbehaviour should be reported to the EV Technical Inspection Captain for investigation.						
					Reinspect	Passed
THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES						
FT.5.1	IMD	A red indicator light marked "IMD" must be installed in the cockpit that lights up, if the IMD system shuts down the car. The indicator light must be visible to the driver in bright sunlight. <u>Point out light on MoTeC, that turns on when the IMD OK High Signal goes bad. Offer to turn the light on for them by turning the car off, disconnecting the accumulator LV loom and turning the car back on. Switch the car off to reconnect.</u>	Inspect			
FT.5.2		The team must provide a test resistor manufactured with insulated banana plugs with an appropriate resistor in place. The resistor must have an <u>appropriate</u> voltage and power rating and be encased in a <u>hard shell</u> enclosure. The resistor should have appropriate power rating and be installed in a <u>hard shell</u> enclosure (heat shrink/tape insulation is not acceptable). Check the resistance of the test lead with the <u>Fluke 289 Multimeter</u>. The resistance should be a minimum of: $R_{Test} = (\text{maximum TS voltage} * 250\Omega/V) - (2 * BPR)$ Measured Resistance: <u>Should be $(453.6V * 250\Omega/V) - (2 * 15k\Omega) = 83.4k\Omega$</u> <u>*BPR = the value of the Body Protection Resistors in Ohms</u> <u>BPRs are installed on HIP, present schematic if asked</u>	Inspect	<u>D.G.</u>		
FT.5.3		Activate Tractive System and connect <u>R_Test</u> between HV+ and GLVS ground. TS voltage as measured at the TSMPs must decrease below 60VDC in 5 sec, IMD may take up to 30s to react. Observe with the <u>Fluke 289 Multimeter</u>. <u>Demonstrate whilst explaining what you are doing, if they do not have a stopwatch, use a phone to time</u>	Check Shutdown Time with Stopwatch			
FT.5.4		Activate Tractive System and connect <u>R_Test</u> between HV- and GLVS ground. TS voltage as measured at the TSMPs must decrease below 60VDC in 5 sec, IMD may take up to 30s to react. Observe with the <u>Fluke 289 Multimeter</u>. <u>Demonstrate whilst explaining what you are doing, if they do not have a stopwatch, use a phone to time</u>	Check Shutdown Time with Stopwatch			

EV CLASS - FUNCTIONAL TEST					
FT.5.5		Remove the test resistor. The tractive system may not automatically return to active state after the IMD test resistor was removed or a BMS error disabled it. The Driver must not be able to reactivate the tractive system <u>Demonstrate whilst explaining what you are doing, try to turn the TSMS switch on and off, and explain that the only way to reactivate the TS is to power cycle LV or press the Hard Fault Reset button on the CEN which is mounted behind the firewall. Display the HFR board schematic</u>	Ask team to demonstrate	<u>p.g.</u>	

EV CLASS - FUNCTIONAL TEST						
SHUTDOWN SYSTEM TEST This section proves the correct function of all shutdown devices. The tractive system must fall below 60V within 5 seconds when the shutdown system is activated. With the Fluke 289 Multimeter monitoring the voltage at the TSMPs, have the team turn on the vehicle's LV and Tractive System then activate the shutdown mechanism. While completing this test, the items previously must continue to operate. Watch that the TSAL, Accumulator Active Indicator, precharge and discharge systems all continue to work correctly.						
					Reinspect	Passed
THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES TAKE CARE NOT TO OVERHEAT THE PRECHARGE OR DISCHARGE RESISTORS DURING TESTS						
FT.6.1	Shutdown System Test	Vehicle with tractive system live --> GLVS master switch off Demonstrate whilst explaining what you are doing	Ask team to demonstrate			
FT.6.2		Vehicle with tractive system live --> TS master switch off Demonstrate whilst explaining what you are doing	Ask team to demonstrate			
FT.6.3		Vehicle with tractive system live --> Left shutdown button off Demonstrate whilst explaining what you are doing	Ask team to demonstrate			
FT.6.4		Vehicle with tractive system live --> Right shutdown button off Demonstrate whilst explaining what you are doing	Ask team to demonstrate			
FT.6.5		Vehicle with tractive system live --> Cockpit shutdown button off Demonstrate whilst explaining what you are doing	Ask team to demonstrate			
FT.6.6		Vehicle with tractive system live --> Brake over travel switch off Demonstrate whilst explaining what you are doing	Ask team to demonstrate			
FT.6.7		Vehicle with tractive system live --> GLVS master switch off Demonstrate whilst explaining what you are doing	Ask team to demonstrate			
FT.6.8		Unmount inertia switch. Activate TS and shake the switch and check if TS is shutdown. TS is not allowed to reactivate without a manual reset e.g. by the driver Demonstrate whilst explaining what you are doing Use ratchet and allen key to remove bolts to demount the inertia switch and shake it, explain that the only way to reset it is to push the switch down and is out of reach from the driver. Remount with bolts.	Ask team to demonstrate			

EV CLASS - FUNCTIONAL TEST						
READY TO DRIVE TEST						
This section tests the on-board systems which are intended to prevent the vehicle running away if a component fails. The plausibility systems are expected to have a high degree of reliability and robustness as they may be the last line of defence from a collision. This section will require that the team put the vehicle into drive, so the wheels and hubs will turn under power. Keep all unnecessary people outside of the test bay and ensure that people are well clear of the any rotating components. The team may elect to have a driver inside the vehicle for this section if required. The driver needs to be in full race gear and must be able to egress the vehicle safely in the event of a fire, so pay attention to the height and stability of the vehicle and any obstructions in the inspection bay.						
					Reinspect	Passed
THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES						
FT.7.1	Ready to Drive	Only closing the shutdown circuit must not set the car to ready-to-drive mode. The car is ready to drive as soon as the motor(s) will respond to the input of the torque encoder/ acceleration pedal Demonstrate whilst explaining what you are doing, put the car into HV and press the accelerator pedal and explain there is no motor response as RTD is not enabled.	Ask team to demonstrate			
FT.7.2		Additional actions are required by the driver to set the car to ready-to-drive-mode e.g. pressing a dedicated start button, after the tractive system has been activated. One of these actions must include the brake pedal being pressed as ready-to-drive-mode is entered. It must be possible for the vehicle to be powered on, with the high voltage on, but with the vehicle not in drive. It is not acceptable for the action of the driver putting the vehicle in drive to simultaneously turn on the HV System and engage drive. Demonstrate whilst explaining what you are doing, explain that pressing the brake and the RTD button on the DEN will enable the car, ask that you can turn on HV and go into RTD, gentle spin the wheels.	Ask team to demonstrate			
FT.7.3		The car must make a characteristic sound, once but not continuous, for at least 1 second and a maximum of 3 seconds when it is ready to drive The sound level must be a minimum of 70dBA, fast weighting, in a radius of 2m around the car The used sound must be easily recognizable. No animal voices, song parts or sounds that can be interpreted as offensive will be accepted Demonstrate whilst explaining what you are doing. Put the car into ready to drive to display the noise. Offer to time if they do not have a stopwatch. If they wish to see a datasheet for the sounder, it can be found in the datasheet folder.	Ask team to demonstrate			
FT.7.4		Pressing the accelerator gently must cause the vehicle's wheels to rotate. Check all of the vehicle's driven wheels turn in the correct direction. Demonstrate whilst explaining what you are doing, explain we have a locked rear axle and gently spin the wheels using the accelerator (wheels off). They will spin in the right direction.	Ask team to demonstrate			

EV CLASS - FUNCTIONAL TEST					
DRIVER INPUT PLAUSIBILITY CHECKS				Reinspect	Passed
THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES					
FT.8.1	Torque Encoder / Brake Pedal Plausibility Check	Torque encoder is at more than 25% and brake is actuated simultaneously. The motors have to shut down. The motor power shut down has to remain active until the torque encoder signals less than 5% pedal travel, no matter whether the brake pedal is still actuated or not. 1. Ask the team to spin the motors by pressing the accelerator pedal. 2. While the wheels are spinning, ask the team to depress the brake pedal. 3. The wheels must stop turning and the torque (current) to the motors should immediately fall to zero 4. Ask the Team to release the brake pedal while continuing to hold the accelerator pedal down. The wheels must not turn. 5. Ask the team to completely release the throttle pedal and press it again. 6. The wheels may now turn. Demonstrate whilst explaining what you are doing, follow the above instructions	Ask team to demonstrate		
FT.8.2	Torque Encoder Plausibility Check	If an implausibility occurs between the values of two torque encoder sensors the power to the motor(s) has to be immediately shut down completely. It is not necessary to completely deactivate the Tractive System, the motor controller(s) shutting down the power to the motor(s) is sufficient Implausibility is defined as a deviation of more than 10% pedal travel between the sensors If three sensors are used at least two sensors have to be within 10% pedal travel, etc. 1. Check that driven axles turn by pressing the accelerator pedal 2. Disconnect at least 50% of the sensors and check that the power to the motors is shut down The sensor should be disconnected while the axles are turning. There must be no uncommanded throttle surges when the sensors are unplugged. Demonstrate whilst explaining what you are doing, follow the above instructions	Ask team to demonstrate		
FT.8.3		Repeat the above disconnecting the other throttle sensors. Demonstrate whilst explaining what you are doing, follow the above instructions	Ask team to demonstrate		

EV CLASS - FUNCTIONAL TEST					
DRIVER INPUT PLAUSIBILITY CHECKS				Reinspect	Passed
THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES					
FT.9.1	Brake System Plausibility Device (BSPD)	A standalone non-programmable circuit must be used on the car such that when braking hard (without locking the wheels) and when a positive current is delivered from the motor controller (a current to propel the vehicle forward), the AIRs will be opened. The current limit for triggering the circuit must be set at a level where 5kW of electrical power in the DC circuit is delivered to the motors at the nominal battery voltage. The action of opening the AIRs must occur if the implausibility is persistent for more than 0.5s. The team must devise a test to prove this required function during Electrical Tech Inspection. However, it is suggested that it should be possible to achieve this by sending an appropriate signal to the non-programmable circuit that represents the current to achieve 5kW whilst pressing the brake pedal to a position or with a force that represents hard braking. The preferred method for testing the BSPD system involves passing actual current from an external supply through the tractive system current sensor while the driver presses the brake pedal. Alternative test methods will require approval from the EV Technical Inspection Captain. The test must be an <u>end to end</u> test, meaning it is not acceptable for the team to simulate the 5kW by simulating the output of the current sensor, or using a substitute current sensor. To demonstrate, use the BSPD test loom and a power supply to put current into the coil plug on the side of the HIP. Display the BSPD circuit in the electrical schematic booklet and the image of the current sensor coiled and the current sensor. When at 1.25 A (with 10 turns on the sensor, will be 12.5 A through the sensor) $12.5 \text{ A} * 388.8 \text{ V} = 4.86 \text{ kW}$ and the brake is pressed, the BSPD will fault, indicating on the <u>MoTeC</u>. This test can be done in or out of HV. <u>EXPLAIN EVERYTHING YOU ARE DOING</u>	Visual Inspection If required, ask team to show BSPD Schematic. Ask team to demonstrate	D.g.	
FT.9.2		The Brake Plausibility Device may only be reset by power cycling the GLVS Master Switch or via a RESET button, located out of reach from the driver Display that the car will not go into HV now without power cycling GLVMS or pressing the HFR on the CEN which is mounted on the rear left of the chassis, inaccessible from the driver.	Ask team to demonstrate		

EV CLASS - FUNCTIONAL TEST					
ENERGY METER TEST					
The energy meter test needs to be done by the Energy Meter Captain.					
Reinspect Passed					
THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES					
FT.10.1		Energy Meter Captain has laptop connected Follow instructions	Ask team to demonstrate		
FT.10.2		Switch on the GLVMS and check the datalogger will communicate with the laptop Follow instructions	Ask team to demonstrate		
FT.10.3		Switch on the tractive system and rotate the wheels such that sufficient power is delivered to the motors to register on the energy meter. Follow instructions	Ask team to demonstrate		
FT.10.4	Energy Meter Test	The energy meter must show both positive voltage and correct current flow direction. Follow instructions	Ask team to demonstrate		
ASK THE TEAM TO TURN OFF THE VEHICLE, DISCONNECT THE MSD AND RE-LOCK THE TSMS LOCKOUT.					
SEAL TS COMPONENTS					
Reinspect Passed					
FT.11.1	Seal Tractive System components	Seal all important parts with tamper proof tape after the IMD test was passed successfully: i.e. Accumulator container, motor controller housings and any other enclosures containing Tractive System Voltage. Record items sealed:	Seal Items		
NOTES					

EV CLASS - FUNCTIONAL TEST

INSTRUCTIONS FOR SCRUTINEERS - END OF EV FUNCTIONAL TEST

If the team have passed all items in this section, go to the second page of this document and initial the relevant box, noting the date and time the team passed this section. Give the team their EV FUNCTIONAL sticker - the team may proceed to the next inspection station.

If the team have items requiring reinspection (did not pass all items), go to the second page of this document and initial the relevant box noting the time and date the team was released from the inspection. The team will need to rectify the reinspection items and may return for reinspection when they are ready. Reinspection is completed on a first come, first served basis, and is dependent on [Technical Inspection lane](#) availability. Teams with Scheduled Inspections times will be given priority over teams presenting for reinspection.

Once the team have been released from the inspection, make note of the team's progress on the central inspection record board.

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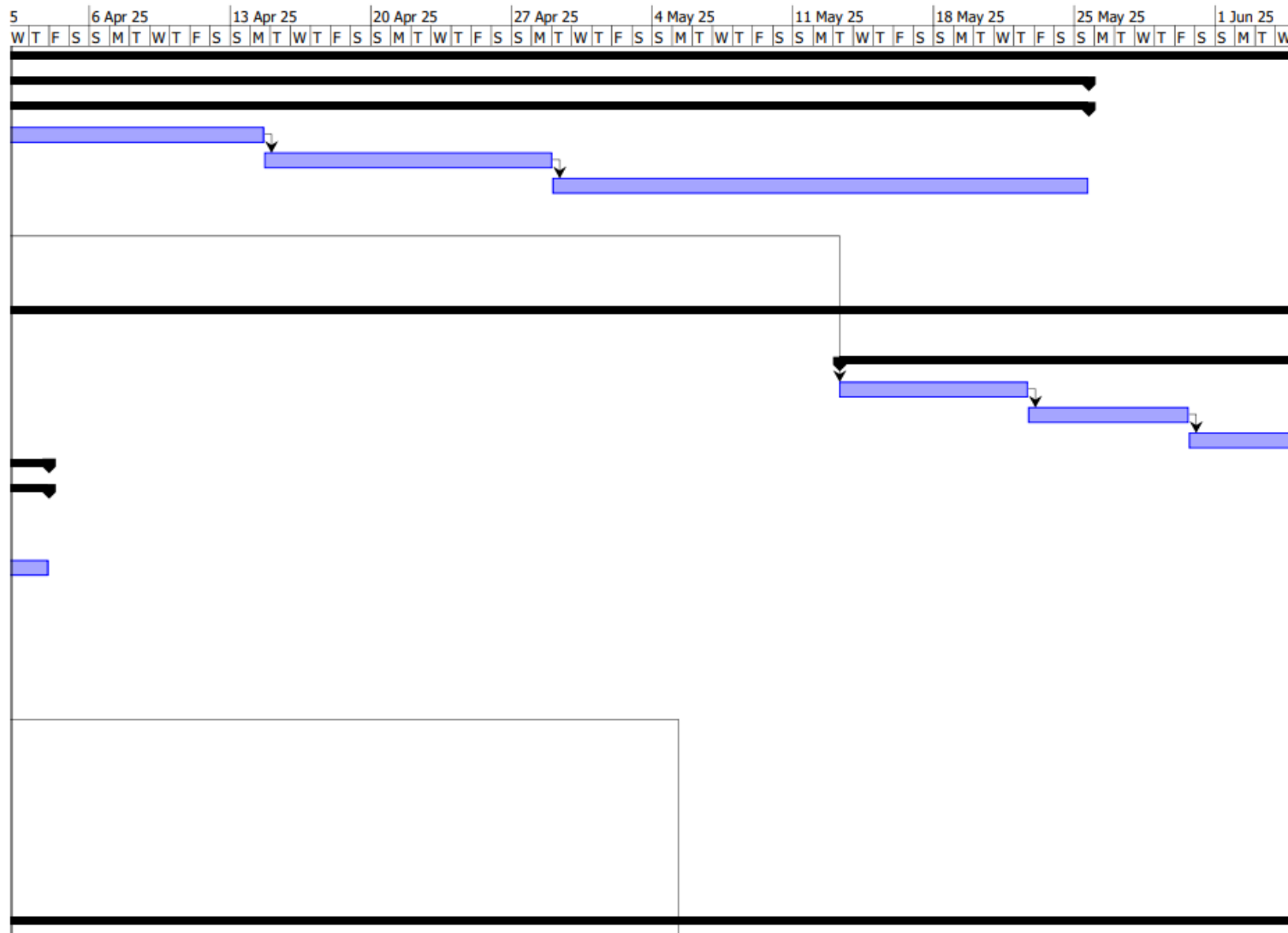
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17.5. Appendix E – MCHA Gantt Chart

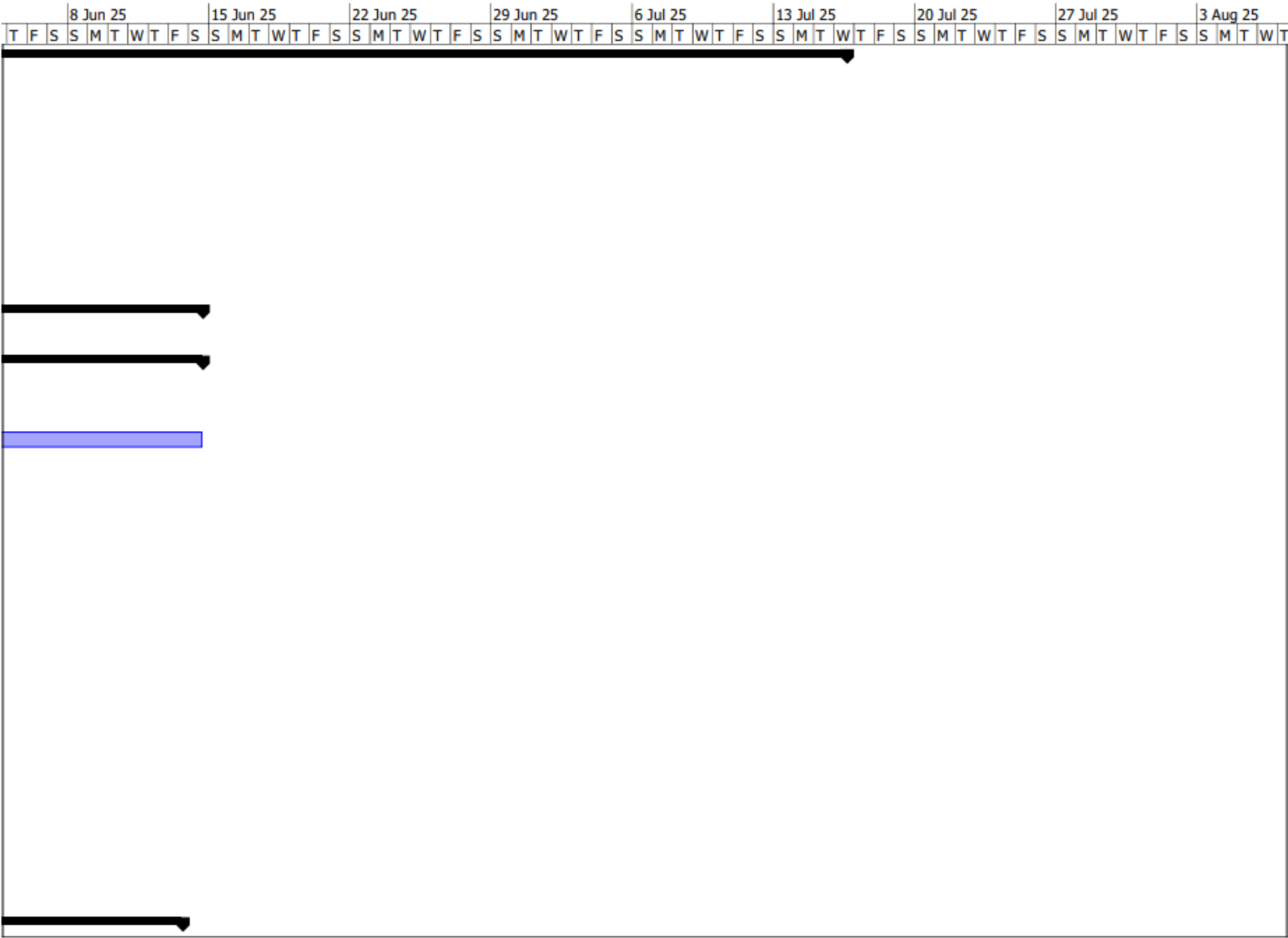
		Name	Duration	Start	Finish
1		☐ Design, Manufacture and Commission of Nodes	427 days?	24/02/25 8:00 AM	16/07/25 4:00 PM
2		☐ CEN	271.125 da...	24/02/25 8:00 AM	25/05/25 5:00 PM
3		☐ MOTHER Board V2	271.125 da...	24/02/25 8:00 AM	25/05/25 5:00 PM
4		Design	148.125 days?	24/02/25 8:00 AM	14/04/25 5:00 PM
5		Shipping	43.125 days?	14/04/25 5:00 PM	29/04/25 2:00 AM
6		Manufacture and Commission	79.875 days?	29/04/25 2:00 AM	25/05/25 5:00 PM
7		☐ BSPD V3	64 days?	26/02/25 8:00 AM	19/03/25 4:00 PM
8		Design	30 days?	26/02/25 8:00 AM	8/03/25 8:00 AM
9		Shipping	14 days?	12/03/25 8:00 AM	17/03/25 12:00 AM
10		Manufacture and Commission	8 days?	17/03/25 12:00 AM	19/03/25 4:00 PM
11		☐ Pedal Box Redesign	331.125 da...	24/02/25 8:00 AM	14/06/25 5:00 PM
12		Pedal Box Redesigned	11 days?	24/02/25 8:00 AM	10/03/25 5:00 PM
13		☐ PENV1.2	97.125 day...	13/05/25 8:00 AM	14/06/25 5:00 PM
14		Design	28.125 days?	13/05/25 8:00 AM	22/05/25 5:00 PM
15		Shipping	24 days?	22/05/25 5:00 PM	30/05/25 5:00 PM
16		Manufacture and Commission	45 days?	30/05/25 5:00 PM	14/06/25 5:00 PM
17		☐ HIP	116 days?	24/02/25 8:00 AM	4/04/25 12:00 AM
18		☐ HIP V1.3 PCB	116 days?	24/02/25 8:00 AM	4/04/25 12:00 AM
19		Design	32 days?	24/02/25 8:00 AM	7/03/25 12:00 AM
20		Shipping	14 days?	19/03/25 8:00 AM	24/03/25 12:00 AM
21		Manufacture and Commission	33 days?	24/03/25 12:00 AM	4/04/25 12:00 AM
22		☐ TSAL_Discharge V3	89 days?	24/02/25 8:00 AM	26/03/25 12:00 AM
23		Design	32 days	24/02/25 8:00 AM	7/03/25 12:00 AM
24		Shipping	14 days?	19/03/25 8:00 AM	24/03/25 12:00 AM
25		Manufacture and Commission	6 days?	24/03/25 12:00 AM	26/03/25 12:00 AM
26		☐ AIL V2.1	73.75 days?	27/02/25 8:00 AM	23/03/25 10:00 PM
27		Design	11 days?	27/02/25 8:00 AM	3/03/25 12:00 AM
28		Shipping	14 days?	13/03/25 8:00 AM	18/03/25 12:00 AM
29		Manufacture and Commission	6 days?	18/03/25 12:00 AM	20/03/25 12:00 AM
30		time till LVD arrives	11.75 days?	20/03/25 12:00 AM	23/03/25 10:00 PM
31		☐ LVD V3.2	68 days?	3/03/25 8:00 AM	26/03/25 12:00 AM
32		Design	18 days?	3/03/25 8:00 AM	9/03/25 8:00 AM
33		Shipping	14 days?	19/03/25 8:00 AM	24/03/25 12:00 AM
34		Manufacture and Commission	6 days?	24/03/25 12:00 AM	26/03/25 12:00 AM
35		☐ EXPANSION Board	307.125 da...	3/03/25 8:00 AM	13/06/25 5:00 PM

NU_Racing_MCHA_Gantt_Chart - page1





NU_Racing_MCHA_Gantt_Chart - page3



NU_Racing_MCHA_Gantt_Chart - page4

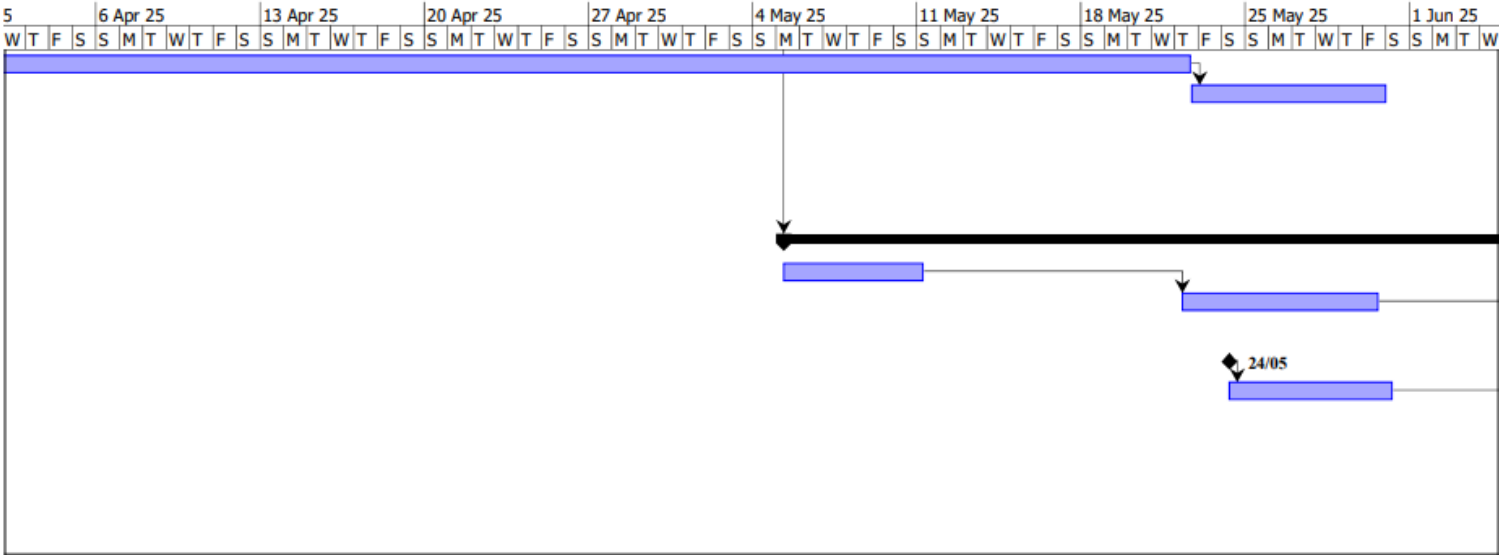
[illegible]

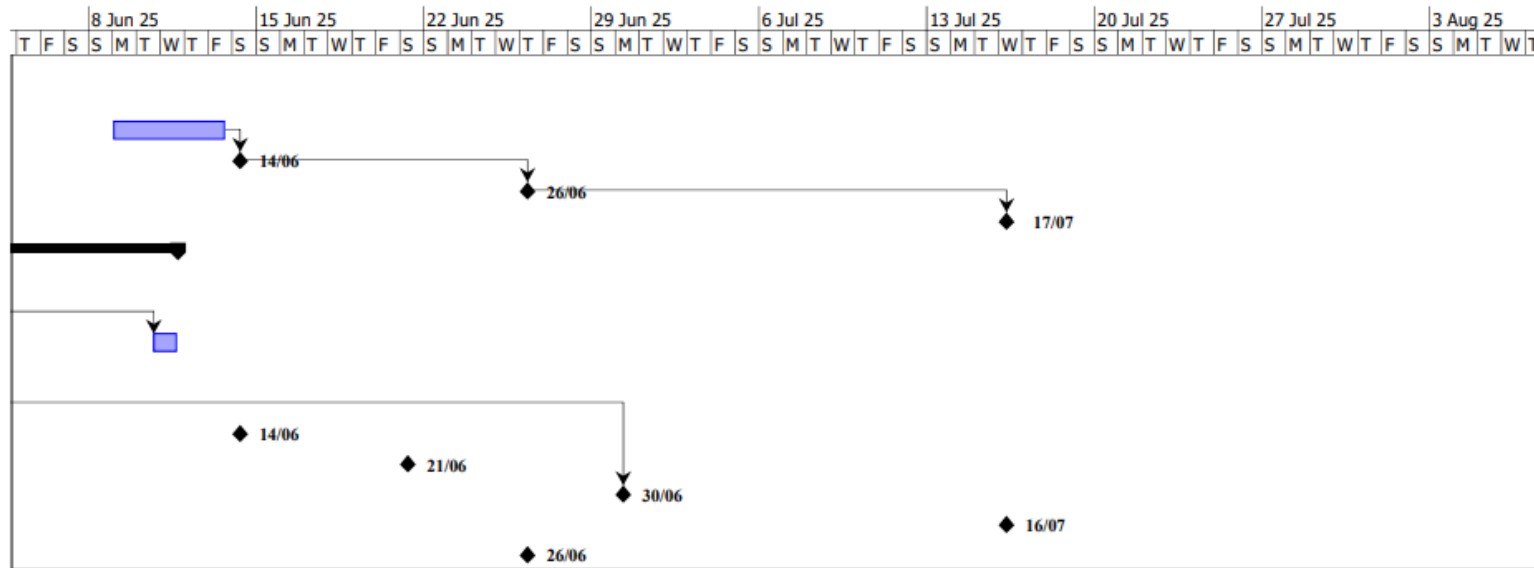
NU_Racing_MCHA_Gant...

		Name	Duration	Start	Finish
36		Design	241.125 days?	3/03/25 8:00 AM	22/05/25 5:00 PM
37		Shipping	25.125 days?	22/05/25 5:00 PM	31/05/25 2:00 AM
38		ManufactureandCommission	13.875 days?	9/06/25 2:00 AM	13/06/25 5:00 PM
39		Wheel turtle test with CANgate	0 days	14/06/25 8:00 AM	14/06/25 8:00 AM
40		Brake Temp sensor test	0 days	26/06/25 8:00 AM	26/06/25 8:00 AM
41		Wheel speed sensor test?	1 day?	16/07/25 8:00 AM	16/07/25 4:00 PM
42		 LVD V3.3	112.125 da...	5/05/25 8:00 AM	11/06/25 5:00 PM
43		Design	18 days?	5/05/25 8:00 AM	11/05/25 8:00 AM
44		Shipping	25.125 days?	22/05/25 8:00 AM	30/05/25 5:00 PM
45		Manufacture	3 days?	10/06/25 5:00 PM	11/06/25 5:00 PM
46		NU24 CEN Integration Test	0 days?	24/05/25 8:00 AM	24/05/25 8:00 AM
47		Fault Finding and Fixing	21 days?	24/05/25 8:00 AM	31/05/25 8:00 AM
48		NU24 LVD & HIP Integration Test	0 days	14/06/25 8:00 AM	14/06/25 8:00 AM
49		Potential Test: NU24 PEN & Expansion Board Integration Test	0 days	21/06/25 8:00 AM	21/06/25 8:00 AM
50		NU24 final drive	0 days	30/06/25 8:00 AM	30/06/25 8:00 AM
51		NU25 first drive (with old power box)	0 days	16/07/25 8:00 AM	16/07/25 8:00 AM
52		New uprights (on NU24), test brake temp sensors	0 days	26/06/25 8:00 AM	26/06/25 8:00 AM
NU_Racing_MCHA_Gantt_Chart - page6					



Predecessors	Resource Names	23 Feb 25							2 Mar 25							9 Mar 25							16 Mar 25							23 Mar 25							30 Mar 25		
		T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T		
36																																							
38																																							
39																																							
40																																							
27	Oscar Rudman																																						
43																																							
44																																							
46																																							
47																																							





[illegible]

Jacob Lukes

17.6. Appendix F – 2025 Tasks Completed by Mechatronics Team

	A	B	C	D	E	F	G	H	I
1	MCHA ITEMS								
2	Section	System	Task	Date Revised: Tuesday, 2 December 2025	Assignee	Goal Date	Completion Date	Category	Complete?
3		PEN	mount components to PEN so that it is shorter	COMPLETED	Jayson	Monday, 27 October 2025	Saturday, 25 October 2025	Compliance	Yes
4		Accumulator	M6 HV Accu Lid crimp	Completed Late	Oscar	Wednesday, 29 October 2025	Friday, 28 November 2025	Compliance	Yes
5		HIP	Fit TSAL Discharge board with 3.3K Ohm resistor to HIP	COMPLETED	Jayson	Wednesday, 29 October 2025	Monday, 13 October 2025	General	Yes
6		Sensors	Shock pots	Completed Late	Bunny	Wednesday, 29 October 2025	Sunday, 2 November 2025	General	Yes
7		Looms	order boots	Completed Late	Bunny	Friday, 31 October 2025	Wednesday, 12 November 2025	Compliance	Yes
8		HIP	Print new HIP enclosure	Completed Late	Ben A	Monday, 3 November 2025	Thursday, 6 November 2025	Compliance	Yes
9		HIP	*Pending Digkey order arrival* Test new HIP PCB on car at track day	COMPLETED	Mia	Monday, 3 November 2025	Friday, 31 October 2025	Compliance	Yes
10	AI.5.8	Accumulator	get photo of accu in car, showing sideward ventilation (tech inspection point AI.5.8)	Completed Late	Oscar/Ryan	Wednesday, 5 November 2025	Tuesday, 2 December 2025	Compliance	Yes
11		Accumulator	Hand cart! Fix brake! And grounding wire! And angle grind nut	Completed Late	Sam S	Monday, 3 November 2025	Saturday, 22 November 2025	Compliance	YES
12	AI.2.3	Tech	HV mat check	Completed Late	Lukes	Wednesday, 5 November 2025	Monday, 1 December 2025	Compliance	Yes
13	AI.2.2	Tech	Gloves check	Completed Late	Lukes	Wednesday, 5 November 2025	Monday, 1 December 2025	Compliance	Yes
14	AI.5.5	Accumulator	Precharge enclosure	Completed Late	Ryan	Wednesday, 5 November 2025	Tuesday, 2 December 2025	Compliance	Yes
15	SI.12.5	Tech	Photo of dash estop accessible when restrained	Completed Late	Lukes	Thursday, 6 November 2025	Wednesday, 3 December 2025	Compliance	Yes
16	SI.12.6	Tech	Video of brake pedal at full extension, showing that it flips the BOTS switch	Completed Late	Jethro	Friday, 7 November 2025	Tuesday, 25 November 2025	Compliance	Yes



17.7. [Appendix G – 2025 MCHA Systems Design Changes](#)

The 2025 MCHA Systems Design Changes document can be found in the NU Teams SharePoint

Design changes for 2025 MCHA system

Table of Contents

Design changes for 2025 MCHA system	1
1. Overarching Changes	2
2. PEN	2
3. DEN	2
4. CEN	3
5. HIP	4
6. LVD	4
7. AIL	5
8. Expansion Board	5

1. *Overarching Changes*

- Shutdown circuit traces
- Keyed connectors where possible

2. *PEN*

- Same
 - o Connector pinout
 - o Majority of circuitry
 - o Functionality of board
- Changes
 - o Zeners removed on analogue, low voltage signals on step down circuits for teensy
 - o Shape and mounting location
 - o Fuses on sensor inputs
 - 0.1 A
 - Chosen from powering sensors with power supply, there was no current seen (at a an accuracy level of 0.00 A), so this value was chosen
 - o 2mm shutdown traces
 - o Low pass filters on brake signal inputs
 - o LED on BOTs on teensy step down

3. *DEN*

- Same
 - o Everything
- Changes
 - o Nothing

4. CEN

- Same
 - o DT pinouts
 - o Teensy step downs
- Changes
 - o Interpose relay (from LVD to CEN)
 - o Shape and mounting location of PCB
 - o MOSFETs for cooling, brake light and sounder switching
 - o Fuse sizes for cooling, brake light and sounder
 - o Teensy pinout
 - o 2mm shutdown traces
 - o Keyed AT connectors for all 8 pins, and 6 pin connectors
- BSPD
 - o Same
 - Functionality of the board
 - o Changes
 - Pot for current sense
 - 1206 capacitor on current sense filter
 - o Recommendations
 - Use multi-turn potentiometer for tuning, surface mount better for vibrations tho
 - Fix back silkscreen
- HFR
 - o Same
 - Everything
 - o Changes
 - nothing

5. HIP

- Same
 - o DT pinouts
 - o Fill area for 12V_RAW
 - o Overall functionality of the PCB
- Changes
 - o EM and DCDC are HV rated connectors
 - o MC, Accu, DCDC, EM all have interlocks
 - o New master switches
 - o HVD naming to MSD
 - o Micro fit molex to Mini fit molex for HV sense connections
 - o Component layout
- TSAL Discharge
 - o Same
 - All of circuitry
 - o Changes
 - Red
 - Move bolt hole for discharge resistor heatsink
 - Traces for switched red and green
 - o Recommendations
 - Thermistor placement

6. LVD

- Same
 - DT mounting position
 - DT pinouts
 - Fits within lid constraints
- Changes
 - Interlock circuit
 - Teensy pinout
 - Teensy code
 - DBC and MoTeC updates
 - 22 pin molex split into 3 molex connectors
 - Fan switching circuit
 - Relay to 2 sets of switching mosfets
 - Precharge, BMS and IMD fuse sizes
 - Teensy pinout
 - Fan control pins
 - Remove interpose circuit

7. AIL

- Same
 - o Turn on a light above 60 V
- Changes
 - o Mounting
 - o Efficient circuitry
 - o Works

8. Expansion Board

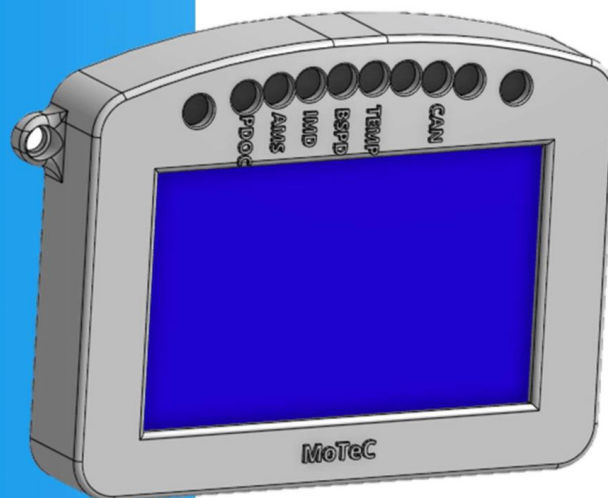
- Same
 - o Logging sensor data
- Changes
 - o fixes to CAN lines and GND plane
 - o Shock pots (have)
 - o wheel speed sensors (order 3 more)
 - o steering angle sensor (have)
 - o SWEN functionality
 - dials for torque
 - switch for regen
 - idk how you'll make sure these things are in the correct position after unplugging the board from the car
 - Programmable button
 - o 3.3v regulator
 - o LCD screen
 - o temp sensors for radiator
 - temp sensors are already in the radiator
 - o DBC file update
 - o MoTeC update

17.8. Appendix H – MoTeC Zero-to-Hero



MoTeC Zero-To-Hero

ZERO-TO-HERO



Contributors:
Jacob Lukes
Jayson Mills

2025





MoTeC Zero-To-Hero



Introduction – PLEASE READ

The information contained within this document is a combination of knowledge over the last 3 or 4 years. From the author's knowledge, this information has not been recorded in any NU Racing reports.

This Zero-To-Hero has been made on a Windows 11 machine. The information should carry to Mac and Linux machines, if Dash Manager, Display Creator, T2 and i2 all have installers for Mac and Linux. The sections requiring folder searching will be the main difference.

As of writing, this Zero-To-Hero is saved to the NU Teams Sharepoint, the author hopes that this Sharepoint still exists. Once this Zero-To-Hero is finished (as finished as the author can get it) it will also be saved to the 'Racing-MoTeC' repository. If edits are made to this file, care should be taken to ensure that these versions remain the same, to avoid new information not being followed correctly.

Contributors: Jacob Lukes

NU Racing

i



MoTeC Zero-To-Hero



Ethernet Access Point 42



Racing-MoTeC Repository

The place for all Racing MoTeC related information. Without this repo installed on the computer that's being used, none of the steps given here can be done

Installing Racing-MoTeC Repository

- Clone from Github (ask Mechatronic Department Leader or Chief Engineer for access)
- Usually, it is encouraged to make a separate folder for all Github repos, that is outside of the OneDrive folder
- Once successfully cloned, the below steps can be followed

Setting up MoTeC for a Track Day

Track Day Setup Checklist

- ☐ Pack MoTeC at TA
- ☐ Set up laptop and clear data from the car
- ☐ Connect the transceiver box and test live data
- ☐ Power the lap beacon
- ☐

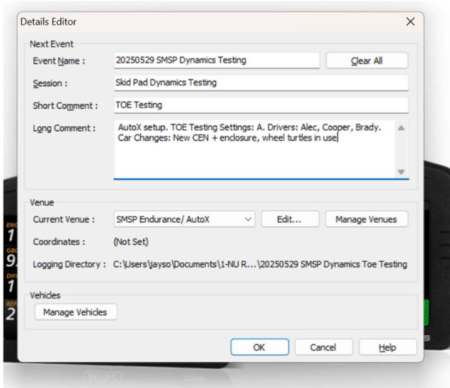
Clearing Data Logger at the start of the Day

Prior to running the car, the logging memory within the dash must be cleared to eliminate any excessive power cycles or TA testing being included within the track day logging files. To do this the data must be pulled (see Data Manager- Pulling Data) and the option for clearing logging memory selected. The directory location of this initial pull may depend on the extent of the prior data. If extensive relevant data is present, it may be preferable to save this data in a separate TA testing folder under with the Racing MoTeC GitHub repo 'Racing-MoTec > Logged-Data > NUXX'. Otherwise, this data may be saved under the track day folder with the session comment "Initial Clear". For each track day or testing event, create a new folder within 'Racing-MoTec > Logged-Data > NUXX' with the convention of YEARMONTHDATE "Track Day Title" (e.g. "20250524 Uon LV Systems Track Day").

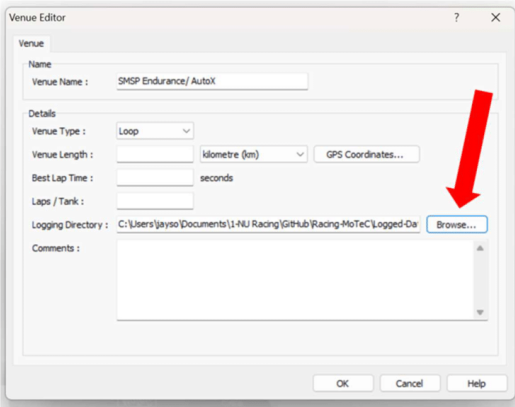
The initial clear is a good opportunity to set up the **event details**, venue and vehicle for the entire day. Every time you get and clear logged data you set the event details of that individual pull but having most of this prefilled will save time. The event details should clearly define the testing that is commencing, any changes and issues with the car, who are the drivers and anything else important to note when looking at the data. An example of filling out these fields is demonstrated below.



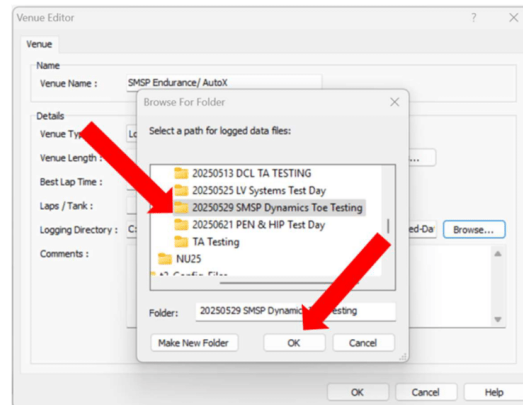
MoTeC Zero-To-Hero



The venue must also be set as shown above. Most importantly the logging directory MUST be selected (sets the folder where the files save). Do this by selecting Edit... in the venue area where the screen below will appear. Hit Browse... in the logging directory field and select the desired folder. Click OK to confirm your folder selection.

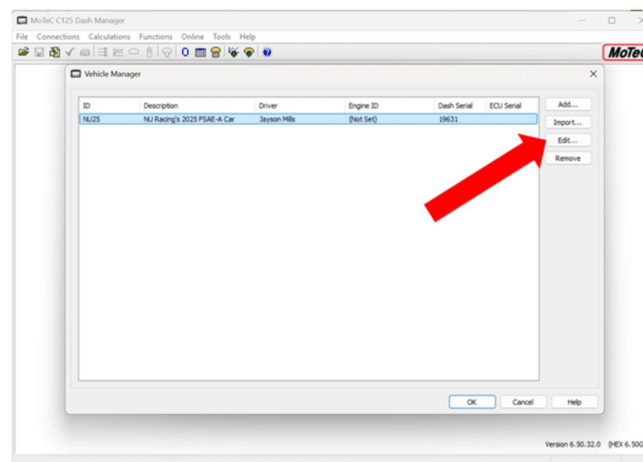


MoTeC Zero-To-Hero



You may now click OK on the venue editor. Double and Triple check this logging directory is properly selected by re-entering the venue editor and checking the file path. You do not want to be saving data throughout the day into a random folder... embarrassing.

In the vehicle section select Manage Vehicles and the menu depicted below will appear. From here a vehicle can be added, imported, edited or removed. If the car you are trying to select it not present, press Add... to create a new vehicle. Otherwise use Edit... to make any changes.



In the edit screen, the car's details may be filled in. Most important fields are the Vehicle ID, the dash serial and driver names. It should be noted that for each data pull, the driver should be set to the first driver and new drivers are added using this menu. This menu is shown below. Hit OK when finished.



MoTeC Zero-To-Hero



Gear	1	2	3	4	5	6	7	8	9	10
Ratio										

With the data log cleared and the event details filled out, the reader may then move on to other steps. It should be reiterated that for each data pull the event details are set for the data logged before that pull. Use this to your advantage by changing any event fields that are useful in discriminating between data files. For example add comments such as “BMS fault POA07 during 2nd enduro run”, “ CEN sounder not operational”, “First Driver: Alec (Regen on)” etc.

Transceiver Box

T2

i2

Dash Manager

Dash manager is used for all communication with the C125 MoTeC Display

Configuring Dash Manager (First Time Set-up)

MoTeC Zero-To-Hero

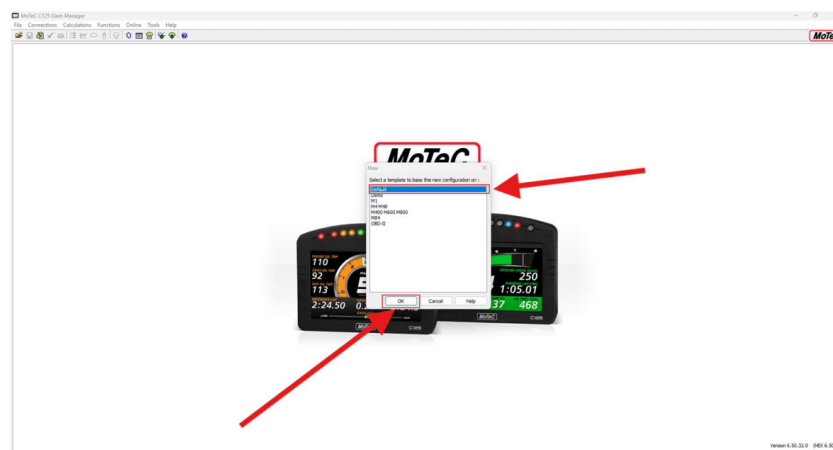


Creating a Configuration File

- Select File in the top right-hand corner
 - o From the dropdown, select 'New'



- Select the desired template, usually Default is fine, then select 'Ok'

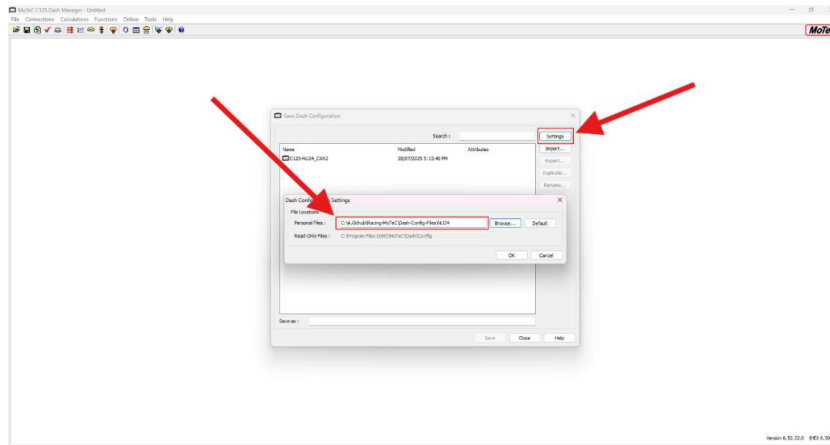


- The new file will be loaded
- To save changes and give the new config a name, select 'File' in the top right-hand corner again
 - o From the dropdown, select 'Save As'. A menu titled 'Save Dash Configuration' should appear
- To ensure that the file path for the new config file is correct, select 'Settings'
 - o From there, ensure that the file path is set to the 'Dash-Config-Files\'car_name'

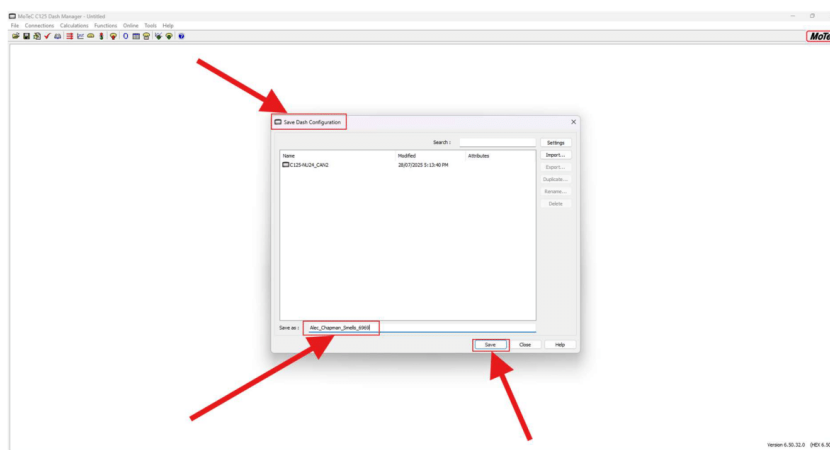
MoTeC Zero-To-Hero



- TODO: the author wants to change this so that it isn't a dash config per car, instead be one main config file that all the NU Racing electric cars use. This requires one main DBC file that doesn't get a new iteration each year (etc, EV3, NU23, NU24, ...) which the author also hasn't done yet



- Give the config file a name, then click 'Save'
 - o This should save it to the correct folder. Double check by going into the folder and checking that it has been saved there



MoTeC Zero-To-Hero



- Then make sure to click 'Save File' in the top left-hand corner
 - o Dash Manager is a funky program, and if this isn't done then changes won't update through the whole program correctly





MoTeC Zero-To-Hero

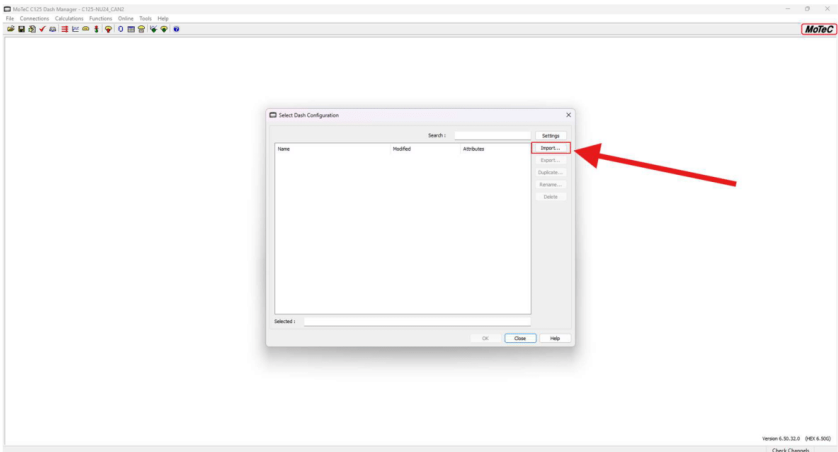


Opening an Existing Configuration File

- Open MoTeC C125 Dash Manager on a laptop/PC
- Select 'Open File'



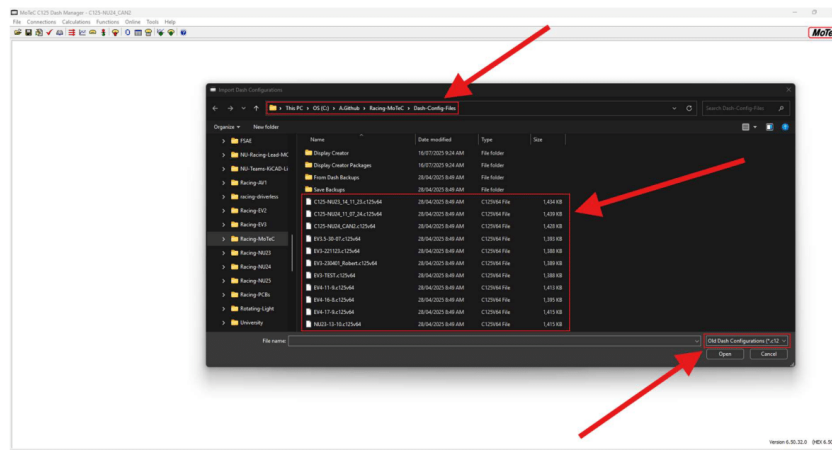
- Select 'Import...'



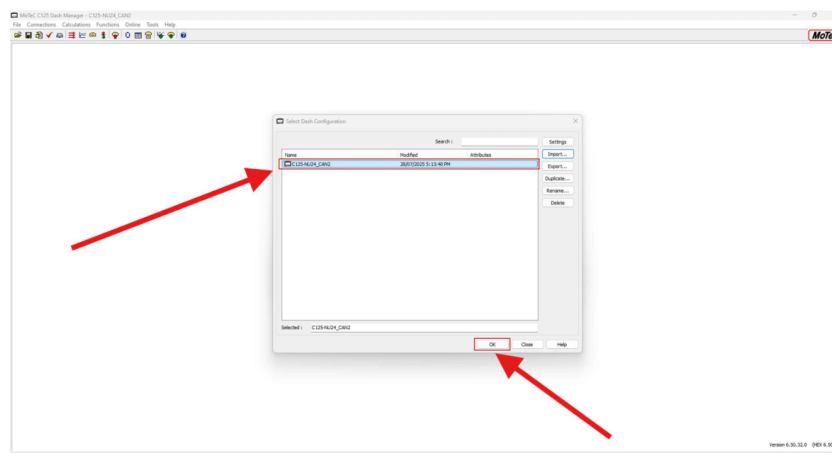
MoTeC Zero-To-Hero



- Navigate to 'Racing-MoTeC > Dash-Config-Files'
 - o See the Racing-MoTeC Repository section if this repo is not yet installed from Github
 - o If there are no files present, the current config file has been made with an older version of the config file type (pre v65), to fix this, ensure that the file type is set to 'Old Dash Configurations (*.c125v15...*.c125v64)' in the lower right-hand corner
 - o Once the files are visible, select the desired config file. For this demonstration, the 'C125_NU24_CAN2.c125v64' will be used



- Once desired config file is selected and imported, it should appear as shown below, then simply select it and click 'Ok', it will now be loaded into Dash Manager



MoTeC Zero-To-Hero



- Then make sure to click 'Save File' in the top left-hand corner
 - o Dash Manager is a funky program, and if this isn't done then changes won't update through the whole program correctly



MoTeC Zero-To-Hero



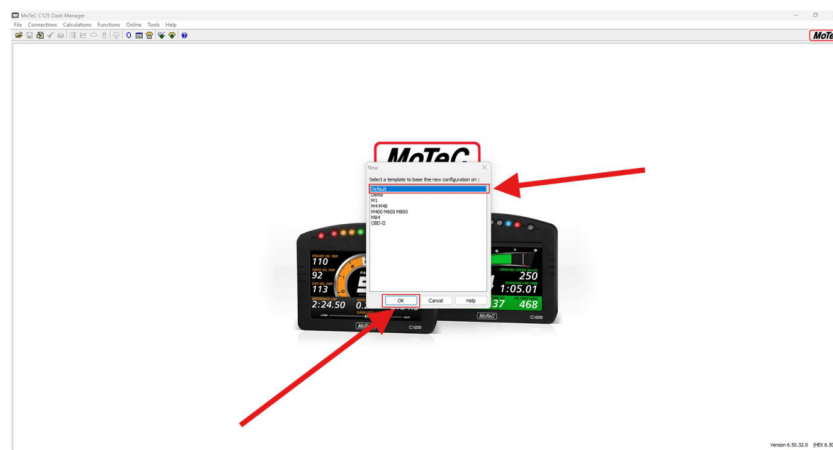
Adding Messages

After opening a configuration file (see the Creating a Configuration File

- Select File in the top right-hand corner
 - o From the dropdown, select 'New'



- Select the desired template, usually Default is fine, then select 'OK'

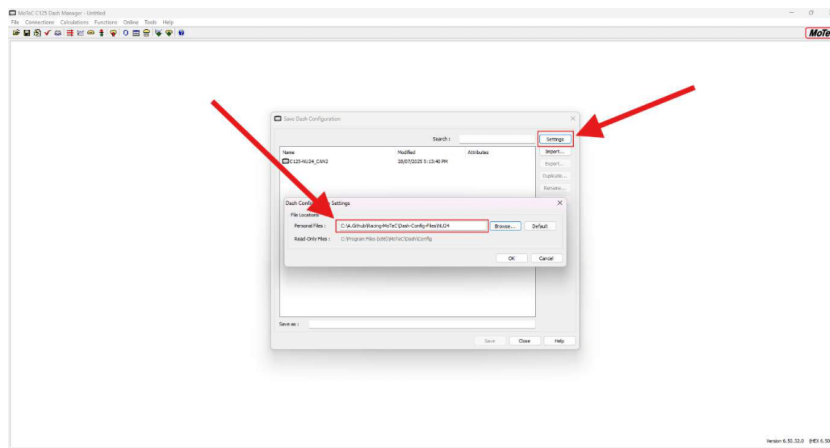


- The new file will be loaded
- To save changes and give the new config a name, select 'File' in the top right-hand corner again
 - o From the dropdown, select 'Save As'. A menu titled 'Save Dash Configuration' should appear
- To ensure that the file path for the new config file is correct, select 'Settings'

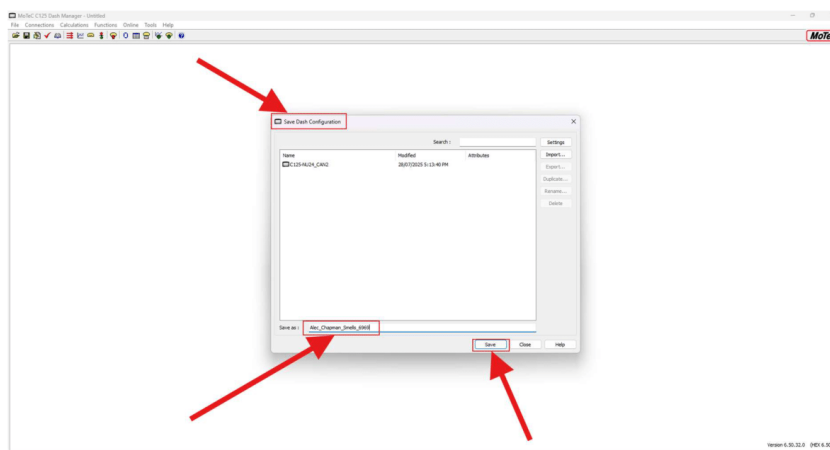
MoTeC Zero-To-Hero



- From there, ensure that the file path is set to the 'Dash-Config-Files\'car_name'
 - TODO: the author wants to change this so that it isn't a dash config per car, instead be one main config file that all the NU Racing electric cars use. This requires one main DBC file that doesn't get a new iteration each year (etc, EV3, NU23, NU24, ...) which the author also hasn't done yet



- Give the config file a name, then click 'Save'
 - This should save it to the correct folder. Double check by going into the folder and checking that it has been saved there



MoTeC Zero-To-Hero



- Then make sure to click 'Save File' in the top left-hand corner
 - o Dash Manager is a funky program, and if this isn't done then changes won't update through the whole program correctly



MoTeC Zero-To-Hero



- Opening an Existing Configuration File section for more information), the following sections can be followed

To a new Config File...

- Follow the same steps as To an existing Config File..., but select all the messages in the DBC that is wanting to be used.

To an existing Config File...

Communications Setup

- Select 'Connections'

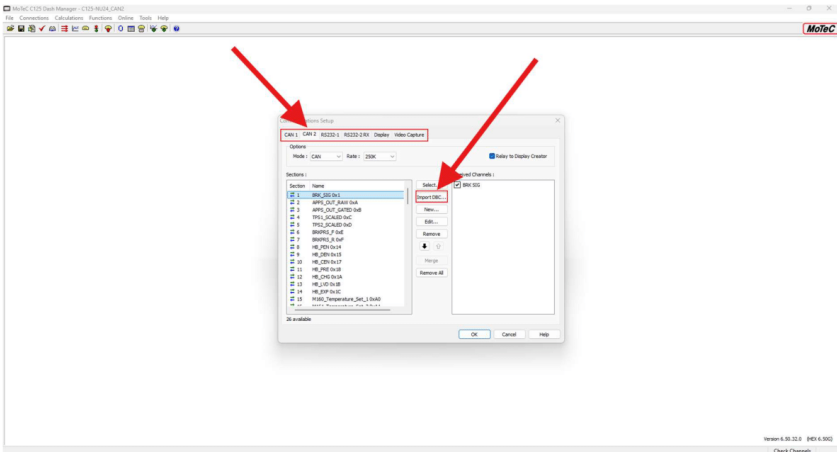




MoTeC Zero-To-Hero



- From the drop down, click 'Communications...'
- Select the desired communication protocol
 - o At the date of writing, NU Racing uses CAN 2 on the C125 due to issues occurring with CAN 1 in 2024, which has now been fixed, just haven't moved back to CAN 1 (as of July 2025)
- Import DBC

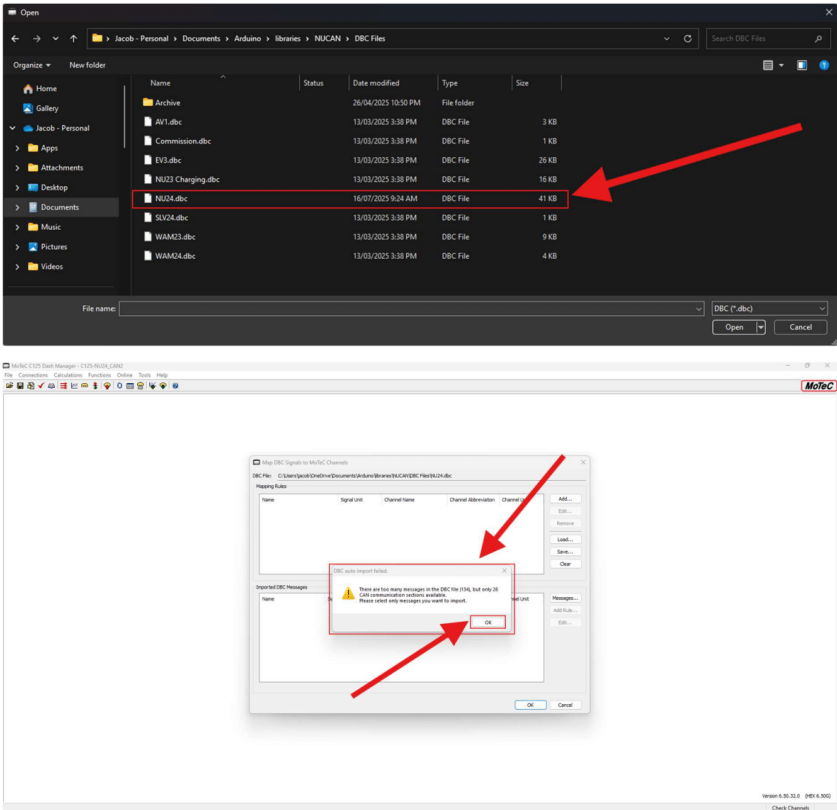




MoTeC Zero-To-Hero



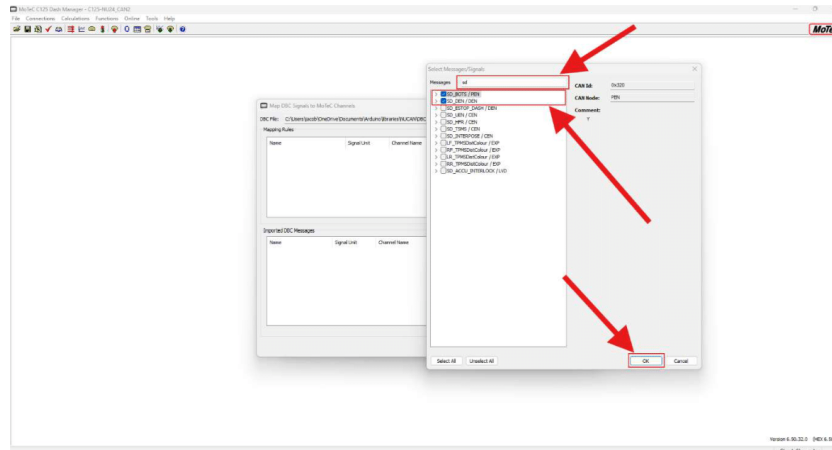
- Navigate to the desired DBC with the new message(s) (e.g. NU24)
 - o There should be an error saying that there are too many messages (if adding a newly added message to the current DBC), just click Ok



MoTeC Zero-To-Hero



- Select the designed message(s) to import
 - o Search button up the top can be used to quickly find the new signal
 - o Just a message or multiple messages can be added at the same time
 - o Once the desired message(s) are selected, click 'OK'



- The message(s) will then be loaded into the 'Imported DBC Messages' box
 - o If there is a double up, then the 'Channel Name' will have a yellow circle with an exclamation mark in the middle. If this is the case, uncheck the box next to either the signal or the message to not import the duplicated message(s). If this is done then there will be message conflicts within the MoTeC (which is not a fun thing to fix)
 - o In Dash Manager, Signals are called Channels
 - o If this occurs when it shouldn't, check the signal names of the message inside the DBC, as the Signal name is what is being duplicated and having issues, not the name of the message.
 - o This can be checked and edited within the DBC. Open the DBC in a text file format (if you hate yourself), or through Kvaser Database Editor. See Alec Chapman's 'NUCAN ZERO TO HERO' (hopefully located either in the Sharepoint, or in the NUCAN Repository) for how to read and open a DBC using Kvaser Database Editor
 - Download link for Kvaser Database Editor (chose not to include a link to a specific version as at the time of writing, the most recent version is V3. So just search for Kvaser Database Editor at the following link): <https://kvaser.com/download/>



MoTeC Zero-To-Hero



MoTeC CAN Manager - CTS-M24-0480

File Edit View Tools Help

MoTeC

Map DBC Signals to MoTeC Channels

DBC File: C:\Users\jacob\OneDrive\Documents\Arduino\libraries\NUCAN\DBC Files\NU24.dbc

Mapping Rules

Name	Signal Unit	Channel Name	Channel Abbreviation	Channel Unit	Abb...
					ADD...
					EDIT...
					Remove
					Save...
					Clear

Imported DBC Messages

Name	Signal Unit	Channel Name	Channel Abbreviation	Channel Unit	Messages...
SD_20V1_100V	Good/Bad	SD_20V1_100V	Good/Bad	Good/Bad	ADD...
SD_20V1_100V	Good/Bad	SD_20V1_100V	Good/Bad	Good/Bad	EDIT...

Messages selected: 2, Sections available: 28

OK Cancel

Version: 6.50.20.0 [2014.5.30]

PN Kvaser Database Editor [C:\Users\jacob\OneDrive\Documents\Arduino\libraries\NUCAN\DBC Files\NU24.dbc]

File Edit View Tools Help

DBC Overview

Messages & Signals

Messages

ID	Name	ID Decimal	Frame Format	DLC	TX Node	xmme
96	M164_Digital_Input_Status	164	Standard	8	INV	
97	M163_Analog_Input_Voltages	163	Standard	8	INV	
98	M164_Temperature_Set_3	162	Standard	8	INV	
99	M161_Temperature_Set_2	161	Standard	8	INV	
100	M160_Temperature_Set_1	160	Standard	8	INV	
101	M174_Firmware_Info	174	Standard	8	INV	
102	M175_Diag_Data_Message	175	Standard	8	INV	
103	BMS_Current_Limit	514	Standard	8	BMS	
104	M177_Torque_Info	176	Standard	8	INV	
105	M177_Torque_Capability	177	Standard	8	INV	
106	SD_80TS	800	Standard	1	PEN	Y

Signals of Selected CAN Message

Name	Type	Byteorder	Mode	Bitpos	Length	Factor	Offset	Minimum	Maximum	Unit	xmme	Values
1 SD_80TS	Unsi...	Intal	Signal	0	1	1	0	0	0	1	Good/Bad	

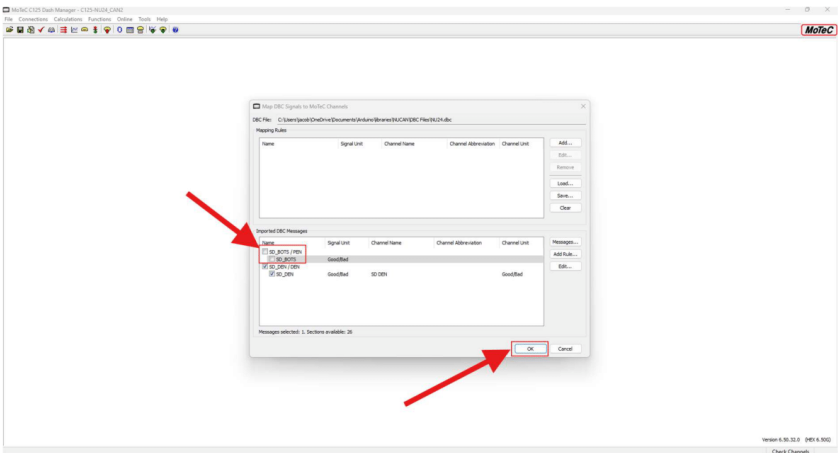
Message Output



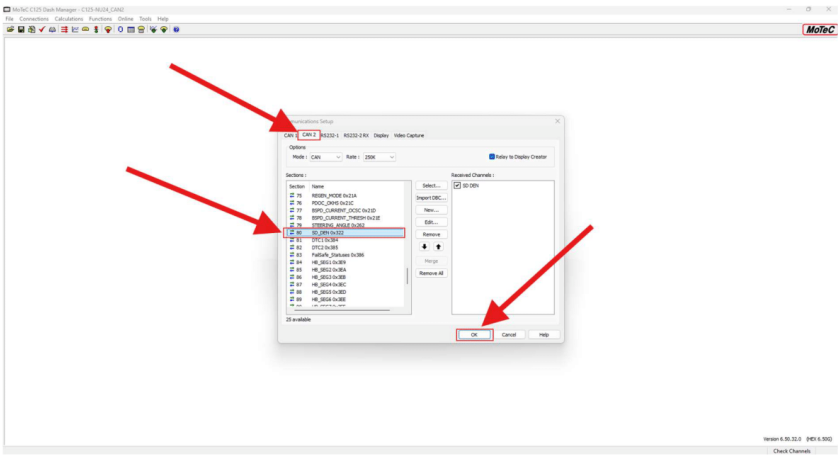
MoTeC Zero-To-Hero



- Once the desired message(s) are selected, click 'Ok'



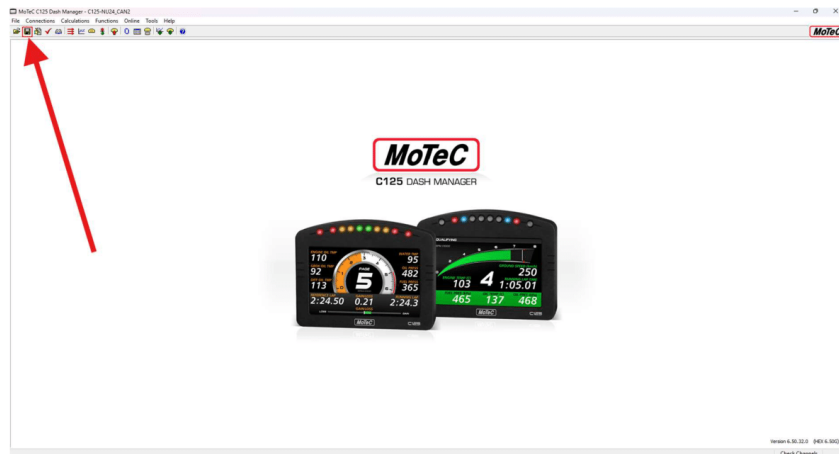
- They should now show up in the Communications Setup page under the desired protocol
- Now click Ok again



MoTeC Zero-To-Hero



- Then make sure to click 'Save File' in the top left-hand corner
 - o Dash Manager is a funky program, and if this isn't done then changes won't update through the whole program correctly





MoTeC Zero-To-Hero

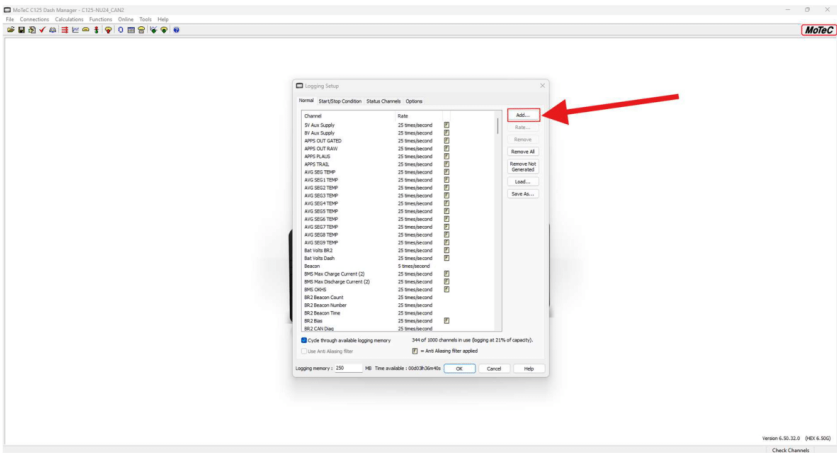


Logging Setup

- Setup Logging



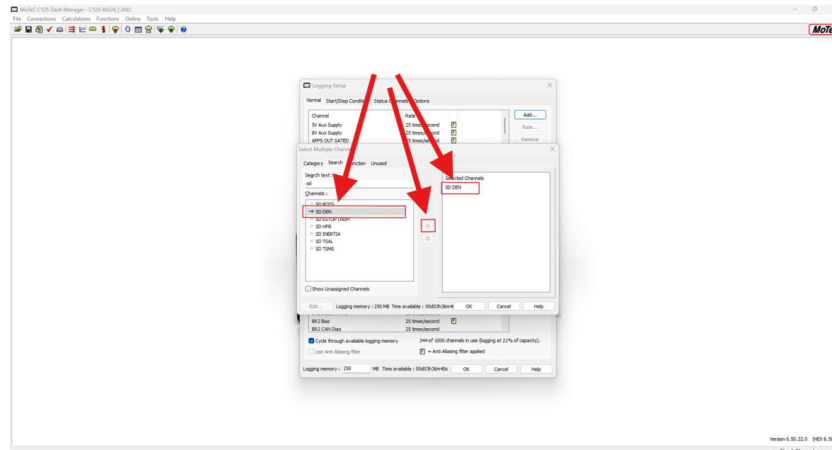
- Add (top right)



MoTeC Zero-To-Hero



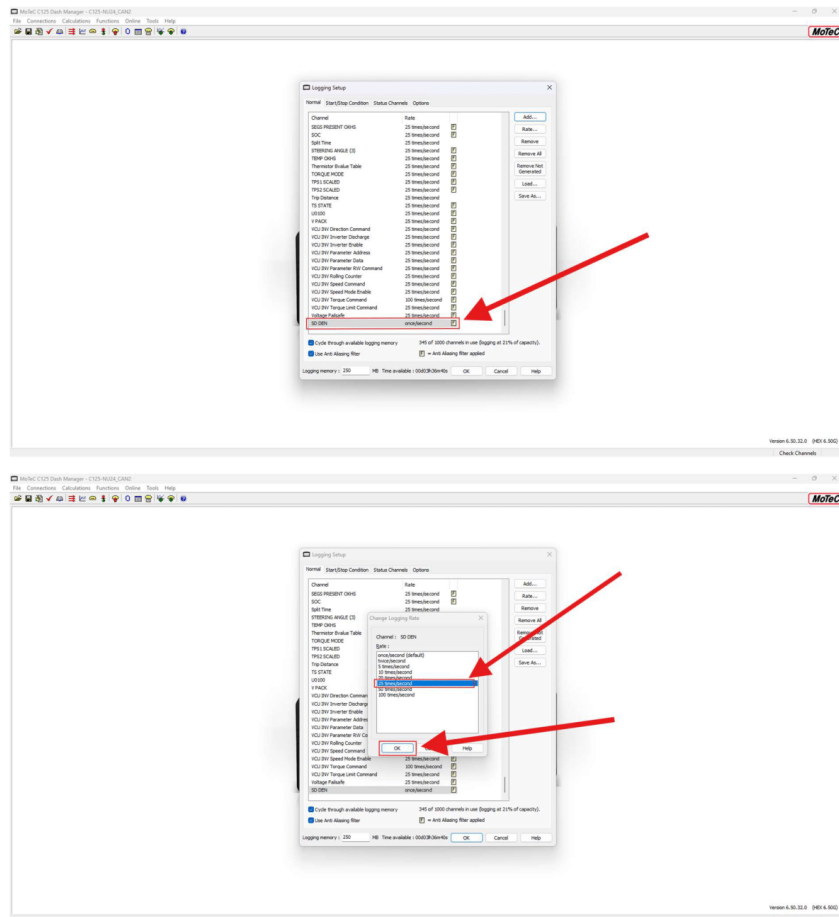
- Find the message(s) that was just added to Communications Setup
 - o New messages will have a blue arrow to the left of them, otherwise just search for the new message
- Click the arrow point to the right in the middle of the window
 - o The message(s) should now be in the 'Selected Channels' box
- Click Ok to close out of that window



MoTeC Zero-To-Hero



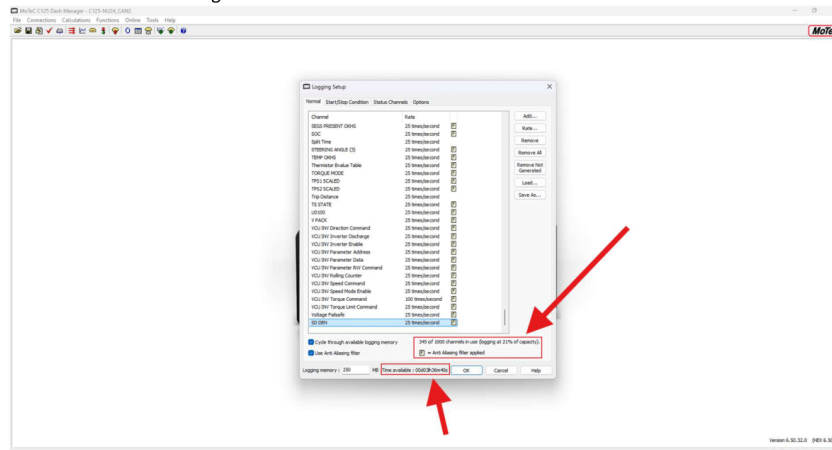
- Scroll to find the new added message(s)
 - o Either double click the message(s) or select 'Rate...' at the top right
 - o Change the rate to the desired value
 - For most messages, a rate of 25 times/second is quick enough for logging purposes. Some OEM components may send these messages quicker, or require them to be sent quicker, for example, the VCU INV Torque Command shown below is set to 100 times/second. If unsure, consult Mechatronic Department Lead or Chief Engineer
 - o Select 'Ok'



MoTeC Zero-To-Hero



- Take note of the number of channels used, capacity, and 'Time available' shown at the bottom. If channels used or capacity are nearing their limit take note and inform the Mechatronic Department Leader or Chief Engineer
 - o If Time available is nearing to zero, inform the Mechatronic Department Leader or Chief Engineer



- Click 'Ok' and to leave the Logging Setup
- Then make sure to click 'Save File' in the top left-hand corner
 - o Dash Manager is a funky program, and if this isn't done then changes won't update through the whole program correctly





MoTeC Zero-To-Hero



Telemetry Setup

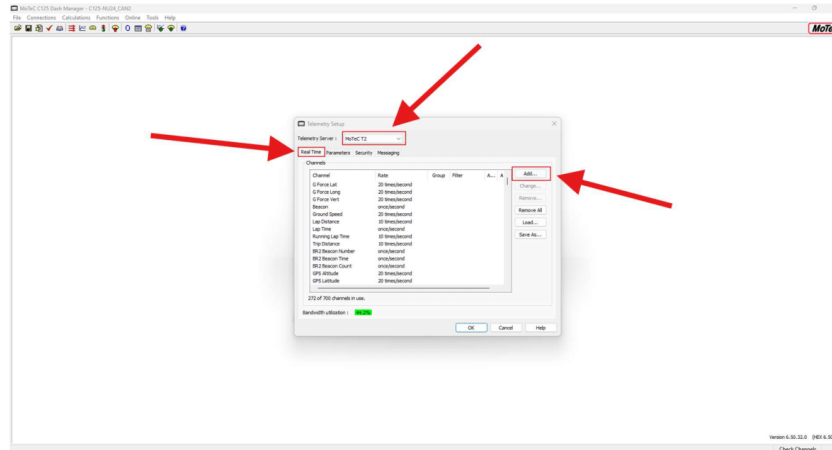
- Select 'Functions', then from the dropdown, select 'Telemetry...'



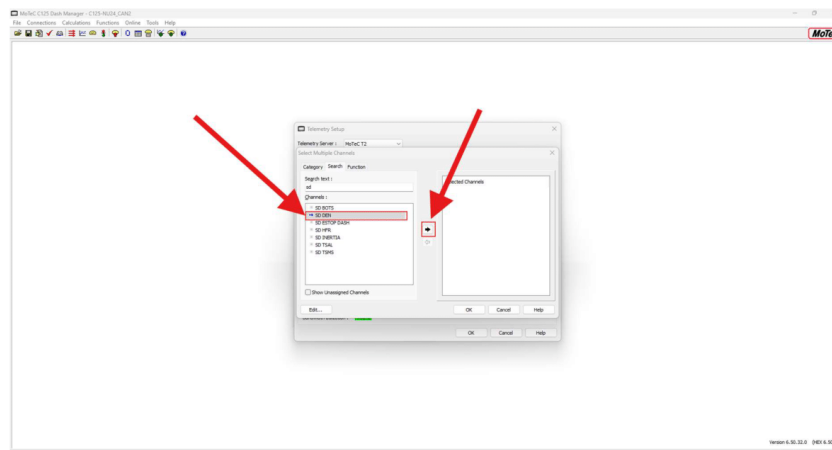
MoTeC Zero-To-Hero



- Ensure MoTeC T2 is selected in the 'Telemetry Server' option
- Ensure the Real Time tab is selected
- Click 'Add...'



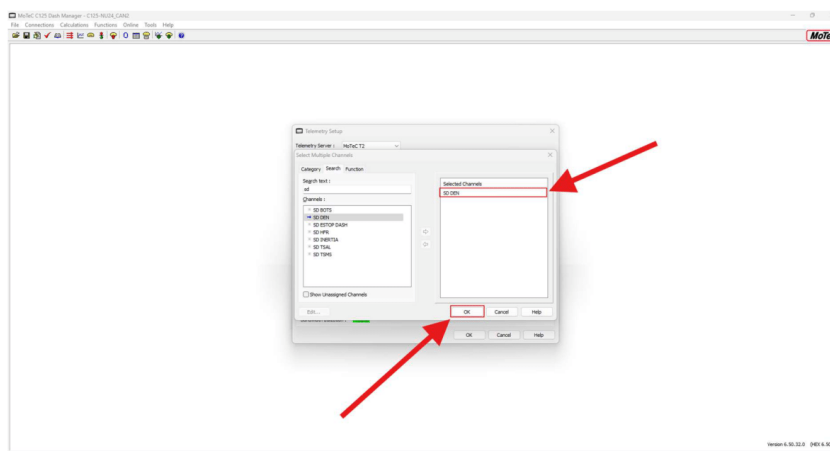
- Find the desired message(s) to add



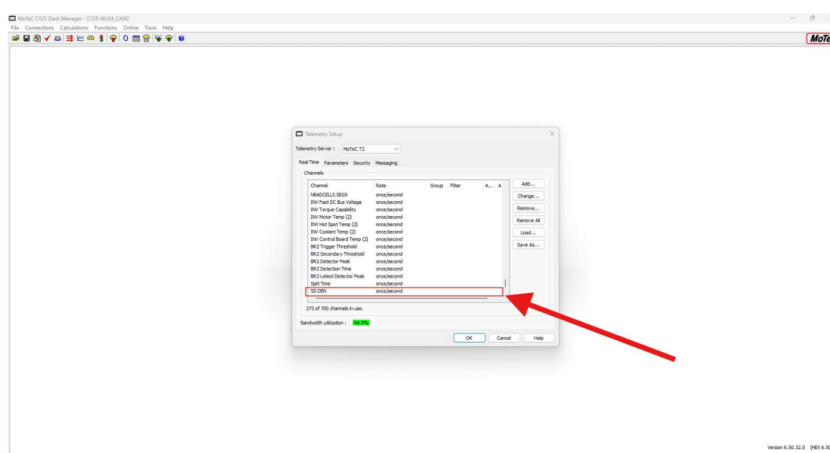
MoTeC Zero-To-Hero



- Ensure the message(s) are in the 'Selected Channels' box, click 'Ok'



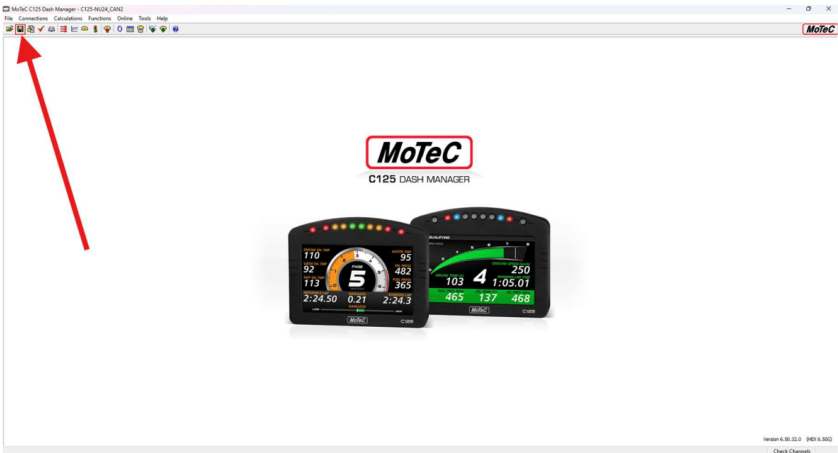
- Scroll to check that the new message(s) has been added correctly
 - o In this section, it is okay to leave it as once/second here, this is how often the signal will update on i2 when on track side



- Select 'OK', then done!
- Then make sure to click 'Save File' in the top left-hand corner
 - o Dash Manager is a funky program, and if this isn't done then changes won't update through the whole program correctly



MoTeC Zero-To-Hero



- Next, upload the config to the Dash. See the Send Config section of this document

MoTeC Zero-To-Hero



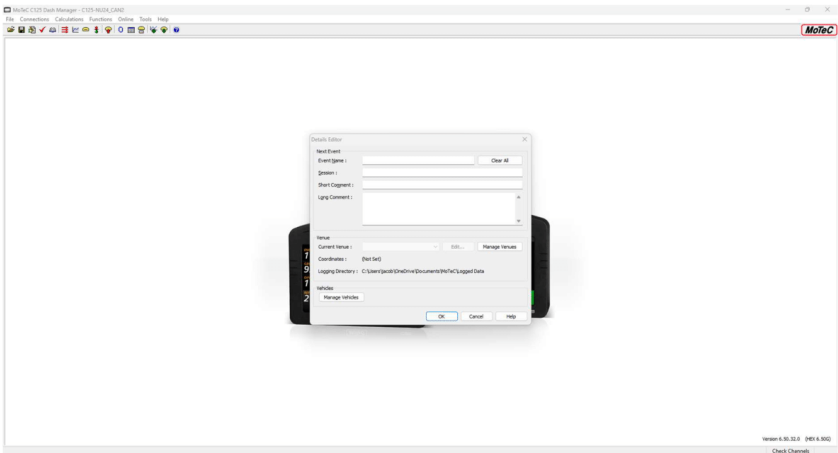
Send Configuration to C125

- Ensure that the desired config file is loaded (see the Opening an Existing Configuration File section for more information)
- Ensure the C125 Dash is connected through an ethernet cable to the computer uploading the config
 - o On NU23, NU24 and NU25, this port is located on the right-hand side of the Dash Electronic Node (DEN)
 - o This connection can be very dodgy, if the cable is connected and it is not connecting, try a different port, reconnect it, wiggle it around a bit, it should connect after a bit of messing about
- Click 'Send Configuration'

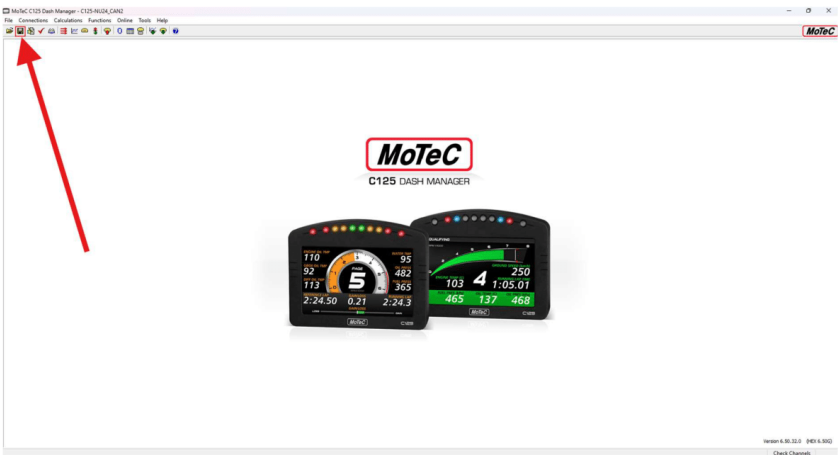


- After a successful connection, the following window should pop up
 - o Fill these out accordingly (on the Chungus, these fields should already be prefilled with whatever was previously entered)
 - o See the Clearing Data Logger at the start of the Day section for more information

MoTeC Zero-To-Hero



- After filling out the fields and selecting 'Ok', the configuration will start uploading. This will take some time to upload so go treat yourself to a snack from TA Snackies (if it hasn't been shut down by the ATO)
- Then make sure to click 'Save File' in the top left-hand corner
 - o Dash Manager is a funky program, and if this isn't done then changes won't update through the whole program correctly



MoTeC Zero-To-Hero



- To see if the new message(s) has been configured correctly, set up the MoTeC transceiver box and Chungus with T2 and i2 set up accordingly (see the Setting up MoTeC for a Track Day section for more information), add the message(s) to the i2 worksheet (see the Editing Worksheets (channels and groups) section for more information), and turn on LV! The message(s) should now be updating! If they are not, try going back through the steps and double check that a step wasn't missed. The author recommends to not use ChatGPT for this stuff (at least at the time of writing, July 2025, as it is not good at knowing what to do, and often gives false information)

MoTeC Zero-To-Hero



Get Configuration from C125

- Ensure the C125 Dash is connected through an ethernet cable to the computer uploading the config
 - o On NU23, NU24 and NU25, this port is located on the right-hand side of the Dash Electronic Node (DEN)
 - o This connection can be very dodgy, if the cable is connected and it is not connecting, try a different port, reconnect it, wiggle it around a bit, it should connect after a bit of messing about
- Click 'Get Configuration'



- It should connect, and the configuration should be able to be edited and changed

MoTeC Zero-To-Hero



Pulling Data

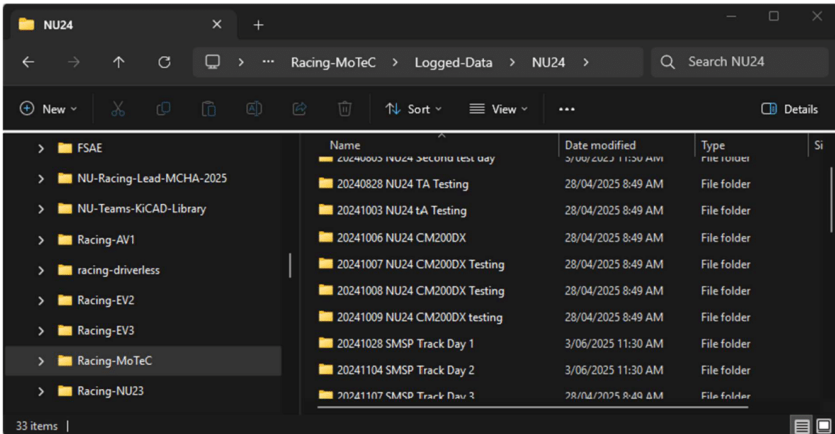
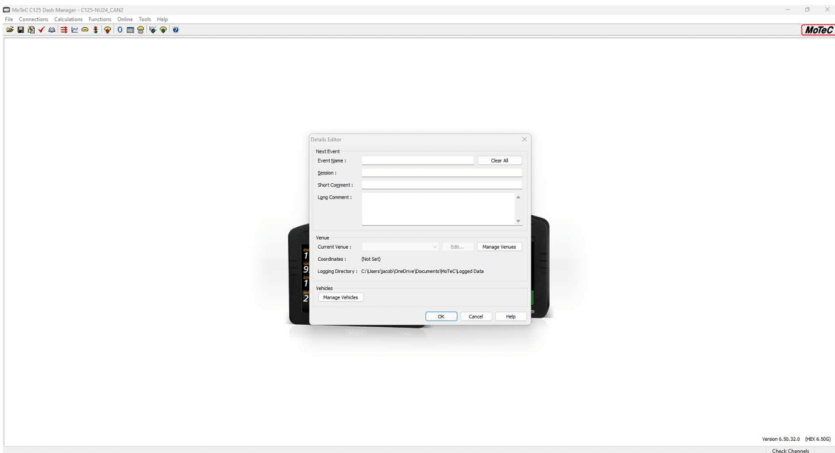
- Ensure the C125 Dash is connected through an ethernet cable to the computer uploading the config
 - o On NU23, NU24 and NU25, this port is located on the right-hand side of the Dash Electronic Node (DEN)
- Select 'Get Logged Data'
 - o This connection can be very dodgy, if the cable is connected and it is not connecting, try a different port, reconnect it, wiggle it around a bit, it should connect after a bit of messing about



- After a successful connection, the following window should pop up
 - o Fill these out accordingly (on the Chungus, these fields should already be prefilled with whatever was previously entered)
 - o Ensure the correct log file has been selected in the 'Current Venue' section. If this is not configured correctly, then after pulling data, the file will be stored in a random location and impossible to find
 - Log files are classically stored in the 'Logged-Data' folder in the Racing-MoTeC Github Repo, in the relevant car name folder, in a folder with the following naming scheme:
 - YYYYMMDD name-of-test (example given below)
 - For this step, the Racing-MoTeC repo is required, see the Racing-MoTeC Repository section for more information
 - o See the Clearing Data Logger at the start of the Day section for more information



MoTeC Zero-To-Hero



MoTeC Zero-To-Hero



Live Signal Monitoring using Dash Manager

- Ensure the C125 Dash is connected through an ethernet cable to the computer uploading the config
 - o On NU23, NU24 and NU25, this port is located on the right-hand side of the Dash Electronic Node (DEN)
 - o This connection can be very dodgy, if the cable is connected and it is not connecting, try a different port, reconnect it, wiggle it around a bit, it should connect after a bit of messing about
- Select Monitor Channels



- Once it connects, a full list of all the messages that the MoTeC transmits should be shown, bit annoying to scroll through and find the message(s) that are wanting to be monitored, but such is life

MoTeC Zero-To-Hero



i2

Live Signal Monitoring using i2

- Data tab (on right hand screen) > 'Telemetry' > 'Add...' (bottom left) > select name of telemetry source (this should show up if the motec transceiver box and T2 is opened and the graphs are green) > Ok > let it connect > Close > data should be updating now

Opening a log file

Editing Worksheets (channels and groups)

Display Creator

To edit the screen with Display Creator, a DBC-style file must be made from within Dash Manager, exported to Display Creator, then a Display Creator Package is exported back to Dash Manager to be uploaded to the C125. After doing this, the desired signals will be displayed on the screen.

WARNING: The MoTeC C125 has been known to have issues with being overloaded with having too many signals being displayed. This can be avoided by only uploading the minimum number of signals to the display, and nothing more. The issues that the C125 has include signal values not displaying correctly and the button to change pages not working.

NOTE: only the signals that are added to be displayed on the dash count within this requirement. The display does not care about all the other signals that are on the CAN Bus

The following steps detail how to:

- export the DBC-style file from Dash Manager to Display Creator
 - o -> Exporting DBC-Style File from Dash Manager to Display Creator
- assign signals to values on the display
 - o -> Assigning Signals to Values on Display
- export the Display Creator Package back to Dash Manager
 - o -> Exporting Display Creator Package to Dash Manager
- upload to the C125.
 - o -> Uploading Display Creator Package to C125 from Dash Manager

MoTeC Zero-To-Hero



Exporting DBC-Style File from Dash Manager to Display Creator

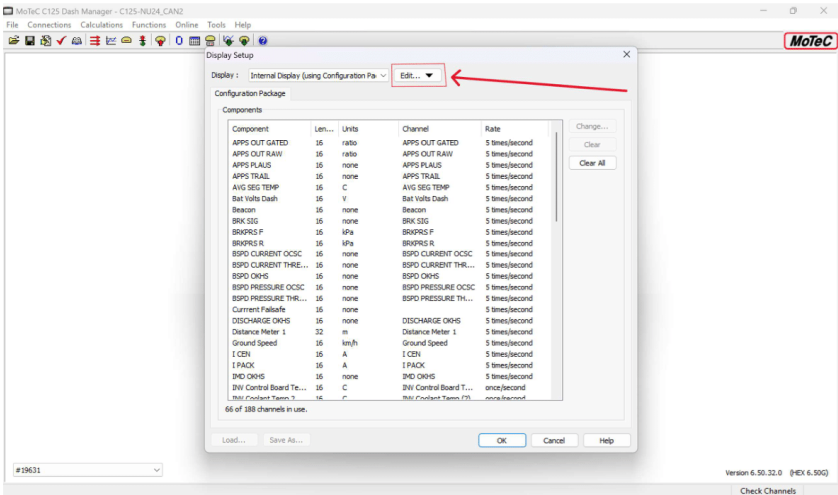
- Open Dash Manger, open the Dash Configuration File (see Opening an Existing Configuration File)
- Navigate to Functions > Display...



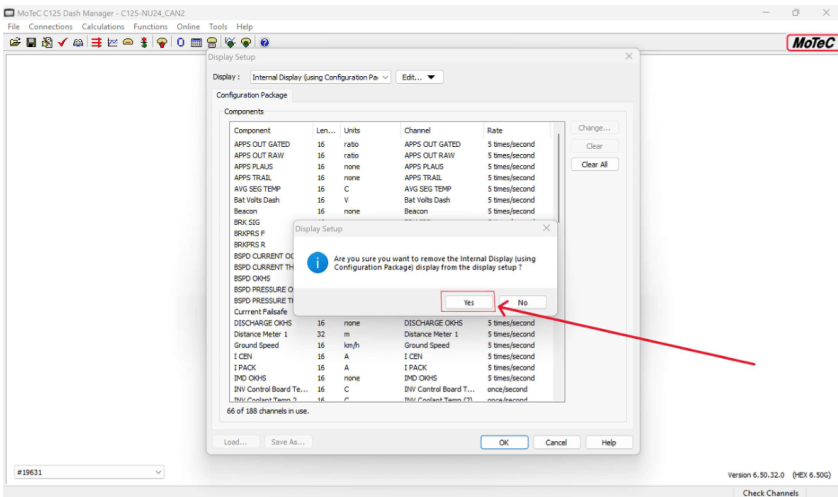
MoTeC Zero-To-Hero



- If there are already channels (signals) added to the Configuration Package, they must be removed to start fresh. To do this, select Edit... > Remove



- Select Yes

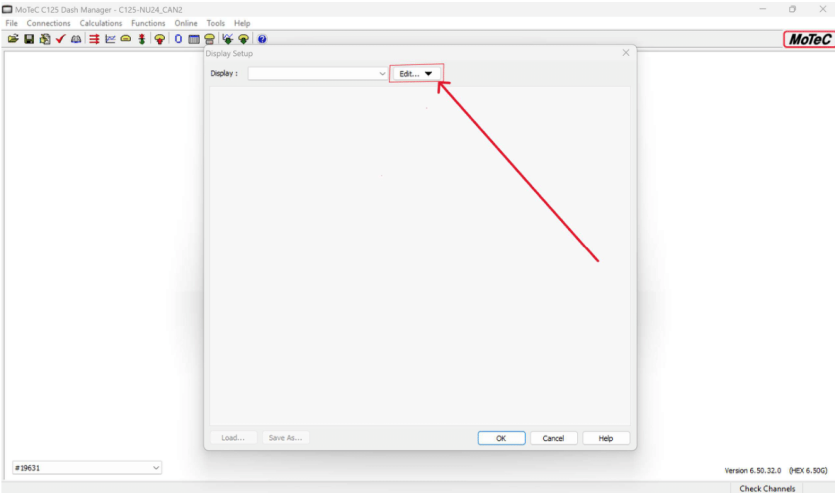




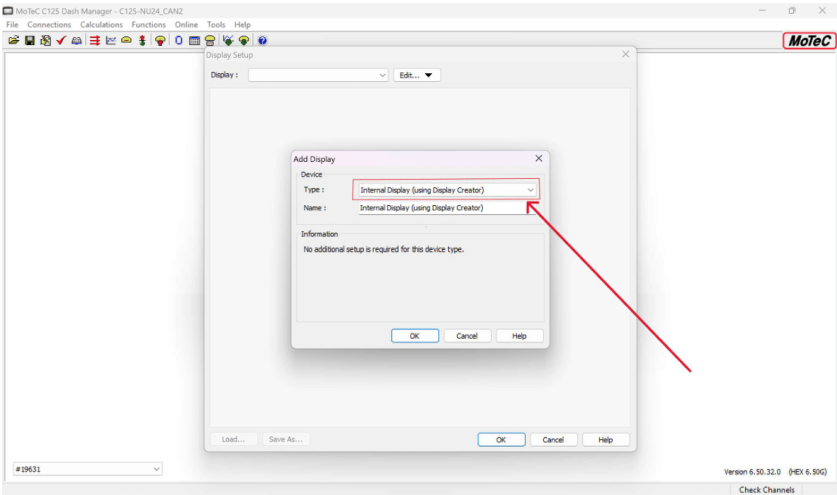
MoTeC Zero-To-Hero



- Now with a blank display, select Edit... again > Add



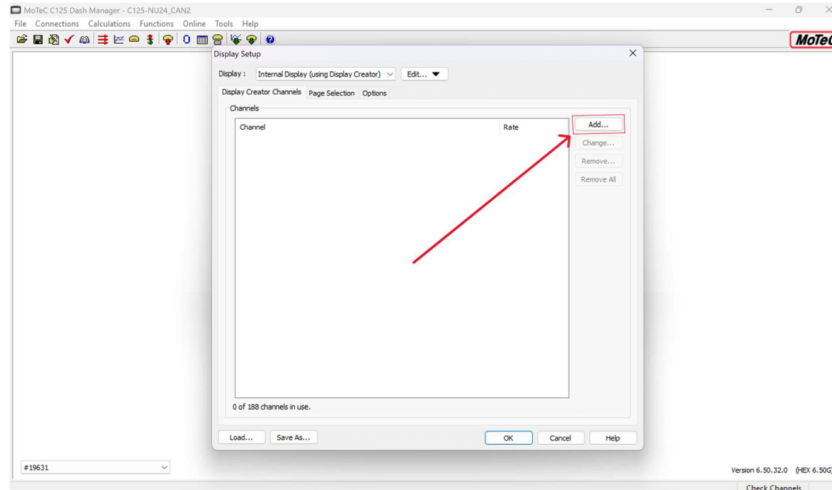
- Select the drop-down box, change 'Internal Display (Fixed Layout)' to 'Internal Display (Using Display Creator)', then select 'Ok'



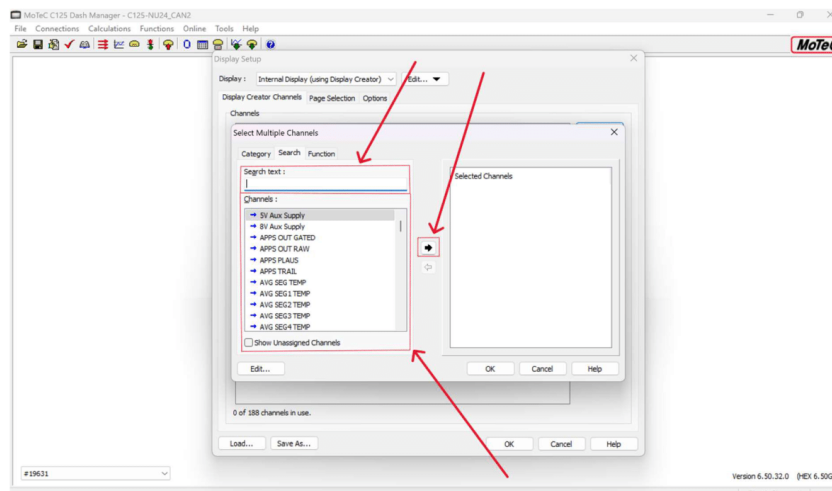
MoTeC Zero-To-Hero



- Now select Add...



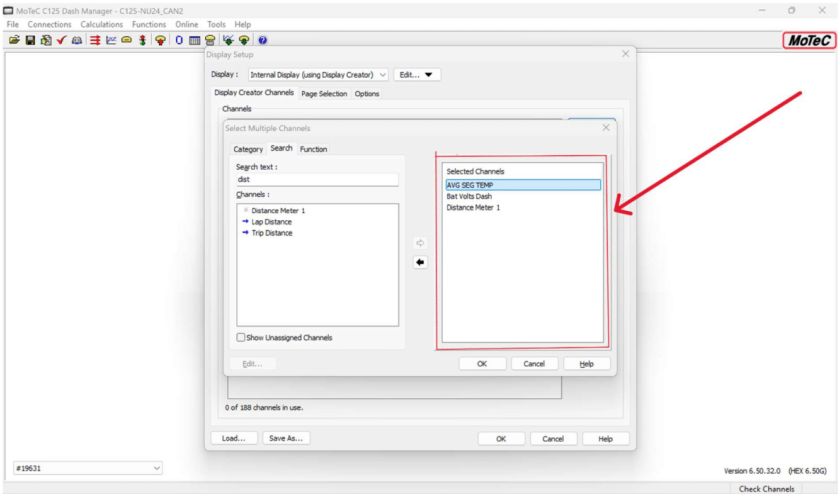
- Then, the signals to be displayed on the dash can be selected, search for signals with the 'Search Text:' box, or, scroll through the list shown. When a signal is selected, the arrow in the middle must be clicked to bring the signal to the 'Selected Channels' Section
 - o NOTE: multiple signals can be selected before clicking the arrow to bring them across



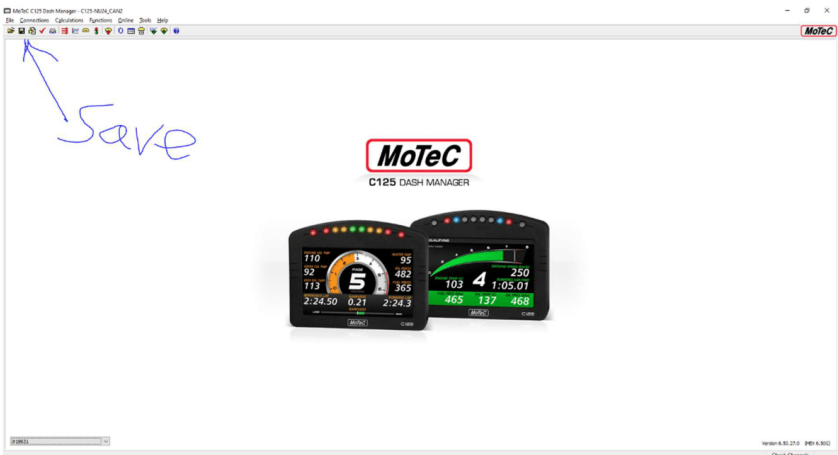
MoTeC Zero-To-Hero



- Signals added to the 'Selected Channels':



- Click 'Ok' to close the Select Multiple Channels window, then 'Ok' to close the Display Setup window. Then click save!



MoTeC Zero-To-Hero



- Then reopen Display Setup (Functions > Display...)

[Assigning Signals to Values on Display](#)

[Exporting Display Creator Package to Dash Manager](#)

[Uploading Display Creator Package to C125 from Dash Manager](#)

[Pages](#)

[DBC](#)

[Other Functions](#)

Settings, alarms, video clips, simulate

T2

- T2 can only be opened when the 'MoTeC Discovery' task is not running
 - o MoTeC Discovery task opens when MoTeC Dash Manager is running and connected to the motec -> happened with matt on the boat

[Using T2 with i2 Pro](#)

[Set up on the Car](#)

[GPS](#)

[Transceiver](#)

[Dash](#)

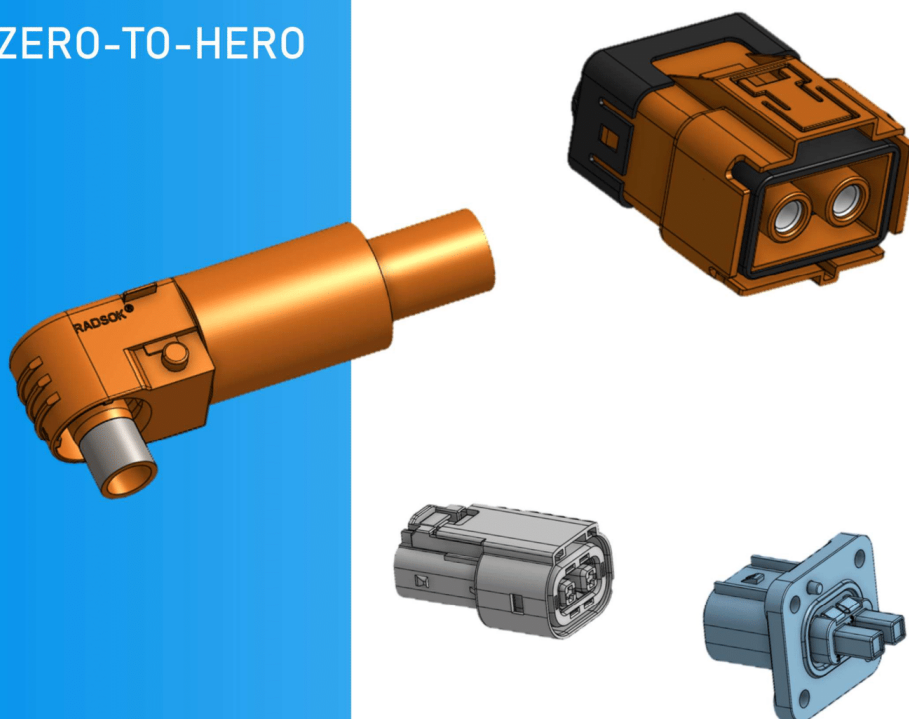
[Ethernet Access Point](#)

17.9. Appendix I – How to Crimp NU25 HV Connectors



How To Crimp NU25 HV Connectors

ZERO-TO-HERO



Jacob Lukes
2025





How To Crimp NU25 HV Connectors



Contents

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How To Crimp NU25 HV Connectors



Sanity Checks

- Ensure all components needed for crimping the connectors are present
- Make double sure the crimp needs to happen
- Ensure correct crimp orientation of connectors

HIP to Motor Controller

SLPPB50BSO0EH / SLPPB50BSO1EH

- **SLPPB50BSO0EH** is a **90-degree** keyway connector
- **SLPPB50BSO1EH** is a **270-degree** keyway connector
- Ensure the following components are present (see Figure 1, part A)
 - Connector
 - Inner copper adaptor
 - Not glue-lined Heat-shrink
 - Glue-lined heat-shrink
 - Rubber gromet
 - Plastic cable ring
 - Cable retainer
- Step 0
 - Check rotation of connector when it is crimped to cable (permanent marker is suggested to get this right)
- Step 1
 - Thread on components in this order (see Figure 1, part B):
 - Cable retainer (ensure facing correct direction)
 - Plastic cable ring
 - Rubber gromet
 - Not glue-lined heat-shrink
 - Glue-lined heat-shrink

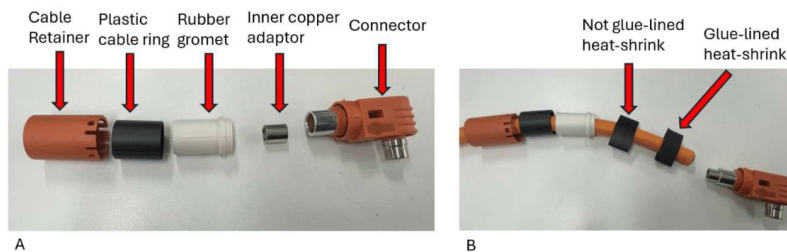


Figure 1

How To Crimp NU25 HV Connectors



- Step 2
 - Strip back 28 mm of outer insulation, using either a knife (refer to: Stanley Knife) or a cable stripping tool
 - Cut off and shorten outer foil shield down to the end of the outer insulation
 - Carefully and slowly unbraid wire shielding
 - Once unbraided, use scissors (refer to: Scissors) to shorten back the wire shielding back to the original 28 mm length (see Figure 2)



Figure 2

- Step 3
 - Using cable shears (refer to: Cable Shears), cut off 10 mm of the inner copper and insulation (as seen in Figure 2 and Figure 3)



Figure 3

How To Crimp NU25 HV Connectors



- Step 4
 - Curl back the wire shielding
 - Strip back 15 mm of the inner copper core insulation, using a Stanley Knife
 - Insert inner copper adaptor
 - See Figure 4



Figure 4

- Step 5
 - Insert and crimp the connector to the copper, after the first crimp, rotate $\sim 45^\circ$ to ensure strong crimp
 - Use the big Blue Handle Cable , on the 95 mm setting, see Figure 5
 - Figure 6 shows what happens if 70 mm setting is used



Figure 5



Figure 6

How To Crimp NU25 HV Connectors



- Step 6
 - Thread the glue-lined heat-shrink over the wire shielding to cover the crimp
 - See Figure 7, part A
 - **Ensure that the crimp and the shielding are not touching**
 - If they do touch after the heat shrink is shrunk down, then when it goes on the car there will be an IMD fault, due to HV and LV ground touching. You'll be chasing your tail for hours
 - Heat-shrink it down
 - Heat gun at ~250 °C, with second or third flow rate setting
 - Figure 7 part B and C show how the crimp and shield wires are not touching
 - The outer insulation and shielding may need to be pulled down to ensure that the wire shield and crimp are not touching



A



B



C

Figure 7

- Step 7
 - Bring up and shrink down the not glue-lined heat-shrink, to cover and secure the wire shielding, refer to Figure 8



Figure 8

How To Crimp NU25 HV Connectors



- Step 8
 - Bring up rubber gromet, see Figure 9
 - You may need to cut and comprise this gromet, depending on how Step 10 goes



Figure 9

- Step 9
 - Bring up plastic cable ring to cover rubber gromet

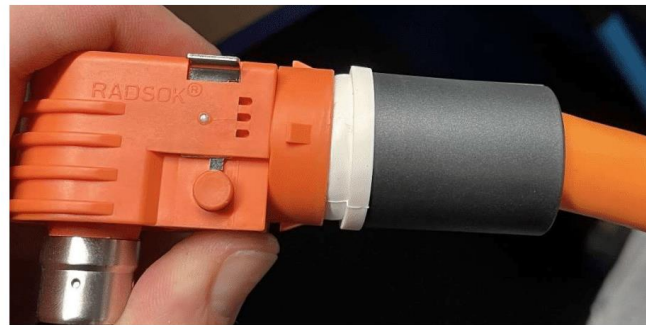
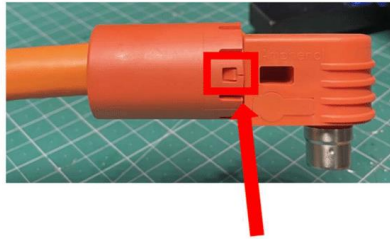


Figure 10

How To Crimp NU25 HV Connectors



- Step 10
 - Bring up the cable retainer, ensure that it comes up far enough for the clips to click in (refer to Figure 11 for the clips in question)

*Figure 11*

- This part will most likely take a lot of force, if there is no hope of it working, go back to Step 8 and cut/'comprise' the gromet, so that when the cable retainer is pushed up, the gromet will break inside of it. This should allow the retainer to easily clip up
- See Figure 12 for the completed assembly

*Figure 12*

How To Crimp NU25 HV Connectors



H1P to DCDC/Energy Meter

2103124-4

- Assembly instructions found in “2103124-4 Assembly Instructions” pdf available on the NU Teams SharePoint shown in the folder shown below. (9/01/2026: I meant to append this document at the end of this one, but do not have time to do this before my report is due 😞)

Documents > NU Racing > Electric Vehicle > 2025 > MCHA > HV Connectors > HIP to DCDC and Energy Meter

Name	Modified	Modified By	Part Number	+ Add column
2103124-4	July 8, 2025	Jacob Lukes		
4-2103177-4	July 8, 2025	Jacob Lukes		
HV Rating compliance	July 8, 2025	Jacob Lukes		
2103124-4 Assembly Instructions.pdf	July 8, 2025	Jacob Lukes		

4-2103177-4

- Assembly instructions found in “4-2103177-4 Assembly Instructions” pdf available on the NU Teams SharePoint shown in the folder shown below. (9/01/2026: I meant to append this document at the end of this one, but do not have time to do this before my report is due 😞)

Documents > NU Racing > Electric Vehicle > 2025 > MCHA > HV Connectors > HIP to DCDC and Energy Meter

Name	Modified	Modified By	Part Number	+ Add column
2103124-4	July 8, 2025	Jacob Lukes		
4-2103177-4	July 8, 2025	Jacob Lukes		
HV Rating compliance	July 8, 2025	Jacob Lukes		
2103124-4 Assembly Instructions.pdf	July 8, 2025	Jacob Lukes		
4-2103177-4 and 2103124-4 Required Components.txt	July 8, 2025	Jacob Lukes		
4-2103177-4 Assembly Instructions - HV RATING.pdf	July 8, 2025	Jacob Lukes		

How To Crimp NU25 HV Connectors



HIP to Accumulator

ELPA2A2510

- Use 25 mm² on Blue Handle Cable Crimpers
- Make sure copper is all the way up to the hole on the pin



- Rotate crimp when crimping



- Use 25 mm² setting





How To Crimp NU25 HV Connectors



- Further instructions can be found in the “ELPA2A2510 & ELRA2A0301 Assembly Instructions” pdf available in the folder shown below. (9/01/2026: I meant to append this document at the end of this one, but do not have time to do this before my report is due 😞)

Documents > NU Racing > Electric Vehicle > 2025 > MCHA > HV Connectors > HIP to Accum

Name	Modified	Modified By	Part Number	+ Add column
ELPA2A2510	July 8, 2025	Jacob Lukes		
ELRA2A0301	July 8, 2025	Jacob Lukes		
ELPA2A2510 & ELRA2A0301 Assembly Instructions.pdf	A few seconds ago	Jacob Lukes		
epower-lite-series-ib-13.pdf	July 8, 2025	Jacob Lukes		

Tools
Scissors



Figure 13 Scissors



Cable Shears



Figure 14 Cable Shears

Blue Handle Cable Crimpers



Figure 15 Blue Handle Cable Crimpers



Stanley Knife



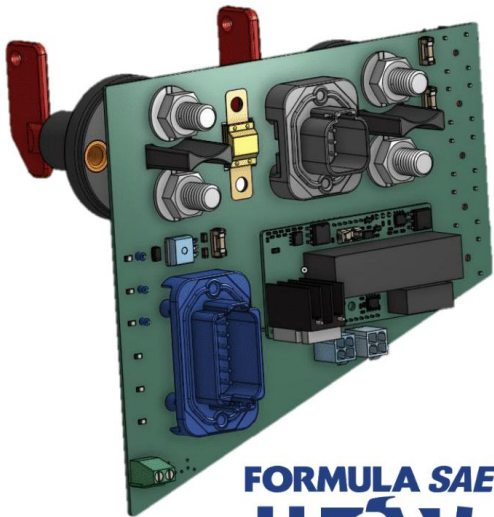
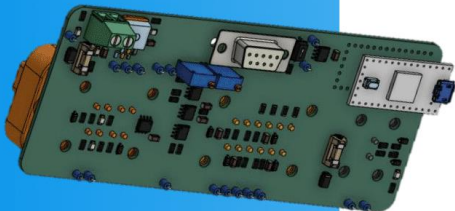
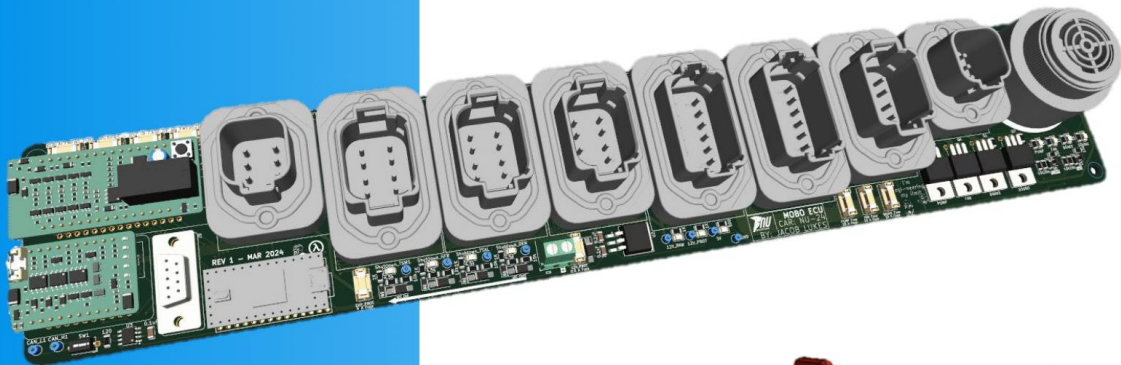
Figure 16 Stanely Knife

17.10. [Appendix J – How To Import PCBs Into OnShape](#)



How To Import PCBs Into OnShape

ZERO-TO-HERO



Jacob Lukes
 2025



How To Import PCBs Into OnShape



Contents

Sanity Checks 2

How To Import PCB Into OnShape..... 2

How To Import PCBs Into an Assembly..... 8

How To Import PCBs Into OnShape



Sanity Checks

- Firstly, ensure **ALL** step files for footprints are present and correctly positioned
 - This includes:
 - All dimensions for components are correct, mainly large components that protrude from the board. It is the PCB designer's responsibility to know this
 - Checking that mating connectors and attached wires are accounted for
 - eg: Micro/Mini Fit MOLEX connectors that attach to the board in any way
 - This can be as simple as editing the original STEP file for the connector, and extruding it
 - Getting this part right saves **LOTS** of time down the line for the individual making the enclosure and the team as a whole
 - Refer to PCB Design Zero-To-Hero on how to make and attach STEP files to footprints

How To Import PCB Into OnShape

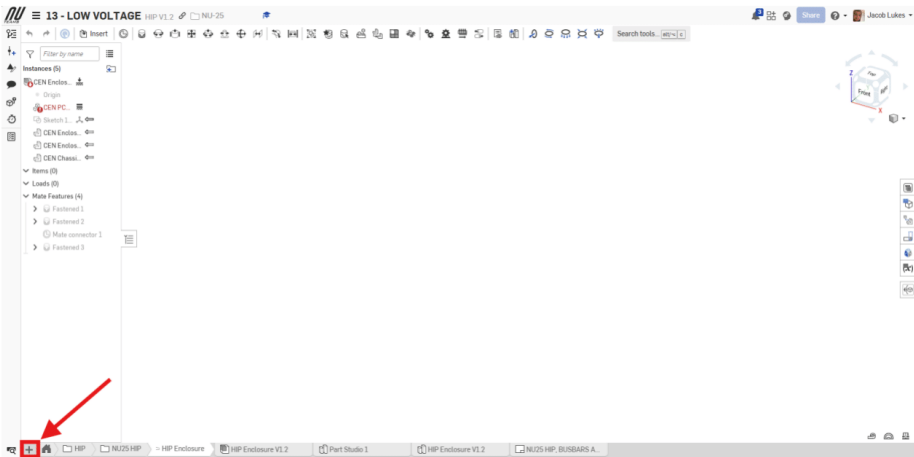
- From within KiCad, export the PCB as a STEP file
- Open OnShape, ensure you are in the correct NU Racing Project Folder for the given year (eg: NU-25)
 - You may need to gain access from the Mechanical Department Leader first before being allowed to complete the following steps
 - If you are not allowed access, please get the enclosure designer to read this document
- Navigate to the desired document (eg: LOW VOLTAGE)



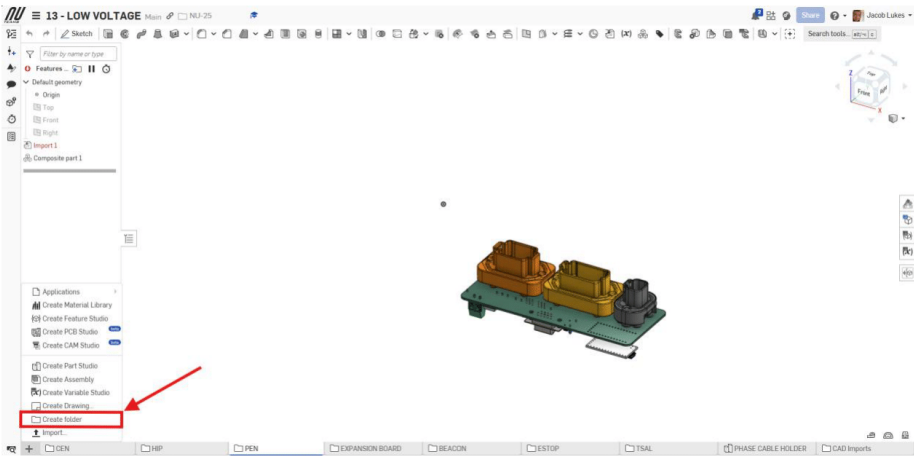
How To Import PCBs Into OnShape



- Select the plus button on the bottom left-hand side of the screen



- Create a folder

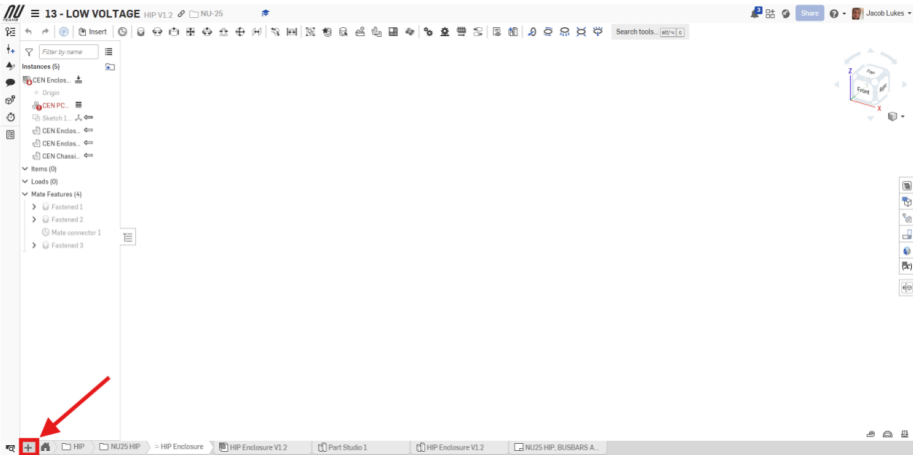




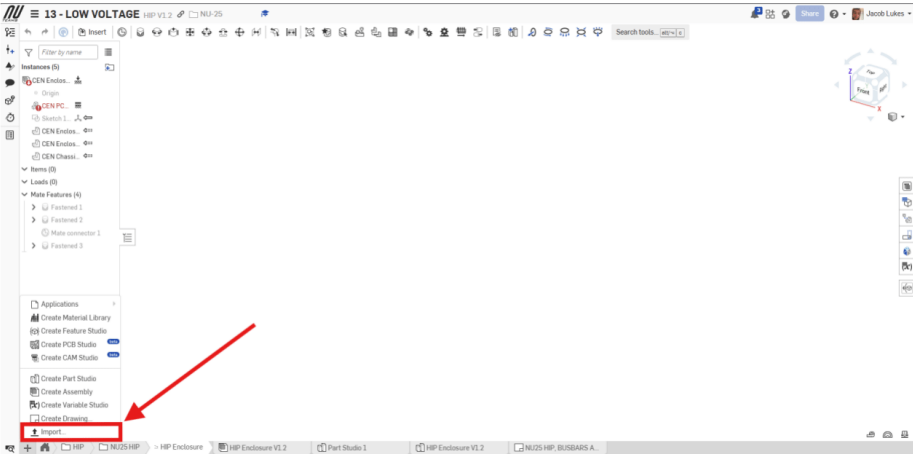
How To Import PCBs Into OnShape



- Again select the plus button on the bottom left-hand side of the screen



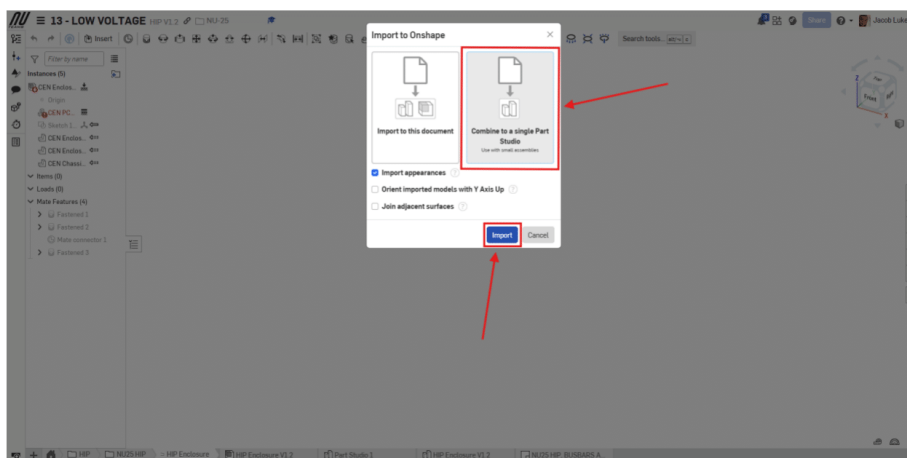
- Select 'Import'



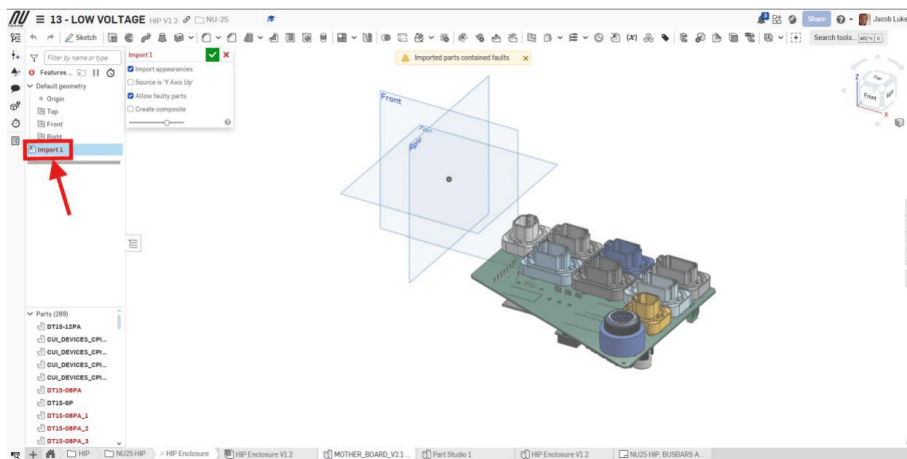
How To Import PCBs Into OnShape



- Navigate to desired step file on PC
- Click '**Combine to a single Part Studio**'
 - **Tick:**
 - 'Import appearances'
 - **Untick:**
 - 'Orient imported models with Y Axis Up'
 - 'Join adjacent surfaces'
- Click 'import'



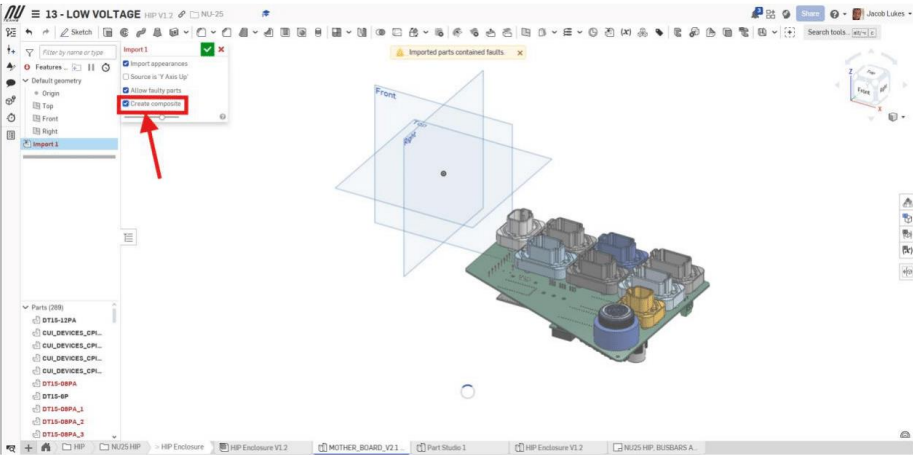
- **Double click the imported part**, usually 'Import 1's'



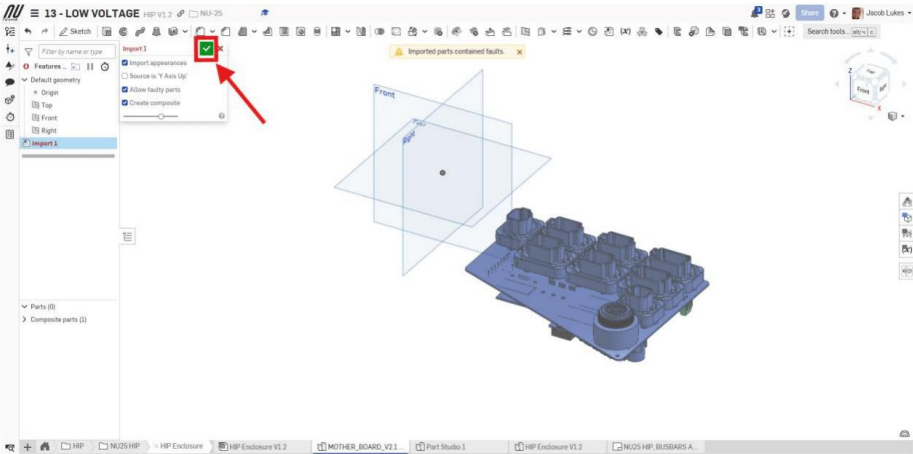
How To Import PCBs Into OnShape



- Tick 'Create composite'



- Click the green check box

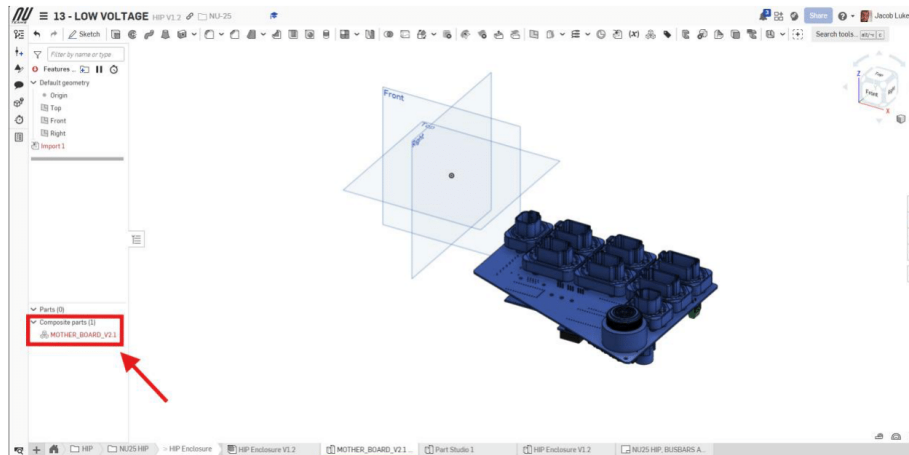


- Wait for it to think

How To Import PCBs Into OnShape



- Ensure the composite part is the name of the given PCB



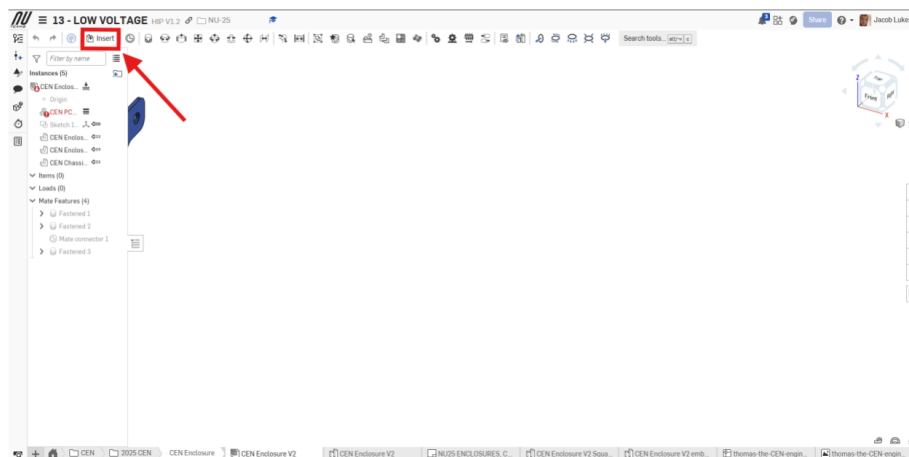
- Done!

How To Import PCBs Into OnShape

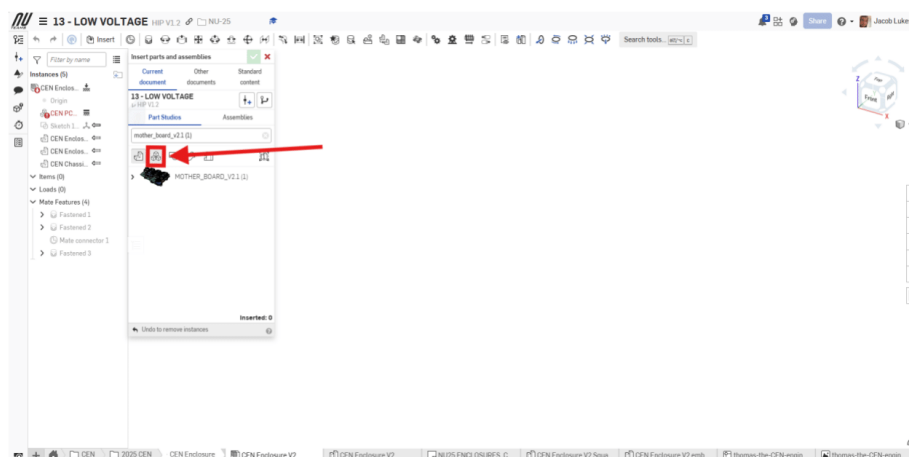


How To Import PCBs Into an Assembly

- **Only complete this section if requested by a member from the mechanical team**
- Navigate to the desired Assembly
- Click '**Insert**'



- Ensure to click '**Composite parts**'



- Search for the desired PCB
- Clicking on the part will bring it into the Assembly